COMPANION TEXT

The Spine You Will Touch

The reading for HOH 3015

Chiropractic History, Theory, and Practice

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Vermont State University · VTSU Online

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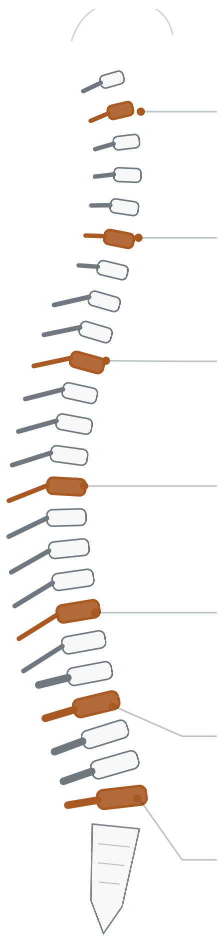


01 WEEK ONE
Origins and First Principles

02 WEEK TWO
The Spine You Will Touch

03 WEEK THREE
The Art of Technique

04 WEEK FOUR
Becoming One

05 WEEK FIVE
In the Clinic

06 WEEK SIX
Alongside Everyone Else

07 WEEK SEVEN
What You Do With This

Contents

This is the print edition. The diagrams in the online reader can be stepped through, switched between layers, and hovered for detail. Here they are shown in their finished state. If a figure looks dense on the page, open the same figure in the Canvas reader and take it apart.

CHAPTER 1

Origins and First Principles

> *Before you can decide what you think about chiropractic, can you state what it claims, in the words it uses about itself, well enough that a believer would nod?*

I want to start this course by asking you to do something that is harder than it sounds. For the next twenty pages or so, I am going to ask you to hold your verdict. Not suspend your judgment forever, and not pretend you have no opinions, but simply postpone the moment where you decide. Most people arrive at a subject like this one already holding a position. Some of you were adjusted as children and have a warm feeling about the whole enterprise. Some of you have a relative who thinks it is nonsense and you half agree. Both of those are ordinary places to start. Neither of them is knowledge yet, because a position you have not been able to state in its own terms is not a position, it is a reflex.

So here is the order of business. First, what was already in the world before 1895, because the founding of chiropractic did not happen on empty ground and you will misunderstand it badly if you think it did. Then the founding itself, which is a genuinely interesting historical mess and which I am going to lay out for you without cleaning it up. Then the two men, the name, the school, and the books the profession wrote about itself. And then the part that matters most for the rest of the semester: the philosophy, presented in its own vocabulary, at its strongest, before anybody evaluates anything. If at the end of this chapter you can explain innate intelligence to a friend in a way a devout practitioner would recognize as fair, you have done what week one asks of you. What you then think about it is your business, and we will come back to it in week seven.

I should tell you where I stand, since you will be reading me for seven weeks. I am a licensed Doctor of Chiropractic. I also hold a master's degree in pharmacology, which means I spent several years learning in fine detail how molecules change what bodies do, which is not the natural training of a vitalist. I adjust patients most days of the week and I believe the work is worth doing. I also think a great deal of what my profession says about itself is not well supported, and I will say so where I think so. If that combination seems inconsistent to you, hold onto that feeling. Sorting it out is more or less the point of the course.

What a healer already had to work with

Human beings have been pushing on each other's spines for a very long time. The clearest early written record we have in the Western tradition is the collection of medical books associated with Hippocrates, a body of roughly sixty texts tied to a lifetime running from about 460 to 370 before the common era. The relevant material is mostly in a treatise usually called *On Articulations*, with related passages scattered through works on fractures and on the nature of bones. What those texts describe is surprisingly concrete. There is a ladder, to which a patient could be bound either upright or upside down and then shaken, a process called succussion, so that the weight of the body itself pulled the spine longer. There is a board. There is a pivoting bench, the scamnum, that combined lengthwise pull with pressure applied at one spot. And there are instructions for the hands: the text directs that the physician, or some person who is strong and not uninstructed, should apply the palm of the hand to the hump, and elsewhere describes using the foot to apply gentle pressure to a spinal deformity.

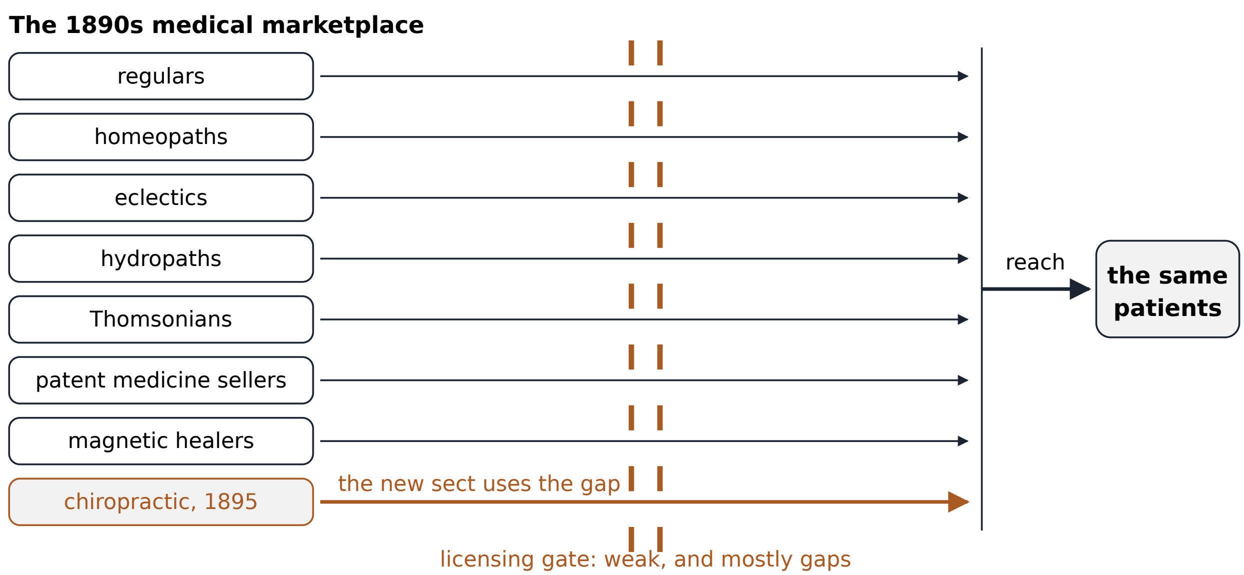
Read that carefully and notice what is there and what is not. What is there is manual treatment of the spine, described in operational detail, by a literate medical tradition, more than two thousand years ago. What is not there is any claim that a displaced vertebra is a general cause of illness. The Hippocratic writer is fixing a visible deformity or a dislocation. He is doing a repair. He is not proposing that the spine is the hidden control panel of the whole organism. That distinction is going to come up again, so keep it: a technique and a theory of disease are different kinds of thing, and having the first does not give you the second. When you see chiropractic marketing that cites Hippocrates as an ancient ancestor, what has usually happened is that a repair manual has been quietly promoted into a philosophy of health. Notice the promotion when it happens.

Closer to Palmer's own world, and much closer in social texture, was bonesetting. Bonesetters were folk practitioners who reduced dislocations, set fractures, and worked on joints that had gone wrong. Before chiropractic and osteopathy existed, they were the people who did this work for most of the population that could not afford or did not trust a physician. What is striking about the trade is how it was transmitted. It ran in families, generation to generation, largely by watching. In Britain there were bonesetting dynasties with names people knew for a hundred miles: the Taylors of Whitworth, the Thomas family of Anglesey, the Matthew family of the Midlands, several of them practising for more than two centuries. There were famous individuals, like Sally Mapp in eighteenth century England and Charles Sweet in New England. Most bonesetters were self taught in the sense that mattered, which is that nobody examined them. The practice is genuinely ancient in a broader sense too. Egyptian management of bone injuries is recorded in a papyrus from about fifteen hundred years before the common era. And bonesetting has not vanished. In parts of South America, Asia, and Africa where orthopedic specialists are scarce, it is still what many people have access to.

Here is the thing to hold onto about bonesetters. They had skill and no apparatus around the skill. No curriculum, no examination, no credential, no journal, no association, and above all no general theory claiming that manual work on the frame could explain illness at large. When a bonesetter died, the practice died with her or passed to one relative. When she was accused of harming somebody, there was no body to answer for her and no written standard she could be measured against and found to have met. Skill that lives only in hands is invisible to a courtroom and invisible to an argument. Much of what follows in this book is the story of a manual practice acquiring exactly the equipment bonesetting lacked, and paying for it in ways nobody at the time anticipated. I should be careful here, because the sources that describe bonesetting are clear that chiropractors and osteopaths eventually replaced bonesetters as the main providers of manual joint treatment, but they do not establish that chiropractic descended from bonesetting. Bonesetting is the weather, not the parent.

The third thing already in place was the American medical marketplace of the 1890s, which was crowded and only weakly policed. Regular physicians competed for patients with homeopaths, eclectics, hydropaths, patent medicine sellers, and magnetic healers, and the boundaries between these groups were argued about far more energetically than they were enforced. It is easy for a modern reader to hear a word like irregular and silently substitute the word fraudulent, but that reading imports a settlement that had not yet happened. Regular medicine at the time had a thin therapeutic arsenal and had not yet consolidated the cultural authority it would hold fifty years later. Patients shopped around. Practitioners advertised. Who counted as a legitimate healer was an open question, and it was answered as much in legislatures and courthouses as in laboratories.

Weak licensure is an abstract phrase, so let me make it concrete. In much of the country, the practical requirements for beginning to treat patients were a room, a sign, and a willingness to be prosecuted if somebody complained loudly enough. Where licensing statutes did exist, they were often narrow, unevenly enforced, and full of definitional gaps about what actually counted as practising medicine. That is the gap a new healing sect walks through. It also explains something students find puzzling later, which is why so much of chiropractic's early history takes place in courtrooms rather than in journals. The gate was legal from the very beginning, so the fight was legal from the very beginning. Chiropractors accumulated something on the order of fifteen thousand prosecutions for practising medicine without a license in the profession's first thirty years. That number comes to us through secondary sources rather than a court census, so treat it as an order of magnitude rather than a count, but even at half that size it tells you what kind of history this is. Chapter 4 takes up what licensure looks like now, and chapter 6 takes up the much later fight with organized medicine.



Several competing healing sects reached the patient of the 1890s through the same weak or missing licensing gate, which is the structural reason a brand new sect could take root at all.

The last piece of the background is Daniel David Palmer himself, who belongs to that marketplace before he belongs to chiropractic. He was born on the seventh of March, 1845, and died on the twentieth of October, 1913, in Los Angeles. From the middle 1880s he practised magnetic healing, first in Burlington, Iowa, and then in Davenport. In his own words: in 1886 I began as a business, and although I practised under the name of magnetic, I did not slap or rub, as others. Magnetic healing was a drugless practice organized around the idea that a vital force runs through the body, that it can be depleted or obstructed, and that a skilled operator can act on it directly with his hands.

That matters more than it looks. Whatever happened in 1895, it did not fall on a blank mind. It fell on a man who already made his living with his hands, who already refused drugs, and who already explained health and illness in terms of an inner force that could be blocked and restored. The basic explanatory move of a magnetic healer is to posit a vital force, locate an obstruction of it, and restore flow by direct action. Change one noun and you have the shape of the argument Palmer would soon be making about nerves, and later of the doctrine of innate intelligence his son elaborated. The therapeutic act changed, from a pass of the hands to a thrust on a vertebra. The grammar of the explanation did not change at all.

> **A note on the sources for this period**
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> The material behind this section is of very uneven quality, and you should know that. Classical texts survive, but working out what ancient practitioners actually did from what they wrote is difficult and argued about. Bonesetting is documented mostly by outsiders, especially physicians who wrote about it disapprovingly, and by reminiscences recorded decades after the fact, so we know a good deal about how bonesetters were regarded and much less about what they did or how their patients fared. Magnetic healing survives largely through advertisements and testimonials, which are evidence of what was claimed rather than of what happened. The thin spot is exactly where a student most wants detail, which is the actual content of the practice. Where I have written a date or a quotation in this chapter, it is because a checkable source carried it. Where I have written that something is disputed, that is not politeness.

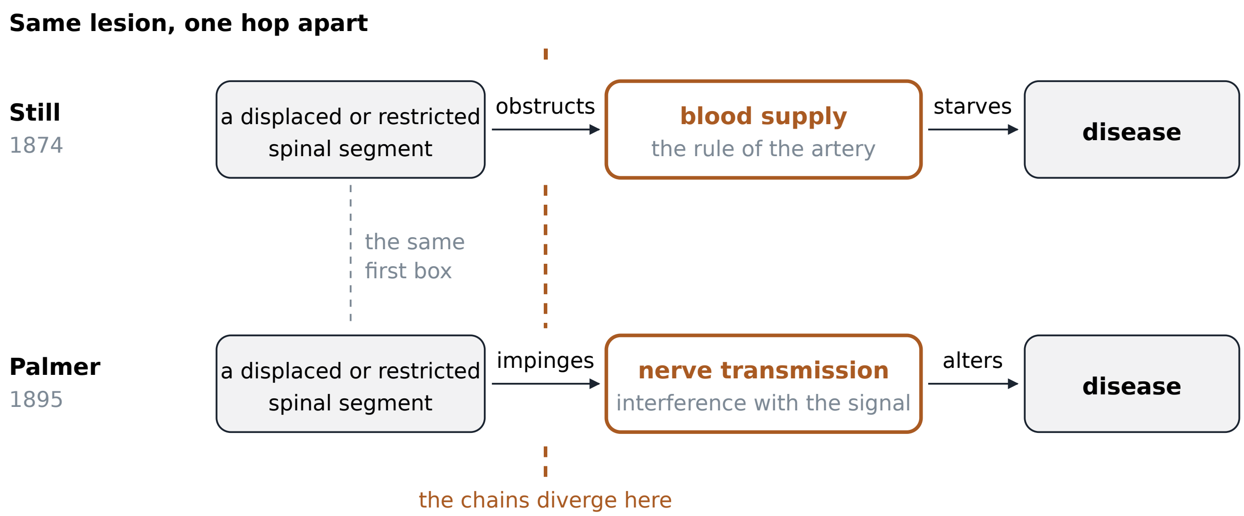
Kirksville, three years early

The layer students most often skip is the most instructive one. Osteopathy began in the United States in 1874, when Andrew Taylor Still, a physician who had lost faith in the medicine of his day, worked out a system of his own. On the twentieth of May, 1892, he opened the American School of Osteopathy in Kirksville, Missouri, with an entering class of twenty one students, and in the same year published a book laying out the philosophy and mechanical principles of what he was doing.

Still's core claim was that disease and physiologic dysfunction were rooted in a disordered musculoskeletal system, so that a physician who could diagnose and treat that system could treat a wide range of diseases. Underneath it sat a premise about the body that will sound extremely familiar in about ten minutes: as Still's son put it, that the body would run smoothly into old age if properly maintained, and that every living organism already possesses the ability to produce all the chemicals and materials needed to cure itself of its ailments. Still routed the mechanism through circulation. Structure governs function, disordered structure obstructs the blood supply that tissues depend on, and correcting the structure by hand restores the conditions under which the body repairs itself.

Two things about Still deserve your attention. The first is that his argument is a real argument, in the sense that each step follows from the one before and the whole thing is short enough to teach in a sentence and long enough to expand into a curriculum. The body contains what it needs for its own repair. Repair depends on delivery. Delivery depends on unobstructed circulation. Structural derangement obstructs circulation. Therefore correcting structure by hand restores the conditions for healing, and the practitioner's job is to find the derangement rather than to supply a remedy from outside. Whatever you conclude about whether it is true, that is a coherent chain.

The second thing is the apparatus. Still did not merely have a theory. He gave his system a name, a founding narrative, a fixed address, a course of study, and a diploma. Three years before anything happened in Davenport, and a few hundred miles to the southwest, somebody had already converted manual treatment into an institution. That is the template. When Palmer built his own school, he was not inventing the idea that a manual practice could have a school.



Osteopathy and chiropractic put the lesion in the same place and then part company at exactly one link in the chain: Still routes the harm through obstructed blood supply, Palmer through interference with nerve transmission.

Set the two chains side by side and the difference reduces to a single link. Both locate the trouble in disturbed structure, usually spinal. Both treat it with the hands. Both refuse drugs. Still then routes the harm through the arteries. Palmer routes it through the nerves. That is one hop apart in mechanism and an enormous distance apart in professional consequence, and the reason is not scientific at all. A distinct mechanism is a distinct territory. It gives a new group its own name for the lesion, its own account of why the neighbours are wrong, and its own claim to be doing something nobody else can do. You will see this logic of distinctiveness at work again in chapter 3, where technique systems differentiate themselves from each other in much the same way.

Did Palmer know about Still? People have argued about this for well over a century, and partisans of both professions have had a stake in the answer. The surviving record does not settle it, and I would be suspicious of anybody who tells you confidently in either direction. Historians generally find priority questions poorly formed anyway. Ideas about structure, obstruction, and force were widely available in that decade. The more useful question is not who thought of it first but who built something durable around it, and the answer to that is that both of them did.

The afternoon that may not have been an afternoon

Now the founding. Almost every account of chiropractic you will ever read opens with the same scene, and I want to walk you through it slowly, because how you handle this story is a decent test of how you are going to handle everything else in the course.

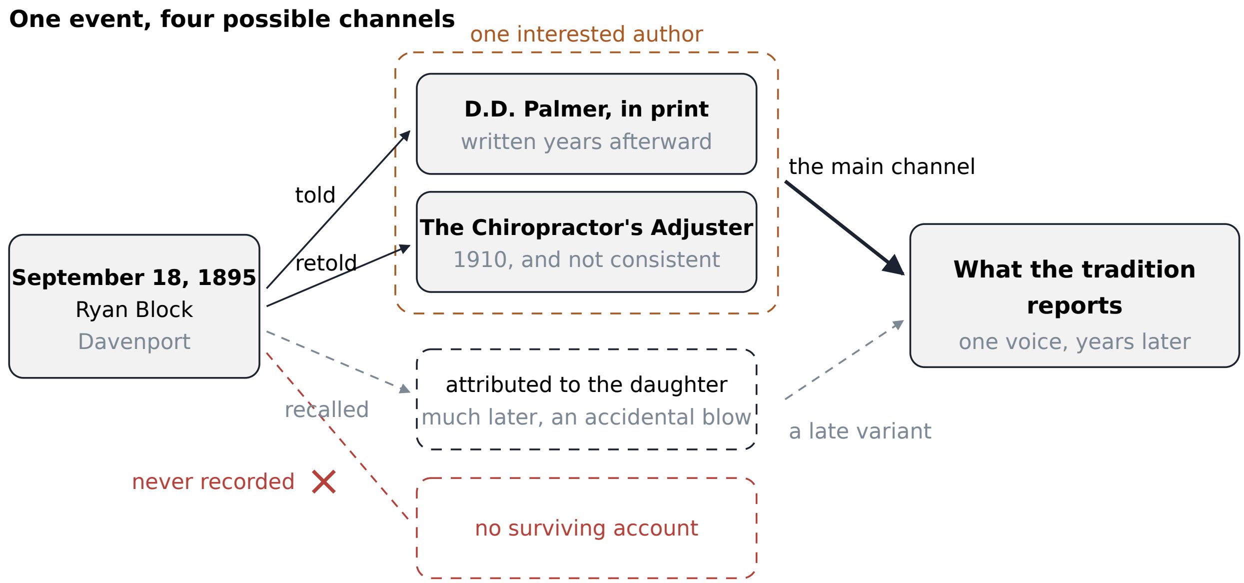
The canonical version runs like this. In September of 1895, in Davenport, Iowa, D.D. Palmer encountered Harvey Lillard, a janitor who worked in the building where Palmer had his office and who was largely deaf. Palmer noticed an unusual lump or prominence on Lillard's upper back. He reasoned that a displaced vertebra there might be pressing on a nerve, and that the nerve might have something to do with the hearing. He manipulated the area. Lillard's hearing improved. Chiropractic was born.

That is the story on the plaque. Here is what the documentary record actually contains, and I am going to give you three accounts and then decline to tell you which one is right, because I do not know and neither does anyone else.

Palmer's own account. In his 1910 book Palmer insisted the manipulation was deliberate, purposeful, and specific. He wrote that there was nothing accidental about this, as it was accomplished with an object in view, and he claimed the adjustment was so precisely done that no chiropractor has equalled it. His telling describes finding a lump between Lillard's shoulders and manipulating it, and curing what Palmer took to be a deafness caused by pressure on a nerve.

Lillard's own published account, from 1897. Two years after the event, in a period publication called *The Chiropractic*, Harvey Lillard was quoted saying this: Last January Dr. Palmer told me that my deafness came from an injury in my spine. Dr. Palmer treated me on the spine; in two treatments I could hear quite well. That was eight months ago. Read the dates in that sentence. Lillard puts the first treatment in January, and the interval he gives puts it around January to May of 1895, not September. He also describes two treatments, not one dramatic afternoon. This is the patient himself, in print, closer in time to the event than any of Palmer's retellings, and he does not match the plaque.

The daughter's account. Valdeenia Lillard Simons, Harvey Lillard's daughter, gave a very different version much later. On her telling, her father was not receiving treatment at all. He was standing in the hallway outside Palmer's office telling a joke to a friend. Palmer, laughing at the punchline, slapped him on the back with the hand that happened to be holding the heavy book he had been reading. Some days afterward Lillard mentioned that his hearing seemed better, and only then did Palmer begin exploring manipulation as a business proposition. Simons also described a profit sharing agreement with her father that she says was never honoured.



Three surviving accounts of the same 1895 event disagree about the month, the intent, the number of treatments, and even whether a treatment occurred, and no later source resolves them.

Those three do not fit together. And the disagreement does not stop there. The date is unsettled even within the Palmer family's own retellings and the secondary literature: dates cited range across the sixth, the fifteenth, and the eighteenth of September, with 1895 the most common year, but with some sources giving months from January to April and even alternative years from 1894 to 1896. There is evidence that B.J. Palmer altered details of month and year in later tellings. B.J. also gave sworn testimony describing a different vertebra being adjusted, the axis, high in the neck, than the shoulder blade area described in other accounts. And the daughter's version sits awkwardly against B.J.'s sworn claim that Lillard had been thoroughly deaf, since a man telling a joke to a friend in a hallway and reacting to a slap on the back is not a man who cannot hear at all.

You will also encounter, in the critical literature, a report that Lillard's widow later said he remained deaf until his death. I am not going to repeat that as a fact, because the source that carries it labels it explicitly as an unconfirmed rumour. It is worth knowing that the claim circulates. It is not worth believing on that evidence.

So I am not going to adjudicate. Not because it would be impolite to, but because the record genuinely will not carry a verdict. What I want you to take from this is not a conclusion about 1895. It is a habit. When a story is told with great confidence and the primary documents disagree with each other, the honest move is to lay the documents out and say so. That is not debunking and it is not defending. It is just reading.

A note on why I did not pick a version

Students sometimes read a passage like the one above as coded criticism, as though laying out three conflicting accounts were a polite way of saying the whole thing is a fraud. It is not. Contested founding accounts are extremely common in the history of institutions, including ones nobody doubts. What the conflict here does establish is narrower and more useful: the canonical story is not settled history, so nothing should be built on its details. If you ever hear somebody argue that chiropractic must work because it cured a deaf man in 1895, you now know enough to say that we do not actually know what happened to Harvey Lillard, and that the case for or against the profession has to be made on other grounds. Chapter 5 is where those other grounds are.

What a founding story is for

Here is the part that I think is genuinely interesting, and that gets missed by both the boosters and the critics. A founding story can do enormous institutional work regardless of whether it is accurate history. That is not a cynical observation. It is a structural one.

Think about what a new healing profession in 1900 actually needed. It needed a theory of disease, so it had something to teach. It needed a school, so it could reproduce itself. It needed a credential, so a licensing statute would have something to license. It needed a defended boundary against its neighbours, so that osteopathy and medicine could not simply absorb it. And it needed an origin story, which is the item that people assume is decorative and which turns out to be load bearing. The origin story fixes a founder, a date, a place, and a lineage. It gives internal disputes a way of being settled by appeal to what the founder intended. It gives graduates a thing to belong to. It gives a courtroom a narrative. Remove it and the disputes of the next fifty years have no vocabulary except evidence, which in 1910 nobody on any side of this had much of.

A founding story can be doing serious institutional work whether or not it survives historical scrutiny, and knowing that is different from knowing whether it is true.

So both of the following are true at once, and holding them together is the intellectual skill this chapter is trying to build. The Lillard story is not reliable history. The Lillard story was and is enormously effective at doing what founding stories do. A student who has actually looked at the documents is in a better position than one who has not, in both directions: better protected against a version of the profession that rests its case on a legend, and better protected against a critic who thinks that discrediting a legend discredits everything downstream of it. It does not. Whether adjusting a spine helps a person with low back pain in 2026 has nothing at all to do with what happened to a janitor in Iowa.

The name, the school, and the son

The founding was a sequence, not a moment, and treating it as a moment is already an interpretive choice. The event, whatever it was, came first. The name came next. It was suggested by a local clergyman, the Reverend

Samuel H. Weed, who assembled it from two Greek words, *cheiro*, meaning hand, and *praktikos*, meaning doing or action. Done by hand. Palmer was using the term by 1896.

The school came after that. Palmer founded the Palmer School and Cure in Davenport, which was eventually renamed Palmer College of Chiropractic. I am going to give you the date as 1897, which is what the standard biographical accounts say, but I want to flag honestly that even here the reference sources disagree with each other: at least one otherwise reputable account puts the school's founding in 1896. This is a small thing and nothing rests on it. I mention it because you should get used to the texture of this material, where even the tidy dates are often less tidy than they look in a lecture slide.

Then the trouble. Palmer was arrested more than once for unlicensed practice, and in 1906 he was convicted for professing to cure disease without a license. He sold the school to his son, Bartlett Joshua Palmer, universally known as B.J., who took control in 1906 and who is the reason there is a profession at all rather than a footnote. Under B.J. the school grew past a thousand students by the early 1920s. He was a relentless promoter, an organizer, a broadcaster, and a writer, and he became known within the field as the Philosopher of Chiropractic. He also elaborated his father's ideas into something far more systematic and far more insistent than his father had left them.

The relationship between the two men was, in the plain language of the biographical record, tenuous and often bitter, especially after the sale. I mention this not for the gossip but because it shapes the doctrine. A great deal of what students are taught as the philosophy of chiropractic is B.J.'s elaboration rather than D.D.'s original, and the two are not identical. When somebody says the founder taught such and such, it is always worth asking which founder.

> **A note on straights and mixers**
>
> Almost immediately the profession split, and the split has never fully closed. The party called straights held that spinal adjustment was the whole of chiropractic and that Palmer's vitalist framework was not an optional decoration but the thing itself. The party called mixers combined adjustment with other treatments, exercise, nutrition, physical therapy modalities, and sought accommodation with scientific medicine. B.J. Palmer led the straights and led them hard. You will meet the descendants of this argument everywhere in this course: in technique choice in chapter 3, in scope of practice law in chapter 4, in which conditions a practitioner claims to treat in chapter 5, and in how a chiropractor writes a referral letter in chapter 6. It is a live disagreement inside a real profession, not a historical curiosity, and you should not expect the practitioners you meet to agree with each other.

The Green Books

A profession that wants to be taken seriously writes its own literature, and chiropractic did, in a series that anybody in the field will recognize by sight. The Green Books are named for their dark green binding; the early editions also had a green vine patterned endpaper, which was discontinued in 1947. The series begins with D.D.

Palmer's foundational 1906 volume, *The Science of Chiropractic*, and continues under B.J. Palmer and the faculty of the Palmer School through 1963, a publication span of roughly fifty seven years.

How many volumes there are is a question with a slightly annoying answer. The Palmer College library lists numbered volumes 1 through 29 and 32 through 39, plus a set of unnumbered works, and no single authoritative total. So the defensible statement, and the one I will use, is roughly thirty nine numbered volumes plus additional unnumbered works. If you see a confident exact count somewhere, it is probably someone rounding off a bibliographic mess.

Three volumes are worth knowing by name. Volume 8 is Harry Vedder's *A Textbook in Chiropractic Physiology*. Volume 9 is *Chiropractic Anatomy*, first published in 1918 and running to at least a fifth edition in 1923, written by Mabel Heath Palmer. It is the first anatomy textbook written specifically for chiropractic students, and its author taught anatomy at the Palmer School for four decades and founded what is described as the first chiropractic sorority. She married B.J. Palmer in 1904 and is sometimes called the First Lady of Chiropractic. Volume 13 is B.J. Palmer's own *A Textbook on the Palmer Technique of Chiropractic*. And volume 14, which is the one that matters most for the rest of this chapter, contains two works bound together: James Leroy Nixon's *The Spirit of P.S.C.*, a narrative history of the school from 1920, and Ralph W. Stephenson's *Chiropractic Textbook*, with editions in 1927, 1946, and 1948, which is where the thirty three principles live.

D.D. Palmer's own major statement came in 1910, in a book usually referred to by its short title, *The Chiropractor's Adjuster*, and published under the fuller title *The Science, Art and Philosophy of Chiropractic*. Note that word order, because it matters and because it is almost always given wrong. Palmer wrote science, art and philosophy, in that sequence. The phrase you will see on modern chiropractic practice websites, and on the catalog description of this very course, reorders it as philosophy, art and science. That reordering is a later convention. It is not what the 1910 title says. It is a small thing and I do not think anybody did it deceptively, but a profession that reverses its founder's own priorities and then quotes him is worth noticing, and the shift from putting science first to putting philosophy first is not a random shuffle.

The philosophy, in its own vocabulary

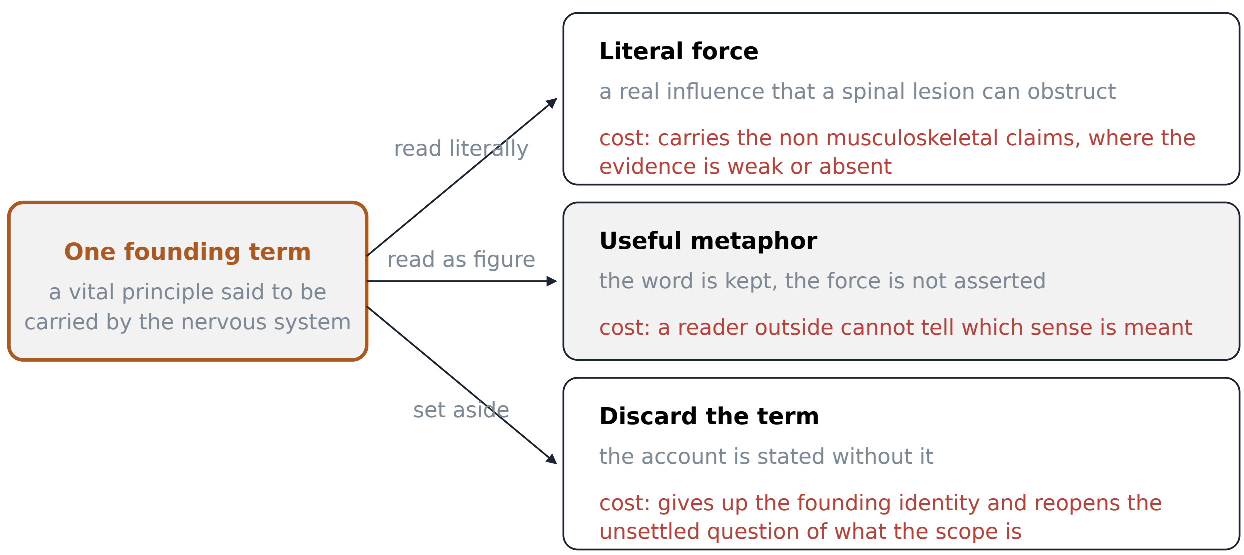
Now the part I asked you to hold your verdict for. I am going to state the philosophy the way a committed practitioner would state it, using the words the tradition uses, and I am going to give it its best hearing. I will not editorialize inside this section. When I have something to add, I will add it afterward, where you can see the seam.

It rests on three moves.

First: the body is self healing and self regulating. This is not a mystical claim in its ordinary form and you already believe most of it. Cut your hand and nobody has to tell the tissue to close. Get too hot and you sweat without deciding to. Eat and your pancreas releases insulin without consulting you. Break a bone, set it, and the healing is done entirely by the patient; the practitioner's contribution was to hold the pieces where they could do their own work. The chiropractic version generalizes this. The body is not a machine that gets repaired from outside. It is an organism with an ongoing, ceaseless capacity to maintain and restore itself, and that capacity, not the practitioner, is what does the healing. Every good outcome in every kind of medicine is ultimately the patient's own tissue doing the work.

Second: the nervous system is the coordinating system. If the body is going to regulate itself, something has to carry the regulation. Something has to tell the vessels to constrict, the gut to move, the muscle to contract, the gland to secrete, and to coordinate all of it in a body where trillions of cells are doing different jobs at once. The nervous system is that something. It is the network that connects the whole organism to itself, it reaches essentially everywhere, and the pathway from the brain to almost everything below the neck runs down the spinal cord and out through openings between the vertebrae. In the chiropractic account, the nervous system is not merely one organ system among several. It is the master system, the one through which the body's self regulating capacity actually operates.

Third: therefore the practitioner's job is to remove interference, not to add something from outside. This is the move that makes chiropractic a distinct profession rather than a technique. If the body already knows how to heal, and if the nervous system is how that knowledge gets delivered, then the most important thing that can go wrong is not a deficiency but an obstruction. Something is interfering with the delivery. The chiropractor's task is to find the interference and remove it, and then to get out of the way. Notice what follows. You do not need to know what disease the patient has in order to know what to do, because you are not treating the disease; you are restoring the conditions under which the body treats it. You do not add a drug, because a drug is something from outside, and outside is the wrong direction. The correction is mechanical, specific, and delivered by hand. The healing that follows is the patient's own.



The vitalist and mechanist accounts of a healing profession answer the same question, what does the practitioner contribute, in opposite directions: one restores a capacity the body already has, the other supplies something the body lacks.

Sit with how coherent that is before you go looking for the holes. It is internally consistent, it is teachable in five minutes, it gives the practitioner a clear job description, and it has a genuine moral appeal: the patient is not a broken object to be fixed but an organism to be unblocked. It also happens to be the reason a lot of patients like going to chiropractors. Somebody puts hands on them, treats their body as fundamentally competent rather than fundamentally defective, and does not hand them a prescription. I have had patients tell me, in almost these words, that they came to me because they wanted to be treated like a person rather than a set of numbers. That is worth taking seriously as a fact about care even if you end up rejecting every word of the theory.

Now the seam, and then I go back to reporting. The word doing the heavy lifting in the third move is *interference*. Everything depends on there actually being an identifiable, locatable, correctable obstruction of nerve transmission, and on our being able to find it reliably. Chapter 2 takes up the anatomy of that question, which is to say what would have to be true about the size and arrangement of the structures for a vertebral misalignment to interfere with a nerve at all. Chapter 3 takes up whether practitioners can reliably agree on where the interference is. I am flagging both now so that you do not think I am hiding them, and then I am going to keep giving the philosophy its full hearing.

Innate intelligence, in Palmer's own terms

Palmer did not merely say the body regulates itself. He named the regulator, and the name is the single most distinctive thing in the whole tradition.

Innate intelligence, in Palmer's formulation, is the soul, spirit, or spark of life, and it controls and directs all the vital functions of the body. It is not a metaphor for physiology in his usage. It is an intelligence, in the ordinary sense of something that knows and directs. And it is not confined to the individual. Palmer's larger claim is that innate is a segment of that intelligence which fills the universe. There is a universal intelligence, present in all matter and organizing everything, and innate intelligence is its individual expression inside a living organism. Your innate is your share of it. Palmer described the idea as arriving like a flash of light, through spiritual insight, and he elsewhere said he received chiropractic from the other world. He and B.J. seriously considered declaring chiropractic a religion and decided against it.

I want you to notice what that does to the structure of the argument, and I want you to notice it without deciding yet whether it is good or bad. It gives the third move, remove the interference, a much stronger foundation. If what runs over the nerves is a form of intelligence directing the body's functions, then interfering with its transmission is not merely a physiological problem, it is an interruption of the thing that runs you. It also explains why straight chiropractic could plausibly claim, for decades, to be relevant to almost any illness. If innate is what maintains the whole body, and if subluxation interferes with innate, then subluxation is upstream of everything. There is no illness the theory excludes in principle.

That last sentence is the strength of the theory and it is also the beginning of its trouble, but I am not going to develop the trouble yet. Hold it.

Thirty three principles and one premise

In 1927, Ralph W. Stephenson published his *Chiropractic Textbook*, which became volume 14 of the Green Books, and in it he did something the founders had not: he turned the philosophy into a formal deductive system. Thirty three numbered principles, each following from the ones before, starting from a single premise and ending at the chiropractic adjustment.

A deductive system is a chain of reasoning where each link is supposed to follow necessarily from the previous one. Geometry is the familiar example: you accept a small number of starting assumptions and then everything else is forced. The appeal of building a discipline this way is obvious. If the chain is valid, then anybody who accepts the starting point has to accept the conclusion. It also does something socially useful for a young profession under attack. It makes the philosophy look like mathematics rather than like enthusiasm.

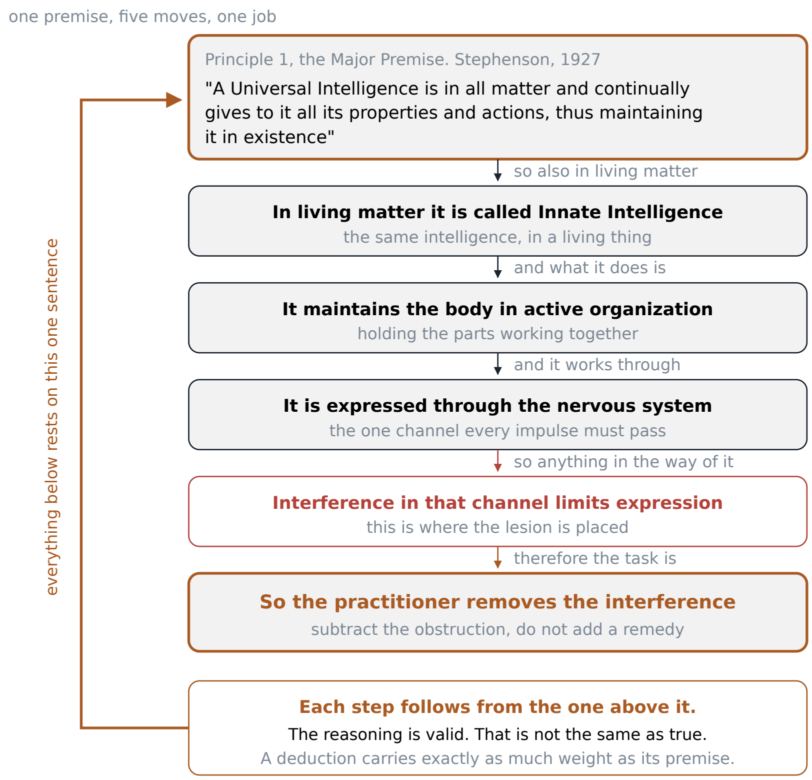
Here is where the chain starts. Principle 1, the Major Premise, quoted exactly:

A Universal Intelligence is in all matter and continually gives to it all its properties and actions, thus maintaining it in existence.

Read that twice. It is a statement about all matter in the universe, not about the human body, and not about spines. It is a metaphysical claim, a claim about the fundamental nature of what exists, and it is the foundation on which the other thirty two rest.

The chain then descends, roughly, like this. Principle 2, the chiropractic meaning of life, holds that the expression of this intelligence through matter is what life means. Principle 4, the triune of life, holds that life is a unity of three necessary factors: intelligence, force, and matter. Principle 20 introduces the term you already know: a living thing has an inborn intelligence within its body, called innate intelligence. Principle 21 gives innate a mission, which is to maintain the material of the body of a living thing. Principle 23 gives it a function, which is to adapt universal forces and matter for use in the body. Principle 24 sets a limit: innate can adapt forces and matter for the body only as long as it can do so without breaking a universal law, which is Stephenson's way of saying that the intelligence in you does not get to overrule physics. Then the sequence turns toward the clinic. Principle 29 states that there can be interference with the transmission of innate forces. Principle 30 states that such interference causes incoordination, or disease. And principle 31 states that interference with transmission in the body is always, directly or indirectly, due to subluxations.

That last one is the payoff. Follow the chain and the chiropractic adjustment is not an optional therapy that sometimes helps. It is the necessary consequence of the premises. If interference causes disease, and if interference is always due to subluxation, then correcting subluxations is the correct response to disease as such. Stephenson also writes the word disease as two parts, splitting it so that it reads as a body out of ease rather than as a thing the body has caught. That is a deliberate piece of vocabulary. Under this system there are no diseases in the ordinary sense, there are only bodies whose self regulation is being obstructed.



Stephenson's thirty three principles form a single descending chain from a metaphysical claim about all matter down to the practitioner's hands, which means every link inherits whatever certainty the first link has and no more.

I promised to give the philosophy its best hearing, and I have. Now here is the observation I owe you, and it is the most important sentence in this chapter.

A deduction carries exactly as much weight as its premise. Not more. Logicians separate two things that students routinely run together. An argument is *valid* if the conclusion really does follow from the premises. An argument is *sound* if it is valid and the premises are actually true. Validity is a property of the reasoning. Soundness requires the world to cooperate. You can build a perfectly valid chain of a thousand steps on a premise that is false, and every step will be flawless, and the conclusion will be worthless. The rigour of the chain does nothing whatever to support the premise, because the premise is what everything else is standing on.

So the question that Stephenson's system does not and cannot answer is the first one: is it true that a universal intelligence is in all matter, continually giving it all its properties and actions? That is not a question spinal anatomy can settle. It is not a question a clinical trial can settle. It is a claim of the kind that people have believed and disbelieved for as long as there have been people, and Stephenson does not argue for it, he assumes it. That is what a major premise is. Notice also that the chain is doing something else along the way: the word intelligence starts out meaning a cosmic organizing principle in Principle 1 and by Principle 29 has become something that can be physically obstructed at a specific joint. The vocabulary stays constant while the kind of thing being talked about quietly shifts from the metaphysical to the mechanical. That shift is where a careful reader should slow down.

None of this makes the system foolish. Deductive systems built on unprovable first premises are extremely common, including in places you respect. What it does mean is that if you accept the chiropractic philosophy because the thirty three principles are so tightly reasoned, you have not actually given a reason. You have admired the carpentry without examining the ground.

Subluxation: four ideas wearing one word

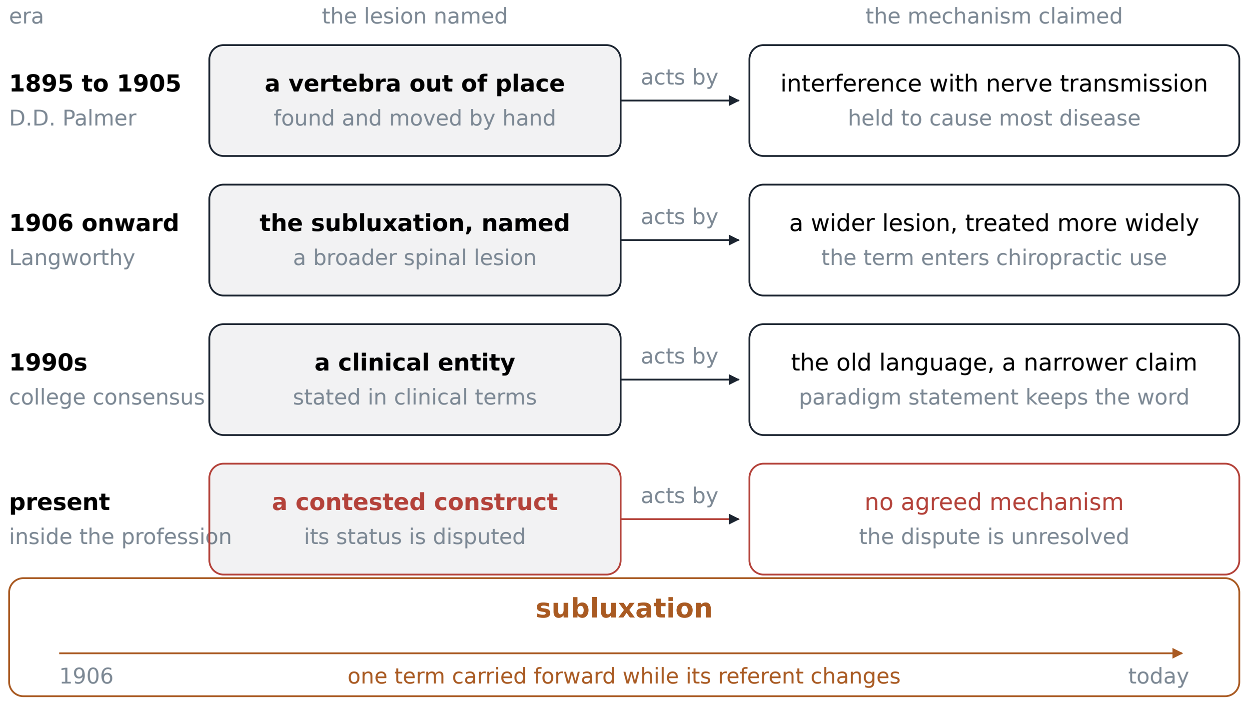
The word subluxation appears at the hinge of that chain, in principle 31, and it is the most overloaded word in this profession. If you take one piece of practical vocabulary out of week one, make it this: when a chiropractor, a physician, a journalist, or a textbook says subluxation, the first question is always which subluxation. There are at least four, and they are not the same idea.

One: the orthopedic subluxation. In general medical usage a subluxation is a partial dislocation, a joint whose surfaces have come out of their normal relationship without fully separating. It is a structural displacement, it is significant, and crucially it is the kind of thing that shows up on a plain radiograph. This is the ordinary meaning of the word in a hospital. Nobody disputes that these exist.

Two: the original Palmer subluxation. D.D. and B.J. Palmer defined it as pressure on nerves producing abnormal function, a lesion affecting the action or the composition of the body. Solon Langworthy, an early figure in the field, is credited with first using the term to describe a vertebral misalignment that narrows the passage through which a nerve travels. This is a specific mechanical claim: a bone is out of its proper place, the opening is narrowed, the nerve is squeezed, and function downstream suffers. Note that this subluxation is defined by its mechanism.

Three: the modern consensus subluxation. In 1996 the Association of Chiropractic Colleges adopted a definition intended to unify the profession. Quoted exactly: a subluxation is a complex of functional and/or structural and/or pathological articular changes that compromise neural integrity and may influence organ system function and general health. The World Health Organization's version is similar in spirit: a lesion or dysfunction in a joint or motion segment in which alignment, movement integrity and/or physiological function are altered, although contact between joint surfaces remains intact. Notice what has happened to the word. It no longer names a bone out of place. It names a complex, with three alternative kinds of change joined by and/or, whose effects *may* influence general health. It has become considerably harder to say what would count as its absence.

Four: the working subluxation. This is the one used in practice, most of the time, by most practitioners, whatever they say in public. It means: this segment feels restricted, or tender, or is not moving the way its neighbours move, and it is where I am going to work today. It is a clinical judgment about a spot on a back. It carries no metaphysical commitment at all, and a physical therapist would recognize the underlying observation immediately, though she would call it something else.



One word has carried four separable meanings, from a partial dislocation visible on imaging to today's working judgment about a stiff segment, and arguments about chiropractic routinely trade on the slide between them.

Those four are related but they are not interchangeable, and enormous confusion follows from treating them as one. A defender can answer a critic by pointing to meaning four, which is modest and defensible, while a practice website makes claims that only make sense under meaning two. A critic can demolish meaning two, which is easy to demolish, and act as though he has said something about meaning four, which he has not. Watch for this move in both directions. It is the single most common piece of bad faith in the public argument about this profession, and most of the people making it are not being deliberately dishonest. They simply have not noticed that the word has drifted underneath them.

You should also know that formal bodies inside and around the profession have said plainly that the classical construct is not evidentially supported. The General Chiropractic Council, the statutory regulator of the profession in the United Kingdom, stated in 2010 that the vertebral subluxation complex is not supported by any clinical research evidence that would allow claims to be made that it is the cause of disease or health concerns. In 2015 a statement by an international group of chiropractic educators said that the teaching of the vertebral subluxation complex is unsupported by evidence. Those are not statements by hostile outsiders. They come from within the profession's own regulatory and educational structure, and many chiropractors have moved away from the historical concept toward language that describes the joint dysfunction they actually treat.

I have not found any survey that tells us reliably what proportion of currently practising chiropractors hold which view, so I am not going to give you a number. What I can tell you from a practice life is that if you walk into ten chiropractic offices in this state you will find a real spread, and that the spread is wider than the profession's public relations would suggest in either direction.

Three ways to read innate intelligence

Which brings us to the question I am going to leave you with. Innate intelligence is still taught. It is still in the literature. It is still on the walls of some offices. But it is read in at least three quite different ways by people currently in the field, and a comparative analysis published in a chiropractic journal sorts them out about as cleanly as anyone has.

The literal reading. Innate intelligence names a real organizing wisdom in the body, something that is not reducible to chemistry and physiology and that genuinely knows what it is doing. This is B.J. Palmer's position and it is held sincerely by a substantial minority today. On this reading, the philosophy is not a historical wrapper around the technique, it is the reason the technique matters, and a chiropractic that abandons it has abandoned itself. The best case for this reading is that it takes the tradition at its word instead of quietly editing it, and that it gives the practitioner something to be that a technician does not have.

The metaphorical reading. Innate intelligence is a nineteenth century name for something real that we now describe differently: the emergent, distributed, staggeringly complex self regulation of a living organism. Homeostasis, immune surveillance, wound healing, endocrine feedback, the whole apparatus. On this reading, Palmer was pointing at something true with the only vocabulary he had, and there is no conflict with contemporary biology because innate was never a separate substance, it was a name for the integrated behaviour of the system. The best case for this reading is that it preserves the founder's insight while letting the profession talk to physiologists. The objection to it is that it is not what Palmer meant, and calling it a metaphor is a retroactive courtesy he did not ask for. He said soul, spirit, spark of life, and a segment of the intelligence that fills the universe. That is not a description of endocrine feedback.

The discarded reading. Innate intelligence is a historical artifact. It belongs in a course like this one, studied as part of understanding where the profession came from, and it has no place in a clinical rationale. On this reading, the useful core of chiropractic is manual treatment of musculoskeletal complaints, evaluated the way any other treatment is evaluated, and the vitalist superstructure is a liability that costs the profession credibility it needs. The best case for this reading is that it lets chiropractic be judged on what it can actually demonstrate. The objection is that it may hollow out the thing that made chiropractic a distinct profession in the first place, leaving a set of techniques that anybody could learn and that other professions are already learning.

I have colleagues in all three camps. The ones in the first camp are not stupid and the ones in the third camp are not traitors, whatever each says about the other at conferences.

Where do you stand, provisionally

So: which of the three are you inclined toward, right now, after one chapter? Literal, metaphorical, or discarded. I want you to actually pick one, and I want you to notice why. It will almost certainly be a mixture of what you already believed before you opened this book and what struck you as reasonable in the last twenty pages, and that is fine. First positions are supposed to be provisional. That is what makes them first positions rather than conclusions.

Write it down somewhere you will be able to find it. In week 7 I am going to ask you the same question, after you have looked at the anatomy in chapter 2, the technique systems and the reliability data in chapter 3, what it actually takes to enter this profession in chapter 4, the clinical evidence in chapter 5, and the long argument with organized medicine in chapter 6. Some of you will move. Some of you will hold your position but for entirely different reasons, which is its own kind of movement and often the more valuable one. What I do not want is for anyone to arrive at week 7 with an opinion they have never had to state carefully. You now have the vocabulary to state one. Use it.

Terms from this chapter

Bonesetting

An informal folk trade of manual joint treatment, reducing dislocations and setting bones, transmitted within families across generations and carrying no school, credential, or general theory of disease.

Magnetic healing

A drugless nineteenth century practice built on the idea that a vital force in the body can be depleted or obstructed and restored by the direct action of a skilled operator's hands. D.D. Palmer's occupation before 1895.

Medical marketplace

The competitive and weakly regulated field of American healing in the 1890s, in which regular physicians and several rival sects reached patients through the same porous licensing gate.

Osteopathy

The system announced by Andrew Taylor Still in 1874 and taught at Kirksville, Missouri, from 1892, holding that disordered structure produces disease by obstructing the blood supply on which tissues depend.

Vitalism

The view that living things are animated and organized by a principle or force that is not reducible to their physics and chemistry. The philosophical family to which innate intelligence belongs.

Innate intelligence

In Palmer's terms, the soul, spirit, or spark of life that controls and directs all the vital functions of the body, and which he described as a segment of the intelligence that fills the universe.

Universal intelligence

The cosmic organizing principle asserted in Stephenson's Major Premise to be present in all matter, of which innate intelligence is the individual expression inside a living thing.

Major Premise

Principle 1 of Stephenson's system, the single metaphysical assumption from which the other thirty two principles are deduced, and therefore the point at which the whole system stands or falls.

Deductive system

A body of claims in which each follows necessarily from those before it, so that accepting the starting assumptions forces acceptance of the conclusions. Valid means the reasoning holds; sound means the reasoning holds and the premises are also true.

Subluxation

A word carrying at least four separable meanings: a partial dislocation visible on imaging; Palmer's bone out of place pressing on a nerve; the 1996 consensus complex of articular changes compromising neural integrity; and the everyday clinical judgment that a particular segment is restricted or tender.

Adjustment

The corrective act of chiropractic practice: a specific mechanical force applied by hand to a spinal segment with the stated intent of removing interference rather than adding a remedy from outside. Its mechanics are the subject of chapter 3.

Green Books

The chiropractic profession's own textbook series, named for its dark green binding, running from D.D. Palmer's 1906 volume through 1963, roughly thirty nine numbered volumes plus additional unnumbered works.

Straights and mixers

The two parties of the profession's founding split: straights hold that adjustment alone, within Palmer's vitalist framework, is chiropractic; mixers combine adjustment with other treatments and seek accommodation with scientific medicine.

Origin story

A dated founding narrative fixing a founder, a place, and a lineage, which does real institutional work for a profession independently of how well it survives historical scrutiny.

Before you move on

1. State the three moves of chiropractic philosophy, the body as self healing, the nervous system as coordinator, and the practitioner as remover of interference, in your own words and without using the word subluxation.
2. Quote the Major Premise and explain, in two sentences, why the logical tightness of the remaining thirty two principles does nothing to establish that it is true.
3. Give the three accounts of the 1895 event and name one specific factual point on which Lillard's own 1897 statement contradicts the canonical version.
4. Name the four meanings of the word subluxation and construct one sentence in which an argument slides from one meaning to another without announcing it.
5. Still and Palmer share almost every link of their reasoning and differ at one. Identify the link, and explain why a difference that small produced two separate professions rather than one.
6. A patient tells you she does not believe in innate intelligence but wants her low back treated. Does anything in this chapter require you to change how you treat her? Defend your answer.
7. Pick one of the three modern readings of innate intelligence, say which, and write two sentences on what evidence or argument would move you off it. Keep what you wrote; you will need it in week 7.

CHAPTER 2

The Spine You Will Touch

> *What do you actually need to know about the spine before you are allowed to put a hand on one?*

Last week we spent our time on ideas: where the profession came from, what it claimed about itself, and how much of that claim was new. This week we put the ideas down and pick up a body. That is a deliberate order. A great deal of argument about chiropractic, in both directions, is conducted by people who could not draw a lumbar vertebra from memory, and it shows. You cannot evaluate a claim about what happens at a joint if you do not know what is at the joint. So before you decide anything about anything, learn the object.

I want to be honest about how this chapter is written, because it is not written the way an anatomy course is usually written. An anatomy course teaches you the spine so you can pass an examination on the spine. Names, counts, foramina, the list of attachments for a muscle you will never think about again. That is a legitimate way to teach it and it is not what we are doing. Everything in this chapter is aimed at your hands. When I tell you that the spinous process sticks backward from the midline, the reason I am telling you is that in a few days you are going to run your thumb down your own back in a warm room, find those bumps, count them, and write down what you found and how confident you were. When I tell you the orientation of the joint surfaces at the back of a lumbar vertebra, the reason is that it explains why your lower back turns so poorly and your neck turns so well, which is something you can verify on yourself in about ten seconds.

There is a limit to this and I will state it now rather than at the end. This course is delivered online. Nobody is going to certify your hands. What I can teach you in seven weeks is a way of looking and a habit of recording what you see, which is worth having whether or not you ever go to chiropractic school. What I cannot teach you is the feel of a joint under a thumb, because that takes a supervisor standing next to you saying no, lower, that is the transverse process, come back to the midline. Keep the difference clear in your mind as you read. Knowing where the seventh cervical vertebra usually is does not qualify you to do anything to it.

One vertebra, part by part

Start with a single bone, out of the column, held in the hand. There are twenty four of these stacked and mobile in an adult, plus the fused sacrum and the fused coccyx below them, and although they differ a great deal in size and proportion from top to bottom, almost every one of them is built from the same short list of parts. Learn the list once and it will serve you at every level.

At the front sits the **body**. It is a squat cylinder of bone, wider than it is tall, with a shell of dense bone around a core of spongy bone that looks, in cross section, like coral or the inside of a cheap sponge. The body is the load bearing part. Almost all of the weight travelling down the column travels down through the bodies, and this is why they get progressively larger as you descend: a cervical body carries the head, a lower lumbar body carries the head and the arms and the chest and the abdomen and everything else above it. Between one body and the next sits the intervertebral disc, which we will come to shortly. If you remember nothing else about the body, remember that it is the column of the column. The rest of the vertebra is scaffolding hung on the back of it.

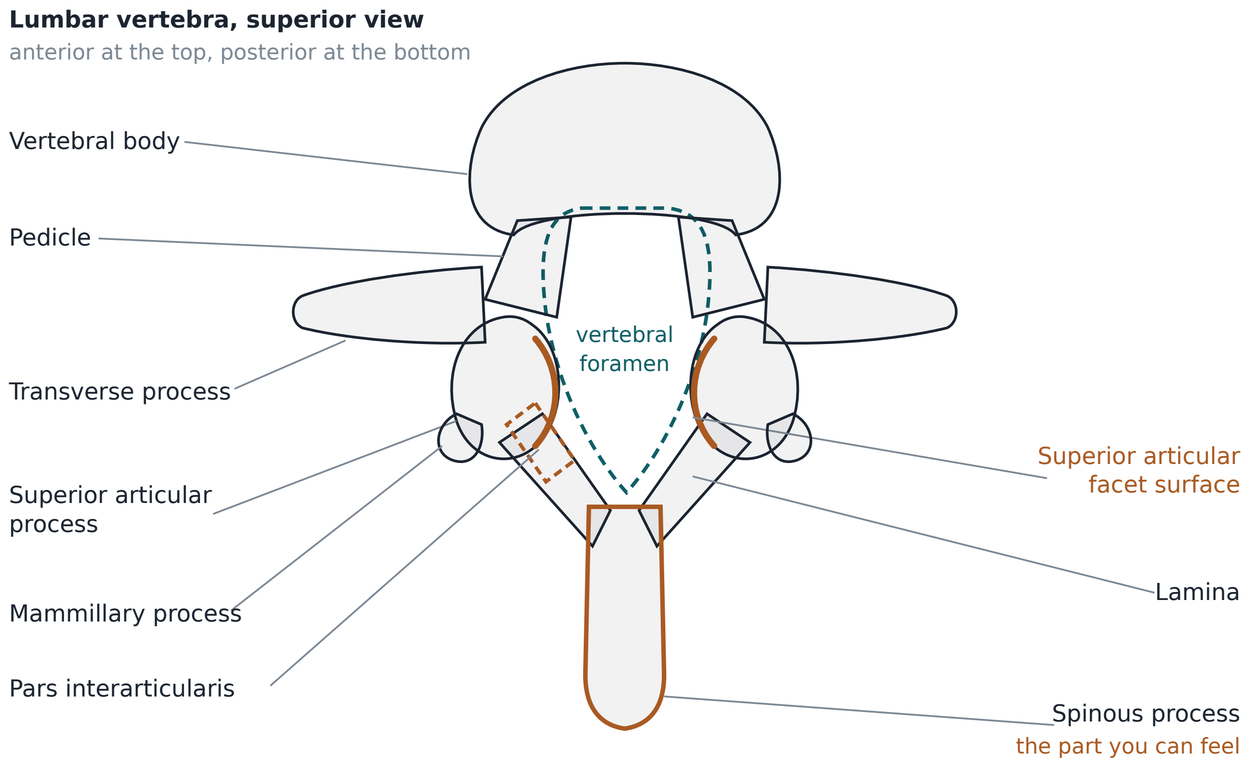
Behind the body, two short thick struts of bone run backward, one on each side. These are the **pedicles**. The word comes from the Latin for little foot, which is not a useful image; think of them as the two bridges that carry everything at the back of the vertebra out from the body and hold it there. Every force generated by a muscle pulling on the back of a vertebra has to pass through the pedicles to reach the body, which makes them small pieces of bone with a large job. They are also, and this matters in a moment, the parts that define the top and bottom of the doorway through which a nerve leaves the spine.

From the back end of the two pedicles, two flat plates of bone angle inward and meet in the midline behind. These are the **laminae**, from the Latin for sheet or plate. They roof over the space behind the body. The body in front, the pedicles at the sides, and the laminae behind together enclose a hole, the **vertebral foramen**, and when the vertebrae are stacked in life those holes line up into a continuous bony canal running the length of the spine. The spinal cord and its coverings sit in that canal. That is the single most important structural fact about a vertebra: it is a weight bearing block with a protective ring built onto the back of it, and the ring exists because the body's main communication cable runs through the middle of the skeleton rather than safely around it.

Now the projections, which are the parts you will actually touch. Where the lamina meets the pedicle on each side, a bar of bone juts out sideways: the **transverse process**. Where the two laminae meet in the midline behind, a single bar of bone juts backward and usually downward: the **spinous process**. These two kinds of projection are levers. Muscles and ligaments attach to them, and because they stand away from the axis of the column, a muscle pulling on the end of one produces rotation of the vertebra rather than simply squashing it. A longer lever gives a muscle more leverage for less force, which is why the spinous processes of the lower thoracic spine, where the long back muscles need purchase, are long and blade like, while in the upper cervical spine, where the head needs room to tip backward, they are short.

The spinous processes are also, for our purposes, the handles. They are the only part of most vertebrae that lies close enough to the skin, in the midline, to be found reliably by a finger. Nearly every landmark method you will read about in this chapter, and nearly every contact a practitioner takes on the spine, is organized around the spinous processes and the transverse processes. When a chiropractor says he contacted a segment, in the great majority of cases he means he placed part of his hand on one of those two bony prominences, or on the small ridge just beside them, and applied force through it. The anatomy of these projections is therefore not an academic matter. It is the interface.

Finally, above and below the junction of pedicle and lamina on each side, there are four small paired projections carrying smooth joint surfaces: two **superior articular processes** pointing up and two **inferior articular processes** pointing down. The inferior pair of one vertebra meets the superior pair of the vertebra below to form two small joints, one on the left and one on the right. These are the **facet joints**, also called zygapophyseal joints, and they are true synovial joints in the same family as a knuckle or a knee: two cartilage covered surfaces, a capsule around them, and a film of slippery fluid between. They are small. In the lumbar spine they are perhaps the size of a fingernail. They will occupy a great deal of this chapter, because the angle at which those little surfaces are set is the single fact that determines what each region of your spine can and cannot do.



Each named part of a single vertebra is labelled with the job it does, so that the body reads as the load bearer, the pedicles and laminae as the ring that protects the cord, and the processes as levers and as the handles a practitioner contacts.

A note on what your finger is actually touching

Students are often surprised, the first time someone corrects their palpation, by how much tissue lies between a fingertip and a bone. On the back you are pressing through skin, a layer of fat that varies enormously between people, a fascial sheet, and in most regions a substantial mass of muscle that runs vertically on either side of the midline. What feels like a hard edge may be a tendon. What feels like a gap may be a soft spot in a muscle. In the midline over the spinous processes the covering is thinnest, which is exactly why the midline is where beginners are taught to work. When you do this week's observation exercise and something feels ambiguous, write down that it felt ambiguous. That is data. Writing down a confident answer you do not have is the beginning of a bad clinical habit, and it is much easier to avoid in week two than in year ten.

Five regions, four curves, and why the column is not straight

Stack the bones and you get regions, and the regions are not arbitrary. Counting downward: **seven cervical** vertebrae in the neck, **twelve thoracic** in the chest, each carrying a pair of ribs, **five lumbar** in the lower back, then **five sacral** vertebrae fused into the single wedge of the sacrum, and finally roughly **four coccygeal** vertebrae, fused and variable, forming the coccyx. Thirty three bones at the start of life, twenty four of them still separate and mobile in an adult. The fusion of the sacrum and coccyx begins around the age of twenty and is not finished until middle age, which means that a college student and a fifty year old do not have quite the same pelvis, a fact worth holding onto when you read about spinal development.

Here is a detail students never forget, and it is worth keeping because it makes a larger point. Nearly all mammals have seven cervical vertebrae, regardless of the length of the neck. A giraffe has seven. A mouse has seven. A whale has seven. The neck of the giraffe is long because each of its seven cervical vertebrae is enormous, not because it has more of them. That count is a fixed mammalian trait rather than something that scales with body size or with need. I raise it because it tells you what kind of thing you are studying. The layout of the human spine is not a design chosen for human convenience last week. It is deep, old, conserved structure that we inherited with very little modification and then stood upright on, and some of what follows in this chapter, including some of what goes wrong, follows from that mismatch.

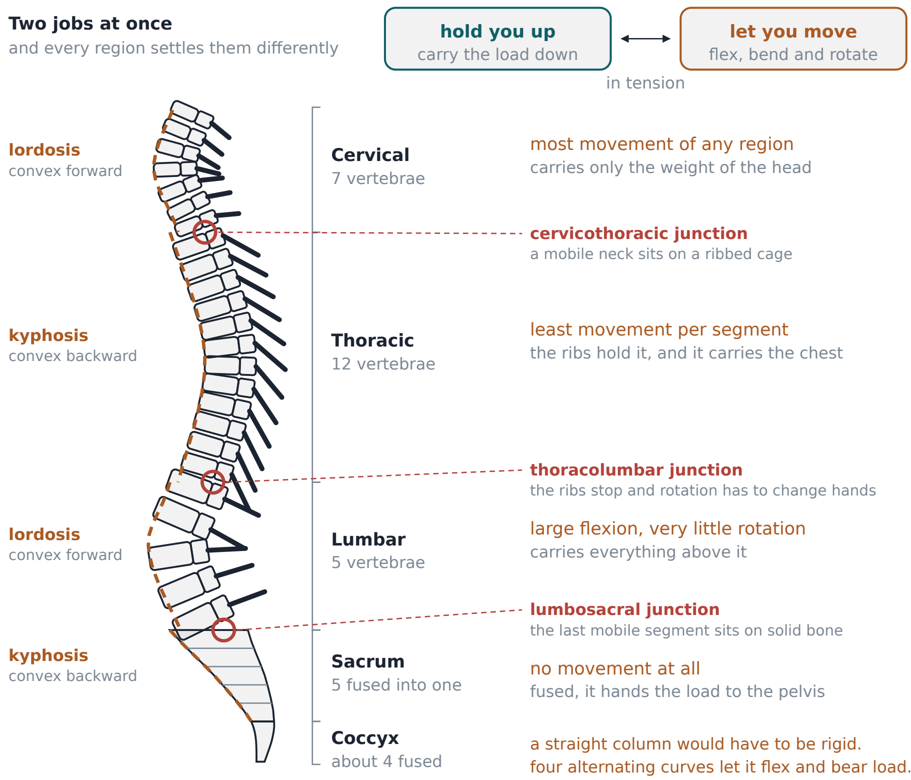
Now the curves, which are where the column stops being a stack and starts being a structure. Seen from the side, a healthy adult spine has four alternating curves. The neck curves forward, meaning the concavity faces backward. The thoracic spine curves the other way, the concavity facing forward, which is why the upper back is rounded. The lumbar spine curves forward again. The sacrum curves backward again. Forward, backward, forward, backward, from skull to pelvis. A curve that is concave backward, as in the neck and the low back, is called a **lordosis**. A curve that is concave forward, as in the thoracic spine and the sacrum, is called a **kyphosis**. Those two words will follow you through every clinical conversation you ever have about a spine, so learn them now.

The obvious question is why. A column carrying weight down to the ground would seem to want to be straight, the way a table leg is straight. The answer is that a straight column is stiff and a curved column is springy. Load a straight rod from the top and the force runs directly down it until something at the bottom fails. Load a curved rod and the curve deflects slightly, storing energy, then returns. The alternating curves let the spine behave a little like a leaf spring in a truck: it accepts a shock at the top, deforms across four curves rather than one, and gives the load back rather than transmitting all of it instantly to the pelvis. Every step you take sends an impulse up from the heel, and the curves are a large part of why that impulse arrives at your skull soft enough that your vision does not blur.

The curves have a second job, which is balance. You are a tall thing with a heavy head standing on two small feet. The forward curve of the neck and the forward curve of the low back exist partly to bring the weight of the head and the weight of the trunk back over the middle of the base, so that standing upright costs you very little continuous muscular effort. If you have ever watched someone with a severely rounded upper back try to look straight ahead, you have seen what happens when one curve changes and the others have to compensate. The column is a system. You cannot alter one part of it in isolation, which is a principle that will come back in chapter 3 when we discuss what an adjustment is supposed to accomplish.

Where the curves reverse, load concentrates. This is a general truth about structures and it holds here. The **cervicothoracic junction**, where the mobile neck meets the stiff rib bearing thoracic spine, the **thoracolumbar**

junction, where the rib cage stops and the lumbar spine begins, and the **lumbosacral junction**, where the last mobile vertebra sits on the fused sacrum, are the three places where a flexible region abuts a rigid one and where the mechanical demand changes abruptly. Those junctions carry a disproportionate share of degenerative change, injury, and clinical complaint, and it is not a coincidence. A hinge between a moving part and a fixed part is always the part that wears.



Working down the column region by region shows how vertebral size, spinous process shape, and the alternating curves each change to meet a different mechanical demand, and where the transition zones between them fall.

A note on variation

The counts above are the standard pattern and most people follow it, but not everyone does. The commonest departure is at the bottom, where the boundary between the lumbar spine and the sacrum is not always clean. The fifth lumbar vertebra can be partly fused to the sacrum, or the first sacral segment can be partly free, and both patterns are grouped under the label lumbosacral transitional vertebra. Reported prevalence across studies ranges from about 4 percent to about 36 percent, with a commonly cited average of roughly 12 percent, and that enormous range mostly reflects genuine disagreement about how to define and count the thing rather than sloppiness in the studies. Sacralization is reported more often in men, lumbarization more often in women. Most of these are incidental findings on imaging in people with no symptoms at all. The practical lesson for you is a counting lesson: if you count up from the pelvis on someone with a transitional segment, you will be one level off for the whole spine, and you will not know it.

The three joint complex

Two vertebrae, one on top of the other, are joined at three places. In front, between the bodies, sits the **intervertebral disc**. Behind, on the left and on the right, sit the two **facet joints** formed by the articular processes. That arrangement is called the **three joint complex**, and thinking of it as one unit rather than three parts is the single most useful mental shift available to a beginner. The three do not act separately. Load the disc badly and the facets take more of the load. Restrict the facets and the disc is loaded differently. Change one and you change all three, permanently.

Take the disc first. It is not a shock absorbing cushion, exactly, although that is how it is usually described. It is more like a small hydraulic bearing. It has a tough outer ring of fibrous rings arranged in crossing layers, laid at angles to each other like the plies of a tyre, and inside that ring a soft, water rich core. Compress it from above and the core, being mostly water and therefore not compressible, pushes outward against the ring, which tightens and takes the load in tension. That arrangement lets the joint bear enormous compressive force while still permitting the two bodies to tilt and twist a little relative to each other. Every disc in a healthy spine is doing that continuously, all day, and getting slightly thinner across the day as fluid is pressed out, which is a large part of why you are measurably taller in the morning than at night.

Now the facets, and here is the part of this chapter I would most want you to remember in five years. The facet joints do not simply permit motion. They restrict it, selectively, and the direction in which they restrict it is set by the orientation of the flat joint surfaces themselves. A synovial joint of this type allows its two surfaces to slide across each other in the plane in which they lie, and strongly resists being pulled apart or driven into each other perpendicular to that plane. So if you know which way the surfaces face, you know what the joint will allow.

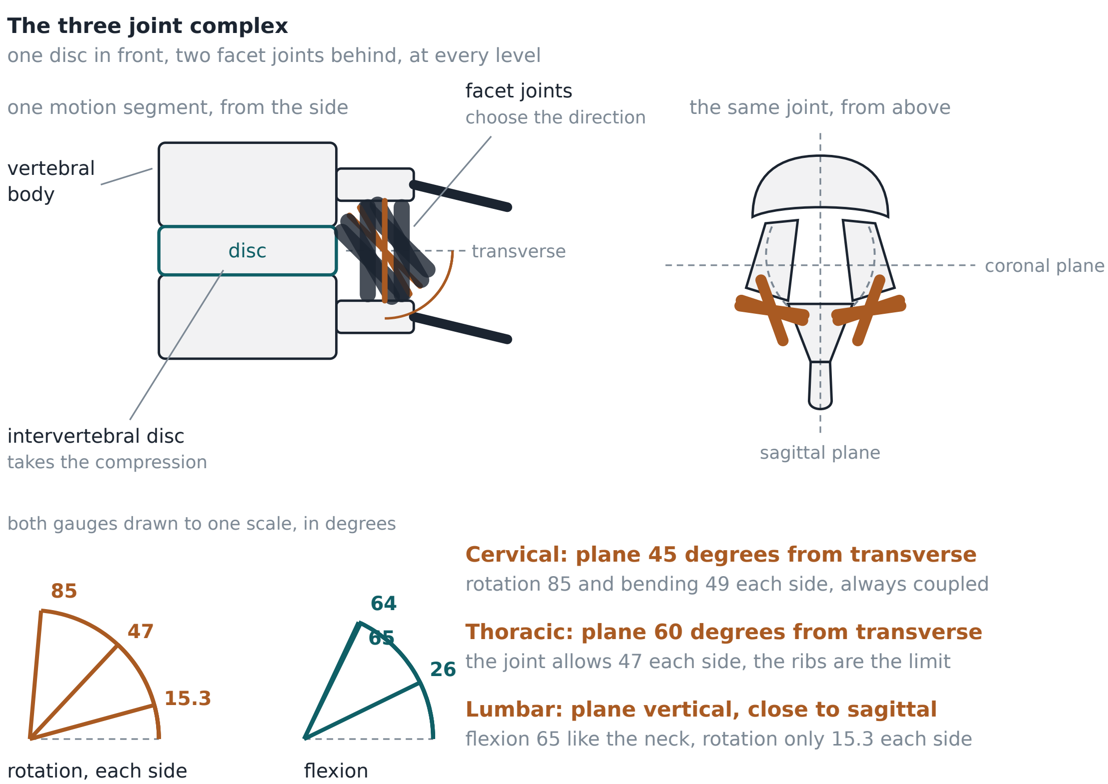
Three quick planes of reference, defined plainly, because I am about to use them. The **sagittal** plane is the vertical plane that divides you into left and right; a knife entering the front of your nose and coming out the back of your skull travels in the sagittal plane. The **coronal** plane is the vertical plane that divides you into front and back, like a wall you could stand flat against. The **transverse** plane is the horizontal one, dividing top from bottom.

In the **cervical spine**, the facet surfaces are set on a slope, tilted between the horizontal and the vertical, facing upward and backward on the lower vertebra and downward and forward on the upper one. Surfaces set on a gentle slope like that can glide across each other in almost any direction, which is why the neck can flex, extend, side bend, and rotate freely and can combine those motions smoothly. The cost of that freedom is stability, and the neck is the least intrinsically stable region of the spine.

In the **thoracic spine**, the facet surfaces sit much closer to the coronal plane, standing nearly vertically and facing backward and slightly outward, arranged along the arc of a circle whose centre lies roughly in the vertebral body in front. Surfaces arranged on such an arc slide happily in rotation about that centre, which is why the thoracic spine is comparatively good at twisting. What it is not good at is bending forward and backward, and the reason for that is not the facets at all. It is the rib cage. Twelve pairs of ribs joining the vertebrae behind and the sternum in front turn the thoracic spine into part of a semi rigid box, and a box does not fold.

In the **lumbar spine**, the facet surfaces turn again and sit much closer to the sagittal plane, curved rather than flat, so that the inferior process of the upper vertebra hooks into a socket like recess on the superior process below it. Picture two hands with the fingers interlocked. Surfaces arranged that way slide freely up and down, so flexion and extension are comparatively free, and they physically block rotation, because rotating requires one surface to pass through the other. That single geometric fact is why the low back is a poor twister and a good bender, and why the last mobile joint above the sacrum, where lumbar facet geometry meets the fixed pelvis, is one of the busiest structures in clinical practice.

The orientation of two joint surfaces the size of a fingernail decides what an entire region of the body is permitted to do.



Laying the cervical, thoracic, and lumbar segments side by side shows the same three joint complex with the facet surfaces set in different planes, which is the structural reason each region moves the way it does.

How much each region actually moves

Numbers make the argument above concrete, so here they are. Before I give them, one plain statement that I want you to carry through the whole book. Published normal ranges of motion for the spine vary a great deal between sources, and the variation is mostly not disagreement about human beings. It is disagreement about method. A range measured with a goniometer, which is essentially a protractor held against the body, does not give the same answer as one measured with an inclinometer, which reads angles against gravity, and neither gives the same answer as one measured from imaging of the bones themselves. Populations differ by age. Techniques differ between examiners. You will find perfectly reputable references that disagree with each other by twenty degrees or more on cervical rotation. I have therefore chosen one internally consistent set, drawn from pooled measurements made in living people across all three mobile regions, and I use it everywhere in this book so that the numbers in this chapter and the numbers in chapter 3 do not quietly contradict each other. Do not mix these with numbers from a different reference set and expect the comparison to mean anything.

In the **cervical spine**: flexion, bending the chin toward the chest, about 64 degrees. Extension, tipping the head back, about 63 degrees, giving a combined forward and backward range of about 127 degrees. Lateral bending, ear toward shoulder, about 49 degrees to each side. Axial rotation, turning the head to look over the shoulder, about **85 degrees to each side**.

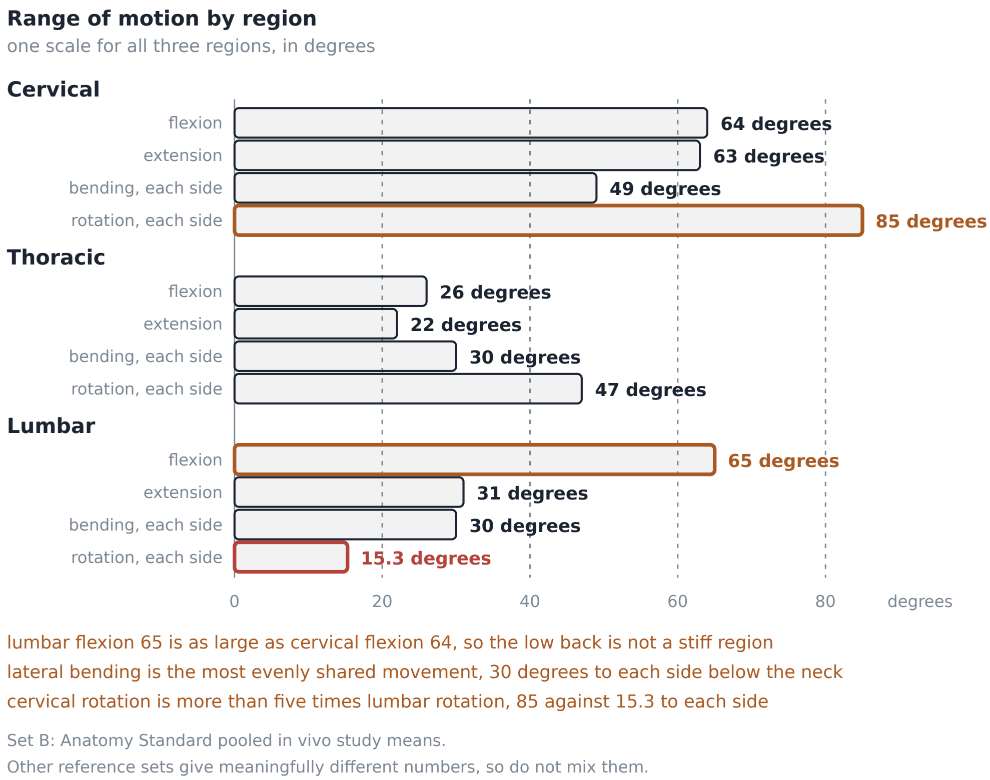
In the **thoracic spine**: flexion about 26 degrees, extension about 22 degrees, a combined range of about 48 degrees. Lateral bending about 30 degrees to each side. Axial rotation about 47 degrees to each side.

In the **lumbar spine**: flexion about 65 degrees, extension about 31 degrees, a combined range of about 96 degrees. Lateral bending about 30 degrees to each side. Axial rotation about **15.3 degrees to each side**.

Now read those numbers against the facet geometry and watch the whole thing lock into place. Cervical rotation, about 85 degrees to each side, against lumbar rotation, about 15.3 degrees to each side. That is not a small difference in degree. It is a difference in kind, and it is produced almost entirely by the orientation of four small pieces of cartilage covered bone at each level. The sloped cervical surfaces let the neck spin. The interlocked sagittal lumbar surfaces refuse. Meanwhile the lumbar spine, which barely rotates, has a combined flexion and extension range of about 96 degrees, roughly double the thoracic spine's 48 degrees, because sagittally set surfaces slide up and down beautifully. And the thoracic spine, which folds poorly, rotates about 47 degrees to each side, better than the lumbar spine by a factor of three, because coronally set surfaces on an arc are built for exactly that.

You can test this on yourself right now. Sit on a chair, hold your pelvis still, and turn your shoulders as far to the right as they will go. Most of what you just achieved came from your thoracic spine and your neck, not from your low back. Now turn only your head, keeping your shoulders forward. Compare. The neck alone will very nearly match everything you got from the entire trunk. That is anatomy you can feel in ten seconds, and it will do more for your understanding than another hour of reading.

Two cautions about these figures. First, they are group averages from healthy adults, and no individual is an average. Second, the total range of a region is the sum of many small contributions from each segment within it, and the contributions are not equal. In practice this means that a person can have a perfectly normal total range for a region while one segment inside it moves very little and its neighbours compensate by moving more. Whether an examiner can reliably detect which segment that is, using his hands, is a serious question with an uncomfortable answer, and we take it up properly in chapter 3.



Comparing flexion and extension, lateral bending, and axial rotation across the three mobile regions shows how sharply the regions differ, most dramatically in rotation, where the cervical spine reaches about 85 degrees to each side and the lumbar spine only about 15.3.

A note on measuring range of motion

When you eventually read a research paper that reports spinal range of motion, look immediately for the method section and find out how it was measured, in what position, on how many people, and of what age. Nothing else in the paper can be interpreted until you know that. This is a general reading habit rather than an anatomy point, and it is one of the three or four most valuable things this course can give you. A number without its method is decoration. Chapter 5 develops this into a full approach to reading clinical evidence, and it is the same habit applied to harder material.

What holds it together

A stack of bones with discs between them and small joints behind them would collapse in a moment if that were all there was. It is not. The column is bound front to back and top to bottom by a set of ligaments, which are tough bands of fibrous tissue running between bones, and it is controlled continuously by muscle.

Two long ligaments run the whole length of the column against the vertebral bodies, one on the front surface and one on the back surface, and together they limit how far the column can bend backward and forward and hold the bodies and discs in line. Between the laminae of adjacent vertebrae runs a short, thick, unusually elastic ligament, the **ligamentum flavum**, named for its yellow colour, which forms part of the back wall of the canal that houses the cord. Its elasticity matters: it stretches as you bend forward and recoils as you straighten, so that it never buckles inward into the canal. Shorter ligaments connect the tips of the spinous processes to each other and run between the transverse processes. Each facet joint has its own capsule, which is itself a ligament in function. Bind all of that together and you have a column that is passively stable through most of its range without any muscular effort at all.

The muscles come in layers, and the distinction between the layers is worth having. The outermost layer contains the large, long muscles that move the trunk and that you can see on an anatomical model: the broad sheets running from the pelvis to the arm and the shoulder blade, and the long columns of muscle running vertically on either side of the spinous processes that you can feel as two ridges when you arch your back. These are movers. They generate the big motions and they generate a lot of force.

Underneath them lies a deeper layer, and it is this layer that matters most for our subject. These are short muscles, some of them spanning only one or two segments, running obliquely from a transverse process up to a spinous process, or between adjacent spinous processes, or between adjacent transverse processes. They are too short and too small to move you anywhere. What they do is control individual segments, resisting unwanted motion and making fine continuous corrections. They are also unusually rich in the sensory receptors that report joint position and muscle length to the nervous system, a sense called **proprioception**, which simply means the body's ongoing internal information about where its own parts are and how fast they are moving. Close your eyes and touch your nose. Proprioception is how you did that.

That density of sensory receptors in the deep spinal muscles is the reason many practitioners and researchers think of the spine as a sensory organ as much as a mechanical one, and it is the mechanism most often proposed for how a manual intervention could have effects beyond the joint it was applied to. I want to be careful here, because this is the boundary of my lane. What the deep muscles are and where their receptors sit is settled anatomy. What follows from that, mechanistically, when a joint is thrust upon, is a live research question, and chapter 3 takes it up along with the evidence for and against. For this week, hold onto the anatomical fact and resist the temptation to build a theory on it.

The cord, the roots, and the doorway

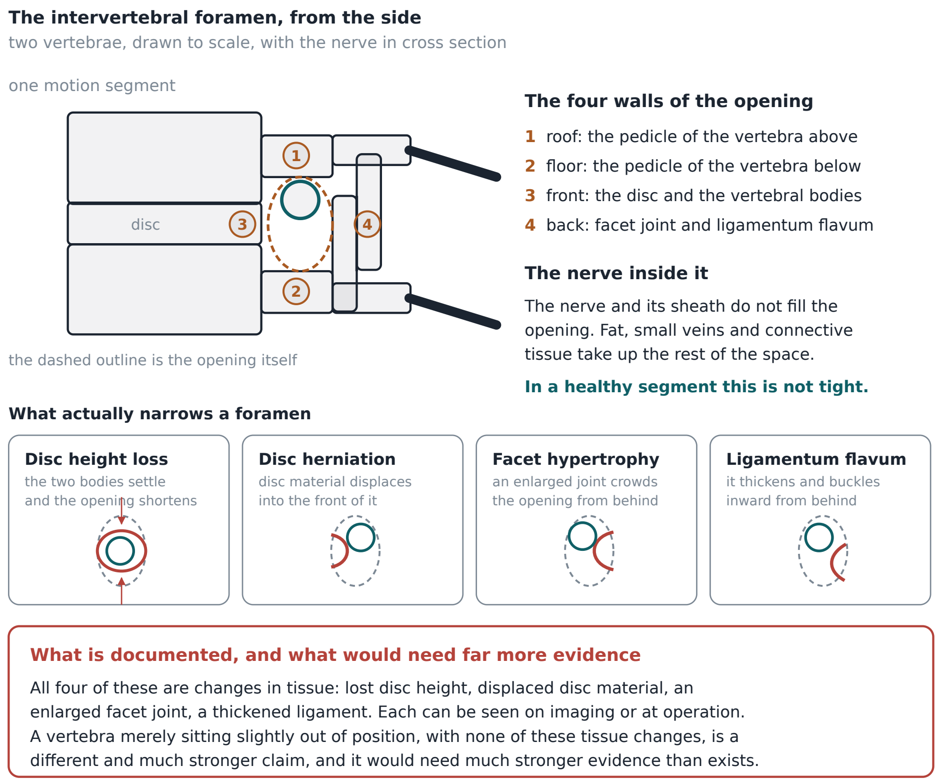
We now come to the part of this chapter that requires the most care, because it is where anatomy meets the founding claim of the profession, and it deserves to be handled honestly in both directions.

The spinal cord runs from the base of the brain down through the bony canal formed by the stacked vertebral foramina. It does not run the whole length. In an adult it ends at around the level of the first or second lumbar vertebra, well above the bottom of the canal, and below that the canal contains a loose bundle of nerve roots descending to their exits, a bundle called the **cauda equina** because someone thought it looked like a horse's tail. This is a fact with immediate clinical consequences, and one of them is why the standard site for a spinal tap is low in the lumbar spine: there is no cord left there to hit. Compression of the cauda equina itself is a genuine emergency, and it appears in chapter 5 as one of the findings that means a practitioner stops and refers rather than treats.

Along its length the cord gives off small rootlets that gather into two roots on each side at each level: one carrying information inward from the body, the other carrying commands outward to muscle. On the incoming root sits a small swelling, the **dorsal root ganglion**, which is a cluster of the cell bodies of sensory nerve cells. The two roots join into a single mixed **spinal nerve**, which then leaves the spine through a gap between two adjacent vertebrae and immediately divides into a branch heading backward to the muscles and skin of the back and a much larger branch heading forward and outward to everything else.

That gap is the **intervertebral foramen**, and you should know exactly what forms it, because everything else in this section depends on the shape of the doorway. Its roof and floor are the pedicles, one from the vertebra above and one from the vertebra below, notched to make the opening. Its front wall is the back of the two vertebral bodies and, crucially, the back of the intervertebral disc between them. Its back wall is the facet joint and its capsule. So the doorway a nerve leaves through is bounded in front by a disc and behind by a joint. Nothing that happens to that disc or that joint is irrelevant to that doorway.

One useful complication while we are here. There are eight pairs of cervical spinal nerves but only seven cervical vertebrae. The first seven cervical nerves exit above the vertebra of the same number, so the sixth cervical nerve leaves above the sixth cervical vertebra, but the eighth leaves below the seventh cervical vertebra, and from there down every nerve exits below the vertebra of the same number. This is the kind of detail that looks like trivia until you are trying to reconcile a level with a symptom, at which point being one segment out is the difference between a right answer and a wrong one.



Following a nerve from rootlets at the cord through the dorsal root ganglion, the joined spinal nerve, and out through the intervertebral foramen shows what forms each wall of that doorway, and therefore what could narrow it.

Now the honest part. Chapter 1 laid out the founding claim in its own vocabulary: that a vertebra out of position interferes with nerve transmission, and that correcting the position removes the interference. Set aside for a moment what you think of that claim, and ask a purely anatomical question. What would have to be true for it to work as stated?

At minimum, three things. First, a vertebra would have to be capable of holding a sustained abnormal position relative to its neighbour, rather than simply moving through a range and returning. Second, that abnormal position would have to reduce the space in the intervertebral foramen enough to reach the nerve and its coverings. Third, the resulting mechanical contact would have to alter the function of the nerve in a way that produces a measurable consequence downstream. Each of those is a separate claim, and each one has to be true for the whole chain to hold. A chain like that is a real proposal and it can be examined, which is more than can be said for many claims in health care.

Here is the size relationship, and it is the crux. The intervertebral foramen is considerably larger than the nerve that passes through it, together with the sheath of connective tissue around it. The remaining space is not empty: it is occupied by small blood vessels, fat, and connective tissue, which cushion the nerve and let it slide slightly as you move. But in a healthy segment, the doorway is not a tight fit. There is room. I am giving you this qualitatively rather than as a percentage because the reference material available to me for this book does not give a well sourced figure for how much of the foramen the nerve occupies, and I would rather tell you that than invent a number that sounds authoritative. The qualitative statement is not in doubt: the space is generous relative to its occupant, and a small positional shift of one vertebra relative to another does not obviously consume it.

What does narrow a foramen? Several things, and they are well documented. **Loss of disc height**, whether from age, injury, or degeneration, lowers the roof and floor of the doorway toward each other, because the vertical dimension of the foramen depends directly on the thickness of the disc holding the two vertebrae apart. **Disc herniation**, in which material from the soft core of the disc pushes out through a tear in the outer ring, can intrude directly into the doorway from the front. **Facet hypertrophy**, an enlargement and thickening of the facet joint and its bony margins that typically accompanies long standing arthritic change, encroaches from the back. **Thickening of the ligamentum flavum**, which also occurs with age and degeneration, encroaches from the back and the canal side. Any of these, alone or together, can genuinely reduce the space around a nerve, and their effects can be seen on imaging and correlated with symptoms.

Notice what those four mechanisms have in common. Every one of them is a change in tissue: something has thinned, torn, thickened, or grown. None of them is simply a bone sitting in the wrong place. That distinction is the heart of the matter. The claim that structural change in a disc or a facet joint can compromise a nerve root is ordinary, well evidenced anatomy and pathology, accepted by everyone. The claim that a positional misalignment of a vertebra, in the absence of such tissue change, produces meaningful interference with a nerve is a substantially stronger claim, and stronger claims require stronger evidence. That evidence is not currently what it would need to be. I am not telling you the claim has been disproved, because that is not the same statement and I do not think it is accurate. I am telling you that it is a bigger claim than it sounds, that it does not follow from the anatomy, and that anyone who presents it as though it were simply an anatomical fact has skipped a step you now know how to spot. The distance between what the anatomy shows and what the founding claim requires is not enormous, but it is real, and you should be able to say exactly where it lies.

I want to be equally clear about the other side, because the version of this argument you will encounter online is usually cheap in both directions. It is not absurd to think that mechanical events in and around the intervertebral foramen matter. The doorway is bounded by a disc in front and a joint behind, both of which change over a lifetime, and the nerve passing through it is not indifferent to its surroundings. What is unwarranted is the leap from that reasonable observation to a general theory of disease. Chapter 5 examines what the clinical evidence actually supports, condition by condition, and it is a mixed picture rather than a rout in either direction. What this chapter can give you is the anatomical baseline against which those claims have to be measured, and now you have it.

Finding it on yourself

The practical half of this week is landmarks. A landmark is a bony feature you can find from the outside and use to estimate the level of something you cannot see. Practitioners of every profession that touches the spine use them, from the anaesthetist choosing a level for an injection to the chiropractor deciding where to place a contact. They are indispensable and they are approximate, and the second half of that sentence is the part that gets forgotten.

Work from the top down. Run a finger up the back of your own skull in the midline until it meets a bump at the base of the skull. That is the **external occipital protuberance**. It is not a vertebra. It is a feature of the skull and it is your starting reference, and its prominence varies between people, so how obvious it feels to you says nothing about anyone else.

Come down from it into the neck. The first firm bony point you meet in the midline below the occipital protuberance is the spinous process of the **second cervical vertebra**, C2. Notice what that sentence quietly tells you: the first cervical vertebra lies between the two, and you did not feel it. C1 does not present a spinous process to your finger the way the vertebrae below it do, which is worth knowing before you start counting confidently downward from the skull.

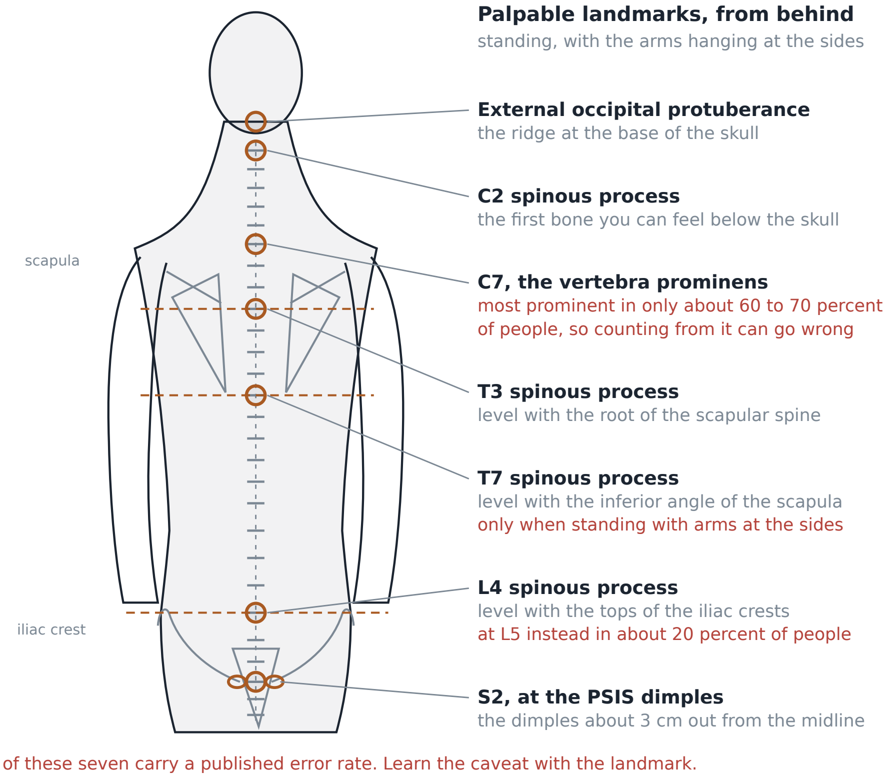
Now go to the base of your neck and bend your head forward. A prominent bump stands up under the skin. This is traditionally called the **vertebra prominens** and identified as the seventh cervical vertebra, C7, and in most people it is. But here is the number I want you to remember from this whole section: C7 is the most prominent spinous process in only about **60 to 70 percent** of people. In the remainder, the first thoracic vertebra or another nearby level is equally prominent or more so. Read that again. The single most widely taught landmark in the upper spine, the one everybody counts from, is wrong or ambiguous in roughly a third of the people you will meet.

Continue down the thoracic spine. Reach across and find the bony ridge running across the back of your shoulder blade, then follow it inward toward the spine until it ends. The medial end of that ridge, called the root of the scapular spine, lies in the same horizontal plane as the spinous process of the **third thoracic vertebra**, T3. Now follow the edge of the shoulder blade downward to its lowest corner, the inferior angle. That corner lies at the level of the **seventh thoracic vertebra**, T7, but only if the person is standing upright with the arms hanging at the sides. Change that and the landmark moves. Lie the person face down, or raise or fold the arms, and the shoulder blade slides across the rib cage and the correspondence shifts, commonly cited as moving up toward T6. The shoulder blade is not attached to the spine by a joint. It floats on muscle. Any landmark taken from it inherits the position of the arm, and if you record a level from a scapular landmark without recording the position the person was in, you have recorded nothing.

I will say plainly that the reference sources I trust give a quantified error rate for the C7 landmark and for the iliac crest landmark below, and do not give one for T3. That absence is not a reassurance. It is an absence, and the honest reading is that we do not have a well sourced figure for how often the T3 correspondence holds, not that it always does.

Down to the low back. Put a hand on each side of your waist and slide it down until you meet the top edge of the pelvis, the crest of the ilium. Now imagine a horizontal line joining the highest point of the left crest to the highest point of the right. That line is called the **supracristal plane**, and it is traditionally said to cross the spinous process of the **fourth lumbar vertebra**, L4. It is used clinically to choose a safe level for a lumbar puncture, and it is safe precisely because the cord has already ended around L1 to L2, well above. And it carries a known error rate: in roughly **20 percent** of people, the supracristal plane sits at L5 rather than L4. One person in five.

Finally, the pelvis itself. Many people have a pair of visible dimples in the skin of the low back, roughly three centimetres to either side of the midline, above the buttocks. Those dimples sit over the posterior superior iliac spines, the two backward projecting corners of the pelvis, and that level corresponds to the second sacral segment, **S2**. If you have them they are among the easier landmarks to find. Not everyone has visible dimples, and where soft tissue is thicker the projections underneath still exist but must be found by pressure rather than seen.



Each surface landmark on the back is paired with the vertebral level it estimates and with the known error rate for that estimate, including C7 being the most prominent spinous process in only 60 to 70 percent of people and the supracristal plane sitting at L5 rather than L4 in about 20 percent.

Now the lesson, which is the real content of this section and not the list. Every one of those landmarks is an estimate with a failure rate, and for two of them we have a measured number. Consider what that means in combination. If you identify C7 by prominence and count downward to find a thoracic level, you have inherited a roughly one in three chance of being wrong at the start, and counting does not correct an error, it propagates it. If you use the iliac crest line to find L4 and then count up, you are wrong one time in five before you begin. If the person has a lumbosacral transitional segment, which may be as common as one in eight, your count from below is off by a whole level and there is nothing in your fingertips that will tell you.

Some students find this demoralizing. It should not be. Landmarks remain the best available starting estimate and they are used by every profession that works on a spine, including in situations far less forgiving than a chiropractic office. The point is not that the methods are useless. The point is that the number attached to each method is part of the method. A practitioner who says this is T7 and means it as a certainty is telling you something false about the world. A practitioner who says this is probably T7, and who knows why probably, and who cross checks against a second landmark from a different direction, is doing the work properly. The habit of holding a finding and its uncertainty at the same time is a clinical skill, and it is learnable in week two of an online course, which is exactly why I have built the exercise this way.

A practitioner who does not know the error rate of his own method is not careful. He is lucky, and luck runs out.

A note on how to do this week's observation

You are being asked to find these landmarks on yourself and record what you observe. Do it in a warm room, in front of a mirror if you can manage one, with something to write on within reach. Record three things for every landmark: what you found, how you decided it was the thing you were looking for, and how confident you are on a scale you define and state. Where two landmarks disagree with each other, write down that they disagreed rather than picking the answer you prefer. Do not press hard, do not attempt to move anything, and do not do this on another person. This is an observation exercise, not a treatment exercise, and the whole point of it is the recording rather than the touching.

Posture, and what it will not tell you

Stand a group of people against a wall and you will see differences immediately. One person's head sits forward of the shoulders. Another's upper back is more rounded. One shoulder sits higher than the other, one hip appears higher than the other, a waist crease is deeper on one side. Posture is the most visible thing about a spine and it is therefore the thing most often over interpreted, in clinics of every description, including mine.

Here is what postural observation legitimately gives you. It gives you a record. If you photograph or describe a person's standing posture carefully today, you have something to compare against in six months, and change over time is far more informative than a snapshot. It gives you a prompt: an asymmetry is a reason to look, to ask a question, to examine a region you might otherwise have skipped. It gives you a way of seeing how the curves relate to each other, since as we said earlier a change in one curve has to be paid for by the others. And in a small number of cases it reveals something structurally significant that needs proper investigation, which means imaging and referral rather than a confident opinion.

Here is what it does not give you. It does not give you a diagnosis, and it does not give you a reliable prediction of pain. This is the place in this chapter where I have to be careful and where the temptation in my profession, and in several neighbouring ones, runs strongly the other way. The intuition that a crooked structure must hurt and a straight one must not is powerful, visually persuasive, and easy to sell. It is also much weaker than it looks. Postural asymmetry of the ordinary kind is extremely common in people with no pain at all, and plenty of people in significant pain have posture that looks unremarkable from across a room. I have examined people whose standing posture I would have described as poor by any textbook standard and who had never had a day of back pain in their lives, and I have examined people who looked like an anatomical illustration and could barely get off the table. If the relationship were as tight as the marketing suggests, that would not keep happening.

So use posture as an observation and not as a verdict. Describe what you see in neutral, concrete language: the right shoulder appears approximately one centimetre higher than the left, the head sits forward of the plumb line, the lumbar curve appears reduced. Resist the sentence that begins therefore. Chapter 5 deals in detail with how to weigh evidence about what causes pain and what relieves it, and one of the things you will find there is that pain is a far less mechanical phenomenon than a chapter on bones can suggest. Hold this chapter's structural knowledge lightly enough to allow that.

Observation as a real skill

Let me close where I started. This is an online course and your hands will not be certified in it. Some of you will find that frustrating, particularly those of you reading this to decide whether to apply to a chiropractic programme. I want to argue that the part we can do here is not the leftover part.

Consider what actually separates a good clinician from a mediocre one in any manual profession. It is not usually the thrust. Technique can be taught, and it is taught, thousands of hours of it, under supervision, with an examiner watching. What is much harder to teach and much rarer in practice is the discipline of observing carefully, describing precisely, knowing the error rate attached to every method you use, recording what you actually found rather than what you expected to find, and noticing when two findings disagree instead of quietly discarding the inconvenient one. Those are habits of mind. They are built by practice, they can absolutely be practised on your own body with a notebook, and they transfer to every clinical thing you may ever do, in this profession or in another.

They also happen to be the habits that make the rest of this course possible. In chapter 3 you will meet reliability studies showing that experienced practitioners agree poorly with each other about which spinal segment is restricted. That finding is very hard to sit with unless you have already discovered, with your own hands and your own notebook, that finding C7 is not as easy as the textbook picture makes it look. In chapter 5 you will read guidelines and reviews that support some claims and decline to support others, and you will need the habit of asking how a number was measured before you decide what it means. This week's exercise is small and it is not glamorous. It is the foundation for the two hardest chapters in the book.

So go find your landmarks. Count down from your skull and up from your pelvis and see whether the two counts meet in the middle. When they do not, and for some of you they will not, you will have learned more from the failure than from a page of memorised anatomy. Write it down honestly. That is the whole assignment, and it is more than it sounds.

Terms from this chapter

Vertebral body

The squat cylinder of bone at the front of a vertebra that carries almost all of the load travelling down the column, growing larger at each level from the neck to the low back.

Pedicle

One of the two short struts of bone running backward from the body, carrying everything at the back of the vertebra and forming the roof and floor of the doorway a nerve exits through.

Lamina

One of the two flat plates of bone that angle inward from the pedicles and meet in the midline behind, roofing over the space that houses the spinal cord.

Vertebral foramen

The hole enclosed by the body, pedicles, and laminae of one vertebra; stacked in life, these holes form the bony canal that contains the spinal cord.

Spinous process

The single bar of bone projecting backward in the midline from the joined laminae, serving as a lever for muscles and as the main bony handle a practitioner can find and contact.

Transverse process

The bar of bone projecting sideways from each side of a vertebra, serving as a lever for muscles and ligaments and as a secondary contact point.

Facet joint

One of the two small synovial joints behind the disc at every spinal level, formed where the inferior articular processes of one vertebra meet the superior articular processes of the one below.

Three joint complex

The functional unit formed by the intervertebral disc in front and the two facet joints behind at a single spinal level, which acts as one thing rather than three.

Lordosis and kyphosis

The two kinds of spinal curve seen from the side: a lordosis is concave backward, as in the neck and low back, and a kyphosis is concave forward, as in the thoracic spine and sacrum.

Intervertebral foramen

The doorway through which a spinal nerve leaves the column, bounded above and below by pedicles, in front by the vertebral bodies and disc, and behind by the facet joint.

Cauda equina

The loose bundle of nerve roots descending inside the canal below the end of the spinal cord, which in an adult finishes around the first or second lumbar level.

Ligamentum flavum

The short, unusually elastic ligament joining adjacent laminae and forming part of the back wall of the spinal canal; its thickening with age is one documented cause of narrowing around a nerve.

Proprioception

The body's continuous internal sense of where its own parts are and how they are moving, supplied in the spine largely by receptors in the short deep muscles between segments.

Vertebra prominens

The prominent spinous process at the base of the neck, traditionally identified as C7, which is in fact the most prominent one in only about 60 to 70 percent of people.

Before you move on

1. Name the parts of a single vertebra and give the job of each, then say which two parts a practitioner would most often make contact with and why those two rather than any others.
2. Cervical axial rotation is about 85 degrees to each side and lumbar axial rotation about 15.3 degrees to each side. Explain that difference using facet orientation, and say what the rib cage contributes to the thoracic figures.
3. Why does a curved column bear load better than a straight one, and what does that predict about the three junctions where the curves reverse?
4. List the four walls of the intervertebral foramen, then list the four documented ways that doorway can be narrowed. Which of the four involves a change in tissue rather than a change in position, and what does that distinction imply for the founding claim you met in chapter 1?
5. You palpate a prominent spinous process at the base of a person's neck and count down four levels. State the probability that your starting point was correct, and say what you would do differently to reduce your chance of being wrong.
6. A person's right shoulder sits visibly higher than the left. Write two sentences: one recording the observation as a careful examiner would record it, and one explaining to that person why you are not going to tell them what it means.
7. This course cannot certify your hands. Make the case, in your own words and to a sceptical friend, that careful structured observation is still a skill worth spending a week on.

CHAPTER 3

The Art of Technique

> *When a chiropractor adjusts a spine, what is the hand actually doing, how much of the work is the part you can hear, and can two practitioners even agree on where to do it?*

D.D. Palmer's major 1910 book carries the phrase *Science, Art and Philosophy of Chiropractic* in its title. That is his word order, and it is worth noticing, because the phrase you will see on clinic websites today, and in the catalog description of this very course, usually runs the other way: philosophy, art, and science. The reordering is a later convention and nothing more, but the middle term stays in the middle either way. Art. In an older sense of that word it does not mean self expression. It means a skill that has to be acquired through practice, that resists being fully written down, and that one person learns mostly by standing next to another person who already has it. That is the subject of this chapter.

I want to be precise about what I am and am not going to do here. I am going to tell you what an adjustment physically is, in the order the hands actually do it. I am going to tell you what that famous sound is, based on the best imaging evidence we have. I am going to lay out the main technique systems in a way that lets you compare them instead of just collecting their names. Then I am going to spend a long section on the least comfortable body of research in the field, which is the research on whether two competent examiners looking at the same patient can agree on where the problem is. Finally I am going to tell you how a practitioner actually picks a technique, which is a more human and less scientific story than students expect.

What I am not going to do here is tell you whether it works for your headaches. Clinical conditions, the trial evidence for each, and the question of harm are chapter 5, and they deserve their own room. I am also going to lean on chapter 2 constantly and not repeat it, so if the words facet joint, spinous process, or posterior superior iliac spine do not yet mean something to your fingers as well as your eyes, go back and reread that chapter before this one. Technique without anatomy is choreography.

What an adjustment physically is

Strip away the vocabulary and a chiropractic adjustment is a sequence of five things, four of which are slow and quiet, and one of which is fast. Students, and honestly most patients, think the fast one is the technique. It is not. The fast one is the last two percent of a procedure whose difficulty lives almost entirely in what came before it.

The sequence starts with **position**. The practitioner puts the patient into a posture that does two things at once: it makes the target region accessible, and it uses gravity and the patient's own limb positions to pre position the joint so that the force, when it comes, will travel where it is meant to and not somewhere else. A patient side lying with the top knee drawn up and the trunk rotated is not simply lying comfortably. That posture has taken the lumbar spine to a particular place in its range and left a specific segment more available than its neighbors. Change the height of the table, change where the patient's shoulder is, change where my own feet are, and I have changed the direction the thrust can be delivered in. I have watched students spend a whole term learning to set up a single lumbar adjustment and still not have it. The reason is that positioning is where the geometry is decided, and the geometry is the whole thing.

Second is **isolation**, sometimes called localization. Having positioned the patient, the practitioner narrows the region until one motion segment is more available to the force than the segments above and below it. This is what stabilization with the other hand or the forearm is for. If the whole lumbar spine moves as a unit, the force spreads and nothing in particular happens at any one level. Chapter 2 explains why isolation is possible at all: the three joint complex, a disc in front and two facet joints behind, means every level has its own permitted directions and its own resistances, and those differ enough by region and by level that a segment can, in principle, be singled out. Whether a practitioner can reliably single out the *right* segment is the question the second half of this chapter is about.

Third is the **contact**. The practitioner places a specific part of the hand on a specific bony landmark. Not on a muscle, not on the general area, on a landmark. The parts of the hand used have their own names in the technique literature, and the ones you will hear most often are the pisiform, which is the small pea shaped bone at the base of the palm on the little finger side, and the hypothenar eminence, the fleshy pad along that same edge. Others use the thumb, the tip of the index finger, or a broad palm. The bony landmarks are the ones from chapter 2: a spinous process, a transverse process, a mammillary process, a facet joint pillar in the neck, the posterior superior iliac spine at the back of the pelvis. The contact is small, deliberate, and chosen so that the line of force can be aimed through the joint rather than across it.

Here is where chapter 2 comes back to bite. You learned there that the C7 spinous process, the one at the base of the neck, is genuinely the most prominent one in only about 60 to 70 percent of people, and that the horizontal plane across the tops of the iliac crests lands on L4 in most people but on L5 in roughly 20 percent. Those error rates are not trivia. They mean that a technique built on contacting a named segment inherits, before it does anything else, the error rate of naming the segment. A very good practitioner with a very specific thrust who has counted one level off has delivered a very specific thrust to the wrong joint. This is a real limitation and it is not a secret; it is measured, published, and taught.

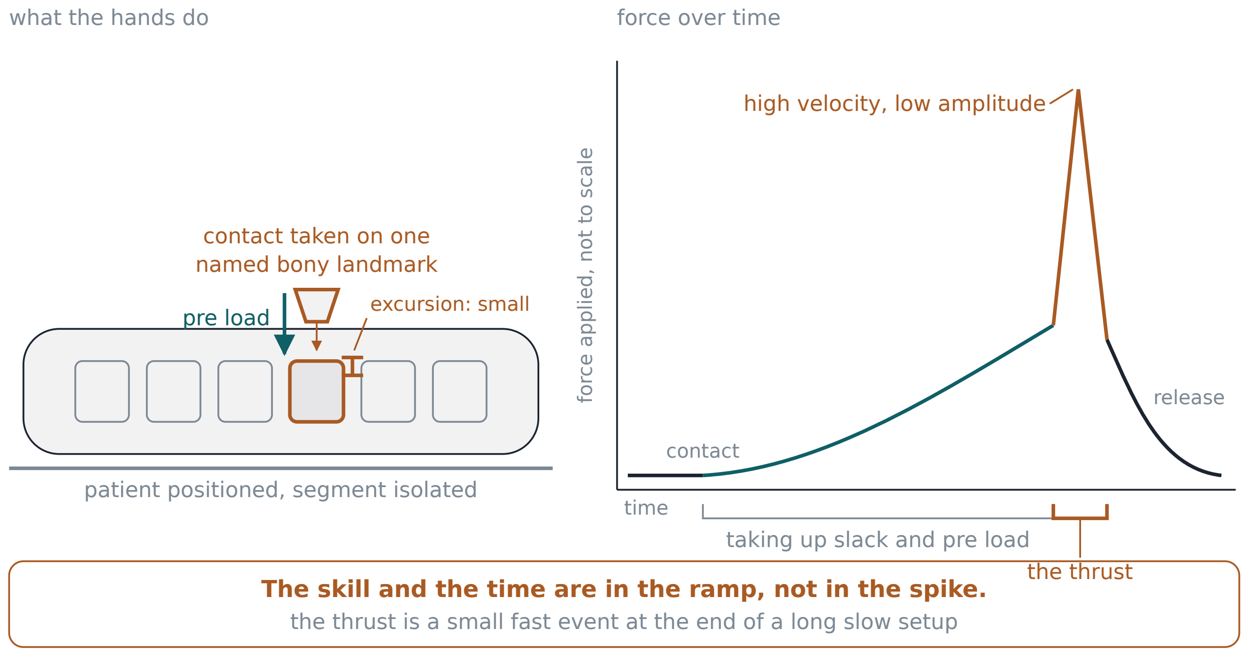
Fourth is **taking up the slack**, followed by **pre load**. Every joint in the body has a little free play in it before the tissues start to resist. Push slowly and gently on a relaxed joint and for a short distance nothing pushes back. Then the ligaments and the capsule and the resting tone of the muscles begin to take the load, and the joint starts to feel firm under the hand. Taking up slack means moving through that free play, unhurried, until the tissue tension arrives. Pre load means then leaning a little further, holding a light, steady pressure at the edge of that tension, so that the joint is already loaded when the thrust comes and no part of the force is wasted on travel through empty space.

Slack and pre load are the parts I cannot teach in writing, and I am not going to pretend otherwise. What the hand is looking for is a change in quality: not a distance, not a pressure reading, a change from give to resistance. It is the same information you get putting your weight on a floorboard that turns out to be loose, or turning a bolt until it stops turning easily. Every student arrives thinking this will be the easy part and that the thrust will be the hard part. It is the other way around. A practitioner who has taken the slack out properly is most of the way there. A practitioner who has not is about to push a patient around the table.

Fifth, and last, is the **thrust**. It is described in the literature as high velocity and low amplitude, which is exactly what those words say: fast, and short. A high velocity low amplitude thrust is a rapid, small displacement force that carries the joint briefly past its passive range of motion, meaning past the point the practitioner could have taken it to by slow steady pressure alone, but not past the point where tissue is damaged. It is not a shove and it is not a wrench. Almost all of it comes from the practitioner's body, not the arm: a small drop of body weight, a short contraction, delivered through an arm that is already set at the correct angle by the setup.

Now the honest part. You would like a number. How far does the joint actually travel, and how many milliseconds does the thrust take? Those numbers exist in the biomechanics literature, and I have seen figures quoted in classrooms and on websites for years. I am not going to give you one, because when I went looking for a source I trusted well enough to put in a textbook that a student would then repeat in an exam, I could not find one I was willing to stand behind. So I will describe both qualitatively and tell you plainly what I have done. The excursion is small, considerably smaller than most people picture when they imagine a joint being moved, and the duration is short enough that the thrust is over before the patient's muscles could have responded to it defensively. That is the functional point, and it is why speed matters: a fast thrust arrives before the body can guard against it, which is what lets the practitioner use less total force. A slow push of the same magnitude would meet a braced muscle.

I would rather hand you an honest gap than a confident number I cannot source. Get used to that move. You will need it in chapter 5, where the temptation to quote a precise figure from an imprecise study is much stronger and the stakes are much higher.



Force over time in an adjustment is a long slow ramp with a brief spike at the end, which is why the setup, and not the thrust, is where the skill and the time actually go.

The thrust is the last two percent of the procedure and the first thing everybody notices, which is exactly backwards from where the skill lives.

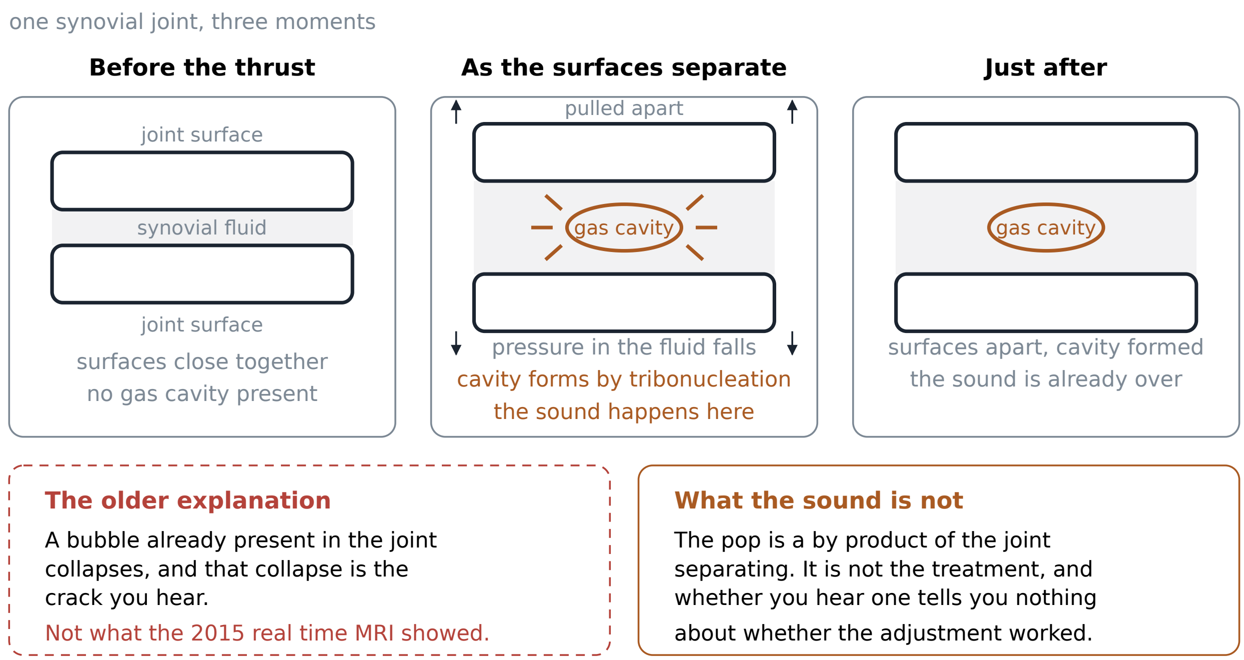
What the pop is

Almost every patient asks. Many of them ask before they ask anything else, including whether it will hurt. The sound is the single most recognizable thing about this profession, it is what shows up in every parody of it, and until fairly recently the profession did not have a settled account of what causes it.

The sound is called a cavitation, and cavitation is a general physical phenomenon, not a chiropractic one: it is what happens when a gas cavity forms suddenly inside a liquid because the local pressure has dropped. Ship propellers cavitate. Pumps cavitate. A synovial joint, which is a joint enclosed in a capsule and lubricated by a slippery fluid called synovial fluid, can cavitate too, and that is the crack you hear when someone pulls on a knuckle.

For decades the dominant explanation was that a small gas bubble already existed in the joint fluid and that the sound was the bubble *collapsing*. That account was taught, it was in textbooks, and it turned out to be backwards. In 2015 Kawchuk and colleagues published a study in PLOS ONE in which they watched a finger joint being pulled apart inside a real time magnetic resonance imaging scanner, so that they could see the inside of the joint at the exact moment the noise occurred rather than reasoning backwards from the noise. What they saw was that the sound coincided with a cavity *appearing*, not disappearing. Their conclusion, in their words, is that "joint cracking is the result of cavity inception within synovial fluid," and not, as the older account had it, the collapse of a bubble that was already there.

The mechanism they name for that inception is tribonucleation. Break the word in half and it explains itself: tribo refers to rubbing or to surfaces in contact, and nucleation is the birth of a new phase, a bubble starting from nothing. When two surfaces that are wetted together and stuck by the fluid between them are pulled apart, the fluid resists separating, the pressure between them drops sharply, and at some threshold the fluid can no longer stay liquid at that pressure. A gas cavity springs into existence in an instant. That sudden formation is the noise. It is the same reason two wet glass plates make a sound when you finally peel them apart.



The 2015 real time imaging work located the crack at the moment a gas cavity forms as joint surfaces separate under tension, reversing the older textbook account that the sound came from a bubble collapsing.

Now the part that matters clinically, and that I would like you to be able to say out loud to a skeptical relative over a holiday dinner. **The sound is a by product of the joint surfaces separating. It is not the treatment.** Nothing in the physics of tribonucleation is therapeutic. A cavity forms because two surfaces came apart under tension, and if the separation is what does the patient any good, then the separation is the mechanism and the noise is the receipt, not the payment.

Several practical consequences follow, and they are the reason I spend real time on this with patients.

The first is that plenty of thrusts that accomplish exactly what they were meant to accomplish make no sound at all. The description of high velocity low amplitude technique in the literature says the thrust produces an audible crack in many but not all cases, and that is my experience across a career. Sometimes the joint is quiet. Sometimes an adjacent joint is what you hear and not the one you were working on, because sound travels and a patient's ear cannot localize a click inside their own back. Sometimes a patient who has just been adjusted well hears nothing and concludes that nothing happened.

The second is the mirror image and it is worse: a loud, satisfying pop is not evidence that anything useful occurred. It tells you a joint separated somewhere in the neighborhood of your hands. It does not tell you which joint, and it does not tell you that the joint that separated was the one you had decided needed attention.

The third is about the practitioner and it is where I get stern with students. A practitioner who chases the noise, who keeps repositioning and re thrusting until something audible happens, who measures his own day by how many cracks he produced, has confused the signal with the work. I have met that practitioner. I was, for an embarrassing stretch of my second clinical year, in danger of becoming him, because the sound is enormously reinforcing. It arrives instantly, the patient's face changes, and you feel like you did something. Almost nothing else in clinical practice gives you feedback that fast. Which is precisely why you should distrust it. Fast, vivid, satisfying feedback that is not actually measuring the outcome you care about is one of the most reliable ways for a competent person to train himself into a bad habit.

> **A note on what patients believe about the sound**
>
> Patients arrive with two opposite convictions and you will meet both in the same week. One group believes the crack is the treatment and feels cheated without it; they will tell you the last chiropractor "got a really good one" as though reporting a score. The other group is frightened of it, often because someone told them it is bone grinding on bone or that it causes arthritis. Neither belief is supported by what the imaging work actually shows, and both are worth correcting gently and early, because a patient who thinks the noise is the point will keep score in a way that has nothing to do with whether they are getting better.

Two questions to ask any technique

There are hundreds of named chiropractic techniques. Some are large systems with their own seminars, instruments, certifications, and vocabularies; some are one practitioner's variation with a trademark on it. A student trying to learn them by collecting names will drown, and worse, will come away thinking the field is more fragmented than it is.

So do not collect names. Ask every system the same two questions, and put the answers side by side:

One: how does this system decide where to work? What does the practitioner examine, measure, or look at, in order to conclude that this segment and not that one needs attention? Call this the assessment question.

Two: how does this system deliver the correction? Is it a manual thrust from the practitioner's own body, an instrument, a table that drops away, sustained low force with body weight, rhythmic traction? Call this the delivery question.

Almost everything you need to compare techniques falls out of those two answers, including a fact I want you to arrive at yourself before I state it at the end of this section: the systems differ from each other far more on the first question than on the second.

The families

Diversified is the default and the closest thing to a common language. It is not proprietary and not owned by anyone; it is eclectic, assembled from many sources, and one report puts usage at roughly 96 percent of chiropractors employing it in some form. Assessment is by palpation, both static and motion, together with the patient history and an orthopedic and neurologic examination. Delivery is a manual high velocity low amplitude thrust aimed at a specific spinal segment with the goal of restoring motion at that joint. That is the plain vanilla adjustment described step by step earlier in this chapter, and if you picture a chiropractor working with bare hands on a table, you are picturing Diversified.

Gonstead, developed in 1923 by Clarence Gonstead, sits close to Diversified on delivery and far from it on assessment. On delivery, it is a hands on, segment specific high velocity low amplitude thrust, distinguished by deliberately removing the rotational component from the thrust, so that the force goes in a cleaner line and the segment is not twisted. On assessment, this is one of the most instrumented systems in the profession. It uses full spine X ray films, a measuring system called the Gonstead Radiographic Parallel, and the Nervoscope, a handheld device that reads the difference in skin temperature between the two sides of the spine as the practitioner runs it down the back. That temperature difference is claimed to mark the level where subluxation and nerve interference are present. Hold onto the Nervoscope; it comes back in the reliability section and it does not come back well.

Activator Methods, developed by Arlan Fuhr, is what the literature calls a mechanical force, manually assisted technique. Delivery is by a spring loaded handheld instrument rather than the practitioner's body. The instrument delivers a small, controlled impulse to the target segment, no more than about 0.3 joules over a pulse of roughly 3 milliseconds. For scale, 0.3 joules is roughly the energy of dropping a golf ball a few inches, and it is delivered in about the time it takes a housefly to beat its wings once. Assessment has the patient prone while the practitioner compares functional leg lengths, meaning how far the two heels reach relative to each other, and uses manual muscle testing and isolation tests to localize the segment. Activator is the technique I reach for most often when the body in front of me will not tolerate a manual thrust, and I will come back to why.

Thompson, usually called drop table technique, is a delivery innovation. The table is segmented, and each section can be cocked so that it drops a short distance under the practitioner's thrust. The practitioner thrusts, the table section falls away beneath the patient, and the momentum of the patient's own body against the sudden absence of support does part of the work. The result is that less force has to come from the practitioner. Here I have to be straight with you: I could not find a clear, well located account of how Thompson practitioners decide *where* to work, in the sources available for this book. That is a gap in what I can document, not a claim that the technique has no assessment procedure. It certainly has one, taught in its seminars. I simply cannot state it with a source, so I will not state it at all.

Sacro Occipital Technique, abbreviated SOT, starts from a premise about the whole system rather than one segment: that the relationship between the pelvis, the spine, and the cranium, specifically between the sacrum at the base of the spine and the occiput at the base of the skull, governs how the spine as a whole functions, and that correcting the position of the pelvis has consequences all the way up. Delivery follows from that premise and looks nothing like a thrust. The practitioner places graduated wedge shaped blocks under the patient's pelvis and lets the patient's own body weight and breathing supply a gentle, sustained corrective force over time. The patient lies there and the correction accrues. As with Thompson, I could not locate a documented account of SOT's assessment procedure in the accessible sources, and I am naming that as a gap in my documentation rather than pretending the technique lacks one.

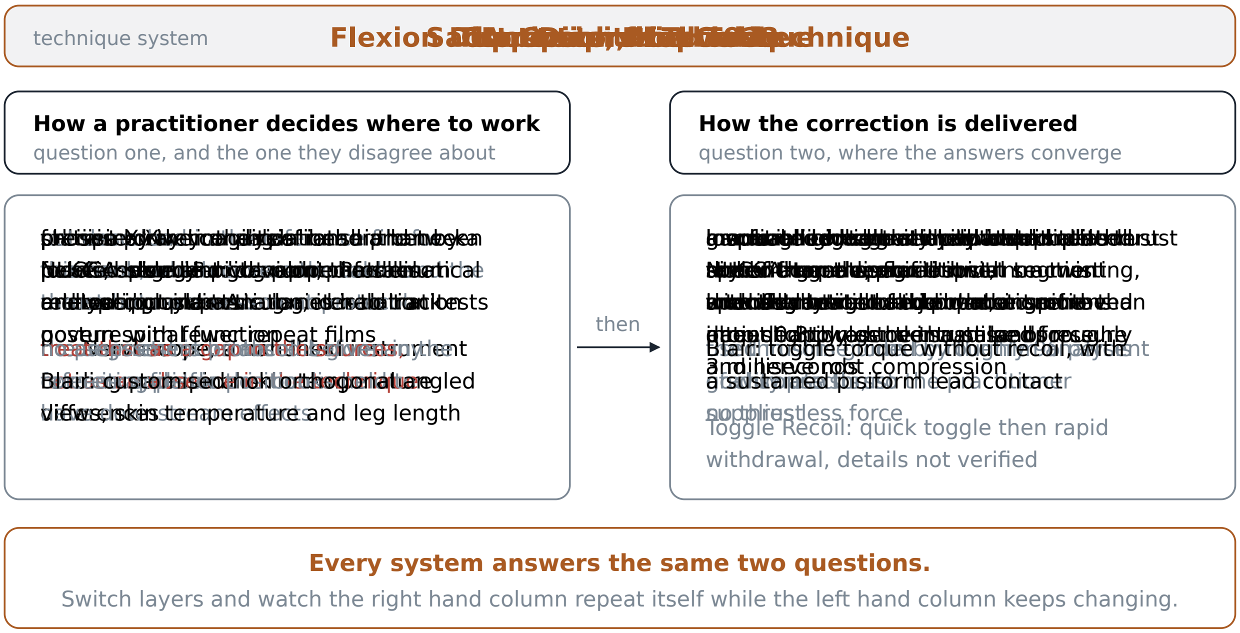
Flexion Distraction, also called the Cox technique, was developed for disc related problems. Delivery again is not a thrust. The patient lies on a specialized segmented table whose lower section moves, and the practitioner applies rhythmic, gentle flexion together with traction, pulling the lumbar spine into a long, repeated stretch. The stated intent is to lower the pressure inside the disc and reduce compression on the nerve root that exits beside it. Chapter 2 gives you the anatomy that makes that intent legible: the disc in front, the nerve root leaving through the intervertebral foramen at the side. This is the technique whose mechanism is easiest to describe to a patient, because you can point at it. The assessment question, again, I cannot answer from a source I trust, and again I am telling you so rather than filling the space.

Upper cervical work is not one technique but a family, and it is the most distinctive corner of the profession. Its common commitment is that the top two vertebrae of the neck, the atlas and the axis, are so mechanically and neurologically consequential that a very precise correction there is worth more than a great deal of work lower down. Three members of the family are worth knowing by name.

NUCCA, the National Upper Cervical Chiropractic Association technique, was founded in 1966 by Ralph Gregory, building on earlier work he did with John Grostic. Assessment is the most elaborate in the profession: precise X ray imaging of the head and neck, analyzed with a mathematical procedure called double pivot point analysis, from which the practitioner calculates an individualized misalignment pattern for that patient. Ongoing status is tracked with a posture measuring instrument called an Anatometer, which reduces how often repeat films are needed. Delivery is described as gentle, non thrust, and non twisting: a specific, low force, precisely aimed contact at the occiput and atlas region rather than a traditional cracking thrust.

Blair technique comes from William G. Blair, a Palmer graduate who practiced from 1949 and began teaching the technique around 1961. Assessment is X ray based like NUCCA but built differently: a customized series of angled views of the atlanto occipital joint, taken with a positioning device called a Blair Protractoclamp, and described by its own literature as non orthogonal, meaning the views are individualized to each patient's own joint angles rather than taken along standard fixed planes. That source calls Blair the only non orthogonal precision spinographic and adjustive cervical technique, distinguishing it from the orthogonally based methods that trace back to B.J. Palmer and Grostic, which includes NUCCA. Ongoing monitoring uses paraspinal dermothermography, meaning skin temperature readings along the spine, and functional leg length. Delivery is the Blair Toggle Torque: a toggle mechanism without recoil, incorporating roughly 180 degrees of torque and a sustained pisiform contact held through the thrust, rather than the quick withdrawal used elsewhere.

Toggle Recoil, historically called Hole In One and associated with B.J. Palmer, is where the whole upper cervical idea comes from. Delivery is a specific, quick toggle contact against the atlas followed by immediate rapid withdrawal of the hands, the recoil, typically with the patient in a knee chest position. I have to flag this one directly: when the research for this book went looking for a substantive, independent description of Toggle Recoil's assessment protocol and detailed mechanics, what came back was directory and navigational material rather than a technical account. So what you have just read is the general shape, and I am not going to elaborate beyond it. If you go to chiropractic school you will be taught this properly from the technique literature and the college archives. Here it stays general, on purpose.



Laid on one frame with assessment along one axis and delivery along the other, the technique systems scatter widely on how they decide where to work and cluster much more tightly on how the correction is delivered.

Now the observation I promised. Look down the delivery column and the variety is real but bounded: a manual thrust, a thrust with a table helping, a small instrument impulse, sustained low force through blocks and body weight, rhythmic traction. Five ways to move a joint, and any competent practitioner can learn several of them in an afternoon of seminars and get decent at them within a year. Look down the assessment column and something else is going on entirely. One system says palpate. One says take a full spine film, measure it with a proprietary grid, and run a temperature instrument down the back. One says compare the patient's leg lengths and test muscles. One says take individualized angled films of the top joint in the neck and compute a vector. These are not variations on a procedure. They are different claims about what a spinal problem *is* and what kind of evidence would reveal it.

That is the finding to carry forward: the technique systems of chiropractic differ modestly on how they move a joint and profoundly on how they decide which joint to move. Which means the assessment question is where the profession's real intellectual disagreements live, and it is also, inconveniently, where the research is least kind.

A note on a technique this chapter deliberately keeps short

Applied Kinesiology, developed by the chiropractor George Goodheart in 1964, assesses by manual muscle testing: the practitioner resists a muscle and judges the quality of its response, which is claimed to reveal organ dysfunction and other conditions elsewhere in the body. A 2003 survey found roughly 37.6 percent of United States chiropractors reported using it. I mention it here for one reason. Unlike most assessment methods, this one has been tested head on, and the result is unambiguous: controlled study found the testing performed no better than random guessing, and reviews of its repeatability across time and across examiners found results consistent with chance. When a student asks me how the profession should handle a technique whose central assessment claim has been tested and failed, my answer is that it should say so, in its own textbooks, and this paragraph is me doing that.

Can two examiners agree?

Everything in the previous section rests on an assumption so basic that it usually goes unexamined: that when a practitioner examines a spine and concludes that L4 is restricted, that conclusion is about the patient. If it were instead mostly about the practitioner, then two equally competent practitioners examining the same patient an hour apart would reach different conclusions, and the finding would be telling us more about who is doing the examining than about who is being examined.

That is a testable question and it has been tested a great deal. The tool for it is a number called **kappa**.

Here is kappa in plain language. Suppose two examiners each check the same twenty patients and each decides yes or no for some finding. You could simply count how often they said the same thing, but that number flatters everybody, because two people who both say no most of the time will agree most of the time by pure luck. Kappa corrects for that. It asks how much better than chance the two examiners agreed. A kappa of 1.0 means perfect agreement. A kappa of 0 means they did no better than two people guessing independently. A negative kappa means they did worse than chance, which sounds impossible until you realize it happens when two examiners are systematically disagreeing about something. The conventional reading, and you should treat it as a rough convention rather than a law, is that below about 0.2 is slight, 0.2 to 0.4 is fair, 0.4 to 0.6 is moderate, 0.6 to 0.8 is substantial, and above 0.8 is very good.

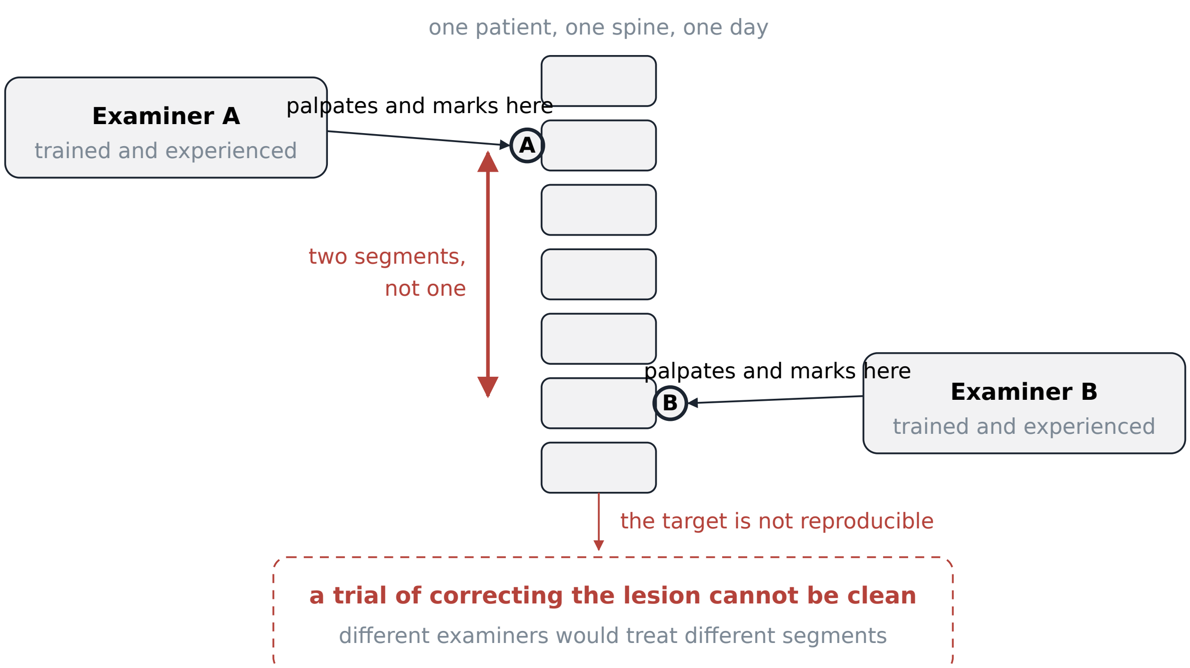
Now the findings, from a systematic review of the methods used to locate a manipulable spinal lesion and a companion review of the reliability of manual palpation in low back pain. I am going to give you the numbers as ranges, because that is how they come: different studies of the same test get different results, and the spread itself is information.

Pain provocation does reasonably well. Pain provocation means the examiner presses on something and the patient reports whether it hurts. For pain at the lumbar facet joints on static palpation, kappa ran from 0.38 to 0.73. For pain over the lumbar spinous processes, 0.21 to 0.57. For sacroiliac joint pain, 0.14 to 0.59. The prone instability test reached 0.46 to 0.87 in its muscle contraction phase, though it fell to 0.30 to 0.41 in the relaxation phase and to 0.10 when the phases were combined into one score. Soft tissue tenderness was inconsistent: sciatic nerve pain reached 0.80 in one study, gluteal tender points 0.51 to 0.68, and lumbar paraspinal muscle pain only 0.34 in others. Read the whole block together and the picture is fair to good, with real variability. Two examiners can find the same tender spot at a rate meaningfully better than chance.

Detecting which segment is stuck does badly. This is the other thing, and it is the thing most technique systems are nominally built on: not where it hurts, but which specific segment is hypomobile, restricted, fixated, not moving as it should. Static palpation for segmental mobility at the lumbar facet joints produced kappa values ranging from minus 0.17 to 0.17. At the lumbar spinous processes, minus 0.02 to 0.26. At the sacroiliac joints, minus 0.11 to minus 0.10, which is to say at or below chance across the board. Motion palpation, where the examiner moves the joint through its range instead of pressing on a still one, did somewhat better for sacroiliac joint motion, at 0.14 to 0.75 between different examiners and 0.23 to 0.73 for the same examiner repeating himself. Sacral base position reached 0.37 on flexion and 0.05 on extension.

The review's own summary sentence is worth quoting rather than paraphrasing: "reliability of soft tissue structures palpation is inconsistent, and reliability of bony structures and joint mobility palpation is poor."

I am not going to soften that and I am not going to let anyone use it as a club either. So let us do the reasoning carefully, because this is exactly the sort of finding that gets misused in both directions.



Plotted on the same kappa scale, agreement between examiners is fair to good for where a patient hurts and drops to poor or below chance for which segment is restricted, which is the finding the profession has to answer for.

What poor reliability does and does not mean

Start with what it does not mean. Poor inter examiner reliability of an assessment method is not, by itself, proof that the treatment does not work. Those are different claims and it is worth seeing exactly why they come apart.

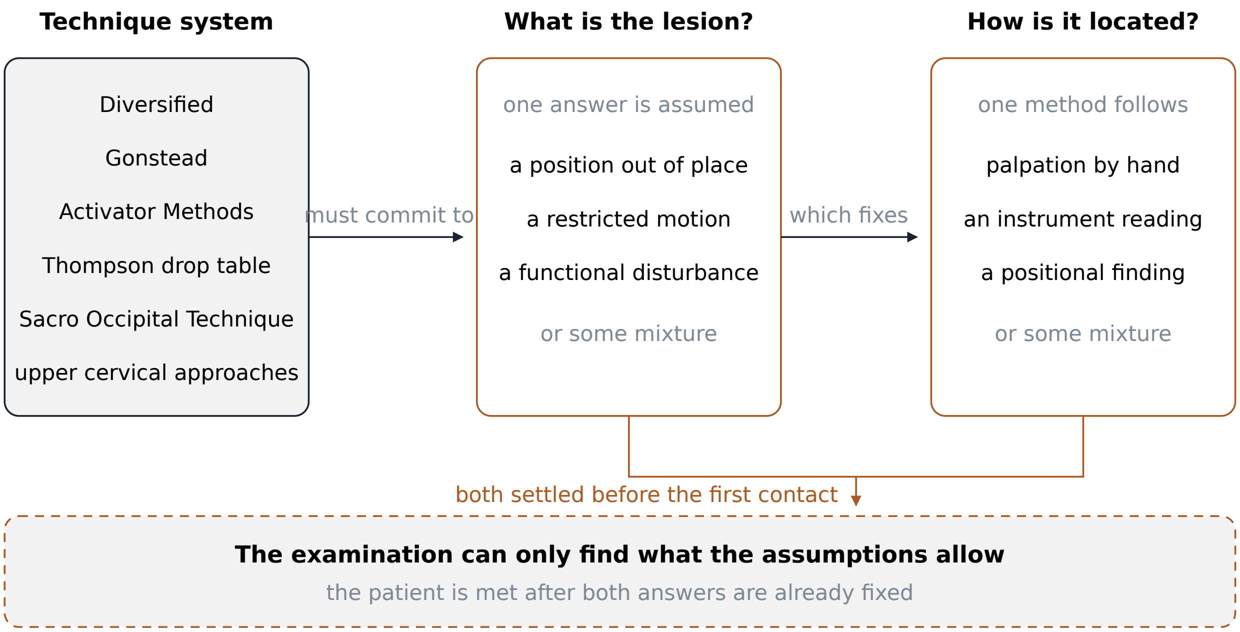
Imagine, purely as a logical exercise, a treatment that helps regardless of which segment you deliver it to, perhaps by acting on the region rather than the level, perhaps through reflex effects on muscle, perhaps through effects that have nothing to do with the specific joint at all. Under that scenario, examiners could disagree wildly about the target and patients could still improve, because the target was never doing the work. That is not a fringe scenario; it is a live hypothesis, and chapter 5 will show you what the trial evidence does and does not say about whether the treatment helps.

Notice, though, that this defense is expensive. It rescues the treatment by abandoning the specificity claim. If it does not matter much which segment you hit, then the elaborate apparatus for deciding which segment to hit, the films, the grids, the temperature instruments, the leg length comparisons, is not doing what it says it is doing. You can have the treatment or you can have the specificity, and the reliability data makes it hard to have both on the current evidence. A student who understands that trade is ahead of a great many working practitioners.

Now what the findings do mean, stated as plainly as I can. Two competent, well trained, honest practitioners examining the same patient on the same afternoon may not agree on which segment to work on. Not because one of them is careless. Because the thing they are trying to detect, small differences in how freely one joint moves compared to its neighbors, in a living body, through skin and fat and muscle, in a patient who is guarding a little because a stranger is pressing on their back, is right at or past the edge of what human hands can resolve consistently.

That is a genuine problem for any system whose central claim is that it locates a specific segment. It is not a fatal problem and it is not an accusation of dishonesty. It is a measurement problem, and measurement problems are ordinary in clinical medicine. Two radiologists disagree about films. Two cardiologists disagree about a murmur. What separates a mature field from an immature one is not the absence of such problems but whether the field measures them, publishes them, and adjusts its claims accordingly. This chapter is part of a course that intends to do the first two. The third is your generation's job.

There is one more consequence worth naming because it will come up in your clinical courses if you go on. If the location finding is unreliable, then a practitioner who tells a patient "your atlas is out" or "you have a fixation at T6" is making a more confident statement than the underlying evidence supports. There is a real difference between saying that and saying "this area is tender and moves poorly to me today, and I am going to work here." The second is defensible. The first sounds more impressive and claims more than we can deliver. I changed how I talk to patients about this maybe fifteen years into practice, and I did not lose a single one over it. Patients handle honest uncertainty better than practitioners expect. They handle being caught in false precision much worse.



Each technique system rests on an assumption about what its assessment method can detect, and lining those assumptions up against the reliability evidence shows which ones are currently supported, which are untested, and which have been tested and found wanting.

Leg lengths and skin temperature

Two assessment methods named in the technique families deserve their own treatment, because both are common and both are contested.

The first is **leg length inequality**, which is central to Activator assessment and used for monitoring in Blair. Here the literature is genuinely split, and I want you to see the split rather than a verdict. The reliability review I have been quoting is unenthusiastic about palpation based location methods generally. But a separate review in the chiropractic and manual therapies literature found high quality evidence supporting the use, with limitations, of static palpation, motion palpation, and measures of leg length inequality as clinically applied tests. That review did not isolate specific kappa figures in the portion available, so I cannot line it up numerically against the values above. Note the phrase "with limitations," which is doing real work in that sentence, and note that the second review is the more favorable framing of the two. The responsible move is to read them together rather than to quote whichever one suits your argument. If you find yourself citing only one of these two reviews, ask yourself which conclusion you had already reached before you started reading.

The second is **thermography**, meaning the measurement of skin temperature differences along the spine, which is the operating principle of Gonstead's Nervoscope and of the dermothermography used in Blair monitoring. Here the evidence is not split. The same review found that indirect assessment methods, such as skin conductance or thermography, tend not to be supported by the available evidence, and concluded that the evidence for these indirect methods was unfavorable relative to direct palpation methods. Direct palpation, as you have just seen, is itself unimpressive for locating segmental restriction. Being rated worse than that is not a good place to be.

I want to model the right posture here, because this is a place where students often overcorrect. The right response is not to conclude that Gonstead practitioners are frauds. Gonstead is a careful, disciplined, widely respected system whose delivery is as precise as anything in the profession, and I have referred patients to Gonstead practitioners and will again. The right response is narrower and harder: this specific instrument, making this specific claim, has been examined and the evidence does not support the claim. A practitioner can hold that finding and keep practicing thoughtfully. What a practitioner should not do is not know it.

A note on why the trials cannot settle this for us

You might reasonably ask why we cannot simply run trials of technique against technique and let the results sort it out. Some of that work exists and chapter 5 covers what it shows. Two problems limit it. First, the Cochrane reviewers assessing spinal manipulation for chronic low back pain rated their certainty low to very low, and they named heterogeneity in technique, dosage, and frequency across trials as a driver of that rating. In other words, the trials pool together things this chapter has just spent pages distinguishing. Second, the Cochrane review of manipulation and mobilization for mechanical neck disorders reported that manipulation and mobilization were not shown to differ from each other in effectiveness, which is a striking result for a profession that argues about delivery. What the literature is not currently able to tell us is whether any one named technique system outperforms another, and you should be skeptical of anyone who tells you their system has been proven superior.

How a practitioner actually chooses

Having read all of that, a reasonable student expects that practitioners select techniques by weighing evidence. I would like to tell you that is how it goes. Here is how it actually goes, and I am including myself.

You practice what your school taught you. Every chiropractic college has a technique culture. Some are Diversified with everything else as an elective. Some are steeped in Gonstead. Some are upper cervical strongholds. You spend four years having one set of hand positions drilled into you until they are automatic under stress, and automatic under stress is what you fall back on with your third patient of a Monday morning when you are tired. Chapter 4 walks through the training road in detail; what matters here is that a technique learned to the point of unconscious competence has an enormous advantage over one you read about, and that advantage has nothing to do with which is better.

You practice what your mentor practiced. This is the part that most resembles the older meaning of the word art. In my last year I stood next to a practitioner for months, and I did not simply learn his adjustments. I learned how he entered a room, how he set his feet, how long he waited before he thrust, how he decided a patient was not ready today. Almost none of that was said out loud. It transferred by watching and by having my hands corrected in the moment. That kind of learning is powerful, it is how skilled trades have always been passed down, and it has one serious defect: it transmits the mentor's blind spots as faithfully as his skills. If your mentor chased the noise, you will chase the noise, and you will not know you are doing it.

Your hands are the size they are. This sounds trivial and is not. Technique instructions are written as though every practitioner has the same equipment. They do not. A practitioner with small hands and a short reach will struggle with contacts on a large patient that a bigger colleague takes for granted, and will get better than that colleague at contacts requiring a small, precisely placed point. Body weight matters too, since much of the thrust comes from dropping your own mass. Practitioners quietly drift toward techniques their bodies do well and away from ones they have to fight. There is a career length version of this as well: hands and shoulders accumulate wear, and it is common for practitioners to shift toward instrument and drop table work in later years for reasons that have nothing to do with the evidence and everything to do with wanting to still be working at sixty.

You practice what your patients need. A practice near a university with a lot of athletes is a different job from a practice in a town with a median age of sixty five. Treat mostly manual laborers with acute low back complaints and you will get very good at lumbar work and rarely open your instrument case. Treat mostly older patients and the reverse. The technique mix in any practice is shaped by who walks in the door far more than by any decision the practitioner remembers making.

And then there is the body actually in front of you today. This is the one that overrides everything else, and it is the reason a practitioner needs more than one tool.

Take two patients I could see in the same hour. The first is twenty five, plays rugby on weekends, has a stiff and painful low back after a bad tackle, and is otherwise healthy. Good muscle tone, dense bone, a body that has absorbed impacts before. If the history and examination find nothing that says stop, he is a candidate for a manual thrust, and he will probably prefer it, because it is decisive and he will feel it work.

The second is eighty one, has thinning bone, is on the fourth year of a medication list I need to read carefully, has had a wrist fracture from a fall in her kitchen, and comes in with the same complaint in the same location. She is not getting the same thrust. She may not be getting a thrust at all. This is where the instrument, the drop table, the blocks, and sustained low force earn their place in the profession: they are ways of doing something useful to a body that cannot tolerate a manual high velocity load. It is also where a practitioner has to know what the red flags are and when the answer is not an adjustment at all but a telephone call to her physician. Chapter 5 covers those red flags and the referral obligation properly, and it is the most important page in this book for anyone who intends to practice.

Even between those two extremes there is a daily judgment about what a body can take. A patient who is guarding, frightened, or in an acute pain flare cannot relax enough for a clean setup, and forcing a thrust past a braced muscle is how you produce a sore patient who does not come back. Sometimes the correct technique today is soft tissue work and a return visit. I have never regretted deciding a patient was not ready. I have regretted the other thing.

The honest account of how practitioners choose techniques is a story about schools, mentors, hands, and the person on the table, and only after that about evidence.

I said I would include myself, so here it is. I practice mostly Diversified because that is what I was trained in and what my hands do without thinking. I use an instrument regularly, mostly for older patients, small children, and anyone whose examination or history makes me cautious. I use a drop table when I want a decisive result with less force from me. I do not practice upper cervical work, not because I have concluded it is wrong, but because doing it properly requires a specific imaging and analysis discipline that I did not train in and would not attempt part time. That last sentence is, I think, the most useful thing in this section: knowing which techniques you are not qualified to perform is a technique skill.

What you can learn without touching anyone

This week you are not going to adjust anybody. You do not have the training, you do not have the license, and you should not want to. What you are going to do is watch, and I want to close by convincing you that watching, done properly, is not a consolation prize.

Here is the argument. Everything this chapter has covered turns on assessment, and assessment is mostly observable. The setup is visible. The order of operations is visible. What the examiner compares to what is visible if you know to look for it. And the logic connecting a finding to the next step is fully available to a careful observer with no hands on skill whatsoever. Structured observation is how you build the analytic half of clinical judgment before you are allowed to build the manual half. It also happens to be exactly what the practical licensing examination scores: the technique station of the national board practical examination described in chapter 4 grades candidates on safety, positioning, contact points, stabilization, and verbalized intent. Four of those five are things a person in the room can see and hear.

So here is the method. Use it on any examination you observe, in person or on video, in a clinic or in a course module.

First, describe before you evaluate. Write down what happened in the order it happened, in ordinary words, with no judgment attached. Patient was seated. Examiner stood behind and to the left. Examiner's thumbs moved down both sides of the spine from the base of the neck. Examiner stopped twice. Examiner asked the patient to bend forward and repeated the movement. Most students skip straight to opinions and lose the data. Describe first, always. You can evaluate afterward and the description will still be there.

Second, name the comparison. This is the heart of it. Every clinical finding is a comparison, and an examination is only as good as what it is comparing to what. When the examiner presses on one side of the spine, is she comparing left to right? When she moves a segment, is she comparing this level to the levels above and below? Is she comparing today to last week? Is she comparing this patient to a normal range from a textbook, and if so, whose normal? Once you start asking this question you cannot stop noticing it, and you will find that a surprising number of confident clinical statements are comparisons with the second half missing.

Third, predict the next step before it happens. This is the part that converts watching into thinking. When the examiner finds something, stop and ask yourself what should follow. If a finding is real, it should change what happens next: it should narrow where the examiner works, or send her to a different test, or rule something out, or stop the examination entirely and send the patient somewhere else. A finding that changes nothing about the next step was not worth collecting. So predict, then watch what actually happens, then account for the difference. When your prediction is wrong, one of two things is true: either you misunderstood the finding, or the examiner did something the finding did not require. Both are worth knowing, and you will learn more from the times you were wrong.

Fourth, ask what would have changed the plan. After the examination, ask what the examiner would have had to find in order to do something different: to choose an instrument instead of hands, to work at a different level, to refer the patient out. A practitioner who cannot answer that question about their own examination is not really examining. They are performing a routine.

Those four moves are not a warm up for real clinical skill. They are a large part of it. The manual half takes years and cannot be learned from a book, and I have told you plainly in this chapter which parts of it I cannot even put a number to. The analytic half starts now, costs nothing but attention, and is the part that separates a practitioner who is reasoning from one who is repeating what he was shown. When you watch an examination this week, you are practicing the thing that everything clinical is built on.

And when you get to chapter 5 and start reading actual trials, you will find that you have already been doing this. Asking what is being compared to what, and what a finding should change, is the same skill applied to a paper instead of a patient.

Terms from this chapter

Adjustment

The chiropractic term for the whole procedure of positioning a patient, isolating a joint, taking a contact, loading the tissue, and delivering a corrective force to that joint.

High velocity low amplitude thrust

A rapid, small displacement force that carries a joint briefly past the point slow steady pressure could reach, without taking it far enough to damage tissue.

Contact point

The specific part of the practitioner's hand, such as the pisiform bone at the base of the palm, placed on a specific bony landmark to deliver the force.

Tissue slack

The small amount of free play in a joint before ligaments, capsule, and muscle tone begin to resist; taking up the slack means moving through it until resistance is felt.

Pre load

The light steady pressure held at the edge of tissue resistance so that the joint is already loaded when the thrust arrives.

Cavitation

The sudden formation of a gas cavity inside a liquid when local pressure drops; in a joint it is the event that produces the audible pop.

Tribonucleation

The mechanism identified by real time imaging in 2015, in which pulling apart two surfaces held together by fluid drops the pressure between them enough for a new gas cavity to form.

Static palpation

Examination by pressing on a still patient to judge tenderness, tone, or position.

Motion palpation

Examination by moving a joint through part of its range while feeling how freely it moves compared with its neighbors.

Segmental restriction

The claim that one specific spinal level is moving less freely than it should; the central target of most technique systems and the finding examiners agree on least well.

Kappa

A number expressing how much better than chance two examiners agreed; 1.0 is perfect agreement, 0 is no better than guessing, and a negative value is worse than guessing.

Inter examiner reliability

How consistently two different examiners reach the same conclusion about the same patient, as distinct from whether that conclusion is correct.

Mechanical force manually assisted

A delivery method in which a handheld instrument, rather than the practitioner's body, supplies a small controlled impulse to the target segment.

Thermography

Measurement of skin temperature differences along the spine, used as an indirect assessment method in some systems and rated unfavorably against direct palpation in the evidence reviewed here.

Before you move on

1. Walk through the five stages of an adjustment in order, and say which stage takes the most time and skill and why.
2. What did the 2015 real time imaging study find about the moment the cracking sound occurs, and how did that reverse the older explanation?
3. A patient tells you the adjustment did not work because there was no pop. Using what this chapter says about cavitation, explain to her what the sound does and does not indicate.
4. Pick any three technique systems from this chapter and state, for each, how it decides where to work and how it delivers the correction. Which of the two questions produced the bigger differences between your three?
5. Define kappa in one sentence without using the word statistic. Then explain why a kappa near zero for detecting segmental restriction is a more serious problem for Gonstead or NUCCA than for a system that makes no claim to locate a specific segment.
6. Judgment question. Poor inter examiner reliability of an assessment method does not by itself prove the treatment does not work. Explain why not, and then explain what the profession gives up if it defends the treatment that way.
7. Judgment question. You are observing an examination. The examiner presses along one side of the spine, pauses at one level, and immediately positions the patient for an adjustment at that level. Name the comparison she appears to be making, name one comparison she did not make, and say what you would have expected to happen next if the finding were solid.

CHAPTER 4

Becoming One

> *What does it actually take to become a chiropractor, and what does the life feel like once you are one?*

Every term, somewhere around week four, a student stops me after class and asks a version of the same question. Not whether the work is legitimate, which is chapter 5 and chapter 7 and the argument this profession has been having with itself for a century. Something more personal. Should I do this. Is it a good life. What is it like on a Tuesday.

This chapter is what I say to that student, written down so I do not have to remember it differently each time. It comes in two halves. The first is the road: prerequisite credits, the accreditor, the hours, the four national examinations plus one optional fifth, the state license, and the money. All of that is public. Anyone can look it up, and almost nobody does before they apply. They look it up after, in the fourth month of the first year, when the size of the thing becomes visible. I would rather you looked now.

The second half is the life, and it is the reason this chapter is worth reading rather than skimming. What an associate position actually costs you in autonomy. What a full day of manual work does to a set of hands over twenty years. How much of your week goes to insurance authorizations and payroll and the person who does not show up at two o'clock. Why the same degree licenses a meaningfully different job depending on which state you sit the license in. Why people leave, and why the ones who stay stay.

One warning about the first half. Numbers move. Standards get revised, boards change their requirements, tuition rises every year without asking anyone. Every figure here is accurate as of the sources this book cites and should still be checked against the actual program and the actual state board before you sign anything. This is a map of the terrain, not a current price list.

What you need before you can apply

Start with the thing most people get wrong. A bachelor's degree is not strictly required to enter a chiropractic program. The Council on Chiropractic Education, which is the accreditor that governs Doctor of Chiropractic programs in the United States, requires admitted students to hold either a baccalaureate degree or a minimum of 90 semester credit hours of prior coursework from a regionally or nationally accredited institution. Ninety semester hours is roughly three years of full time study. The federal Bureau of Labor Statistics describes the same requirement from the outside as at least three years of undergraduate education, which is the same fact in different clothing.

Where a student uses the 90 hour route without finishing a degree, the accreditor attaches grade conditions. The standards specify a minimum cumulative grade point average of 3.0 on a 4.0 scale, or a grade point average between 2.75 and 2.99 combined with at least 24 semester credits in life and physical science courses. So the door exists, and it has a threshold on it, and the threshold gets lower only if you have loaded up on science.

Most people who matriculate hold a full bachelor's degree anyway, and I think that is the better choice even though the shorter route is legal. The science load in the first year of a chiropractic program is heavy and unforgiving, and the students who struggle are almost never the ones who took an extra year of chemistry. A degree is also portable. If you get eighteen months in and discover that you do not want to touch strangers for a living, which happens and which is not a failure, the degree you hold is a place to stand while you decide what is next. Ninety loose credits are not.

The prerequisite science coursework itself is the predictable list: biology, general chemistry, organic chemistry, physics, and coursework in anatomy, physiology, and related life sciences. Specific course lists vary somewhat by program, which matters more than it sounds like it does. Two schools you are considering may not want the same physics sequence or the same number of laboratory credits, and finding that out in your final undergraduate semester is expensive. Pull the prerequisite list from each program you are seriously considering, in writing, and plan your last two years against the union of those lists rather than against the one you like best.

It is also worth knowing how small the field of schools is. Counts vary by source and by whether Canadian and Puerto Rican programs are folded in. One accreditor linked directory listed roughly 18 to 20 programs based in the United States; the American Chiropractic Association describes 19 nationally accredited chiropractic doctoral programs across the United States, Canada and Puerto Rico. Treat the number as approximate, because programs open and close: Palmer College of Chiropractic West closed in 2025. The practical consequence is that you are choosing among a handful of institutions, not hundreds, and geography will do a lot of the choosing for you.

The accreditor, and why it is the gate

Students sometimes treat accreditation as administrative background noise. In this profession it is the gate, and it is worth understanding before you send a deposit anywhere.

The Council on Chiropractic Education was recognized as a nationally recognized accrediting agency by the United States Commissioner of Education on August 26, 1974. That recognition is what allows the accreditor's standards to function as the common floor under every program in the country. Those standards are where the numbers in the next section come from. And because state licensing boards and the national examining board both key their eligibility rules to graduation from an accredited program, accreditation is the thing that converts your four years into a license. A program without it is, for practical licensing purposes, a very expensive hobby. Verify the accreditation status of any program yourself, from the accreditor's own directory, and verify it for the year you would actually enroll.

That 1974 date is not an accident of the calendar. The same year, Congress authorized Medicare payment for chiropractic services, and Louisiana is widely reported as the fiftieth and final state to license chiropractic, though I would want the primary state record in hand before putting that date on a slide. Three doors opened in one year. Why they opened then is chapter 6, and it is a better story than I have room for here.

Four years, 4,200 hours, and a thousand of them with patients in front of you

The Doctor of Chiropractic degree is a four year program in full time equivalent terms. Underneath that description sit the two numbers that actually describe the workload. Accreditation standards require a minimum of 4,200 instructional hours, counting classroom, laboratory and clinical work together, of which at least 1,000 hours must be in a supervised patient care setting.

Sit with the arithmetic, because it tells you what the next four years look like. Four thousand two hundred hours across roughly four years of terms is a schedule with very little slack in it. This is not a program you attend around a full time job. Students do work during it, and many have to, and it is one of the most reliable sources of misery I see in people who are otherwise doing fine.

You will encounter a claim, in recruiting material and in arguments, that this classroom hour total is comparable to what medical and osteopathic students complete. The hour figures are real and verified. The comparison built on them is advocacy framing from a professional association rather than an independently adjudicated equivalence, and I want you careful with it for a practical reason rather than a political one. The content mix differs substantially: a chiropractic curriculum weights basic science, diagnosis, manipulation specific technique and a clinical internship, and a medical curriculum weights differently and continues into residency training that has no equivalent here. Deploy the raw hours comparison in an argument and the first competent person you meet will ask about the content mix, and you will have nothing to say. Use the hours as evidence that the training is substantial and demanding. Do not use them as evidence that it is the same training. It is not, and it does not need to be.

The 1,000 clinical hour minimum is the requirement that reshapes the back half of the program. Somewhere in the second half you stop being a student who studies patients and become something closer to an intern with a caseload and a supervising clinician looking over your shoulder. You take histories. You examine. You write notes that someone else reads and corrects. You have the experience, for the first time, of a person sitting on your table who is not a classmate and who is not going to be impressed by your effort.

This is the part of the program where people find out whether they want the job. I have watched students light up in clinic who were mediocre in the lecture hall, and I have watched excellent students discover in clinic that the actual daily transaction, hands on a stranger's body, forty times, every day, forever, is not what they want. That second discovery is a good outcome, not a failure, and I would rather it happen in the clinic entrance year than in year seven of practice. If you go, pay close attention to how you feel in the first month of clinic. It is the most honest data you will get about your own future.

The boards

Running alongside the program is a sequence of national examinations administered by the National Board of Chiropractic Examiners. Essentially every state board requires passage of these parts for licensure, so they are not optional in any practical sense, and their timing is fixed relative to your progress through the program rather than to the calendar. Here is what each one is and when you sit it.

Part I is basic sciences, computer based, 255 questions delivered across two sessions. One session covers general anatomy, spinal anatomy and physiology; the other covers chemistry, pathology and microbiology. Students become eligible in the second year of the program, with registrar approval. This is the examination that tests whether the foundation the first two years poured is actually load bearing. Everything in chapter 2 of this book, the vertebra part by part, the joint complexes, the nerve roots, lives here.

Part II is clinical sciences, also computer based, and it is organized into six domains with published weights: general diagnosis at 19 percent, neuromusculoskeletal diagnosis at 20 percent, diagnostic imaging at 17 percent, principles of chiropractic at 14 percent, chiropractic practice at 17 percent, and associated clinical sciences at 13 percent. Read that distribution carefully, because it is a compact statement of what the profession thinks a chiropractor is. More than half the weight sits on diagnosis and imaging. The philosophical material that dominates chapter 1 of this book carries 14 percent. Part II is taken later in the program, generally as students move into clinical training; the examining board specifies eligibility rules rather than a fixed month.

Part III is case management and clinical competency, computer based, 130 questions. Those questions come in three shapes: 80 standard multiple choice items, 20 extended case presentations that require two responses each, and 30 diagnostic imaging interpretation items. Nine domains are covered, including case history, physical examination, neuromusculoskeletal examination, diagnostic imaging, chiropractic techniques, and case management. Candidates must be within nine months of graduation and must have already passed Part I. The passing score is 375 on the scaled scoring system.

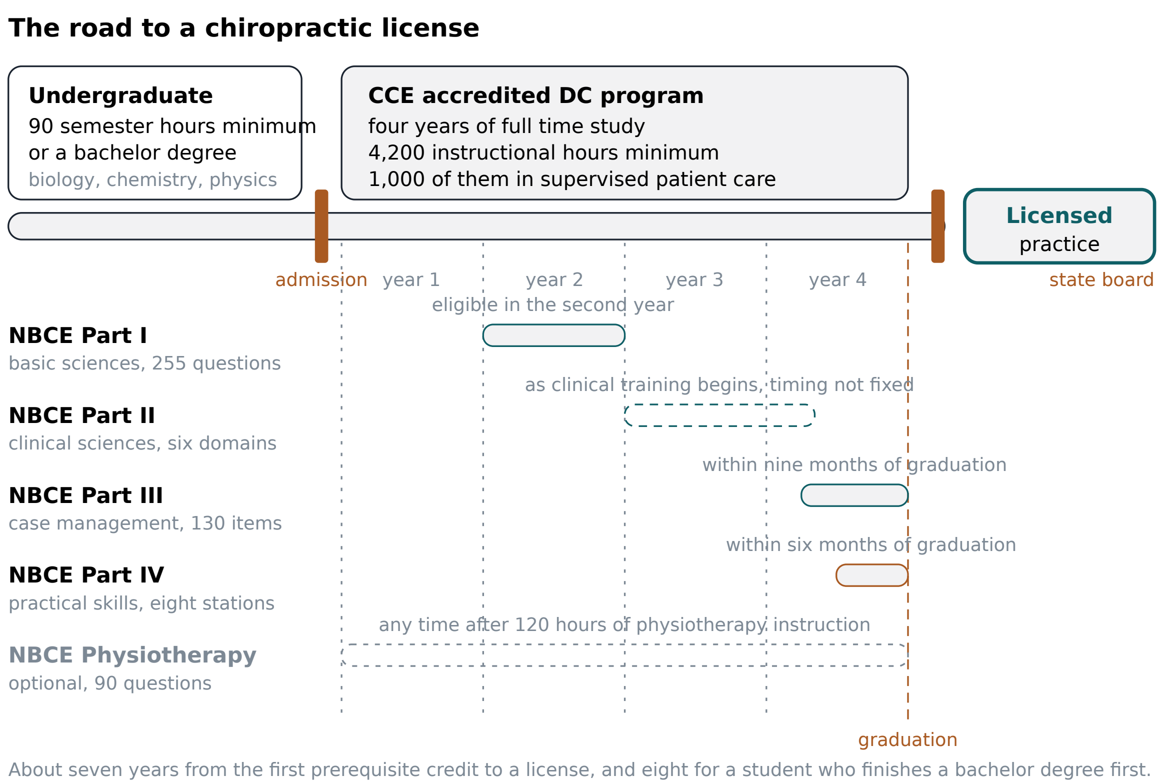
Part IV is the practical examination, and it is the one people remember. It is taken near the end of the program, with eligibility opening six months before graduation. It consists of seven case management stations and one technique station. Each case management station gives you a two minute chart review, a fourteen minute patient encounter, and six minutes to write a note in the standard subjective, objective, assessment and plan format. The technique station gives you a two minute chart review and a fourteen minute encounter in which you perform ten adjustment setups. Every encounter is video recorded, and the recordings are scored on safety, positioning, contact points, stabilization, and verbalized intent.

I want to stop on that last scoring criterion, because students dismiss it as examination theater and it is not. Verbalized intent means saying out loud what you are about to do, to whom, and why, before you do it. That is not a trick for passing a practical. It is the habit that protects a patient and protects you, every day, for the rest of your career, and it is the operational core of informed consent, which chapter 5 takes up properly. The examination is scoring a habit. Build the habit in school and you will never have to remember to perform it.

The Physiotherapy examination, usually just called PHT, is the separate fifth one. It is optional at the national level but required by some states, and not others, for a scope that includes physiotherapy and modalities. It is 90 multiple choice questions in 75 minutes. Eligibility requires that you have completed 120 hours of physiotherapy instruction, and it may be taken alone or alongside the other parts once that coursework is done. State requirements for a passing PHT score vary, which is the first of many places in this chapter where the phrase state requirements vary is going to do heavy lifting.

So the sequence, compressed: Part I in the middle of the program, in year two. Part II as clinical training begins. Part III and Part IV both cluster at the end, Part III inside a nine month window before graduation and Part IV inside a six month window. PHT any time after you have the 120 hours.

The clustering at the end is worth planning for and almost nobody does. In the same few months you are finishing clinic requirements, sitting two national examinations including a video recorded practical, applying to a state board, and either interviewing for an associate position or negotiating a lease. I have watched capable people come apart in that window purely from scheduling, not from difficulty. If there is one piece of tactical advice in this half of the chapter, it is to get Part I and Part II behind you at the earliest eligibility rather than the latest, so that the final year holds two examinations instead of four.



The road from prerequisite credits to a state license is a fixed sequence with the national examinations falling in defined windows, and the last two windows overlap the busiest months of the program.

The license itself

The national examinations are national. The license is not. You are licensed by a state board, under that state's statute, and the board decides what else it wants from you beyond the examination scores.

Here I have to be careful about what I actually know. It is commonly written that all but one state has adopted the national board's Parts I through IV as a licensure requirement. I could not verify that holdout claim for the current year, so I will not repeat it as fact, and neither should you. What is safe to say is that essentially every state board requires the national board examinations, and if you read that a specific state is the exception, confirm it with that state's board rather than with whoever told you.

Beyond the examinations, each state chiropractic board sets its own additional licensure requirements and its own scope of practice rules, and these vary substantially. I am deliberately not listing specifics, because the specifics change and a textbook is exactly the wrong place to look them up. The habit you want matters more than the facts: get in the practice, now, of going to the board's own website for the state you care about and reading the actual rule rather than a summary of it. You will do this for the rest of your professional life. Start while the stakes are low.

The other consequence of a state license is mobility. Carrying it to another state is governed by the receiving board's rules, not by your original board's opinion of you, and that board may want examinations, coursework, or documentation you did not anticipate. If you have any reason to think you will move, and at twenty four most people have more reason than they believe, read the rules in the two or three states you might plausibly land in before you settle anywhere.

The number nobody puts on the brochure

Now the part I promised. A student deciding whether to do this deserves the money figure before they apply rather than after, and recruiters do not lead with it, so I will.

A survey study of United States chiropractors published in 2024 and 2025 found mean student loan debt at graduation of $176,297. The median was $185,000, and the middle half of respondents fell between $120,000 and $240,000. That is debt actually carried at the moment of graduation, self reported by practitioners.

The same study estimated the total cost of a chiropractic education, working from published cost of attendance data at 13 chiropractic programs, at a mean of $260,642, with a median of $277,452 and a middle half between $245,304 and $280,050. Cost of attendance means tuition plus the institution's own estimate of what it takes to live while enrolled.

Those two sets of numbers do not match, and the gap is informative. Total cost runs roughly a hundred thousand dollars above typical graduation debt, and the difference is made up by family money, prior savings, a working spouse, scholarships, military benefits, and work during the program. If you have access to none of those, your number is closer to the cost figure, and you should plan against it.

The finding from that study that I think should worry a prospective student most is not any of the dollar amounts. It is that graduates tended to underestimate their own eventual debt by roughly $50,000 relative to institutional cost of attendance data. These are not careless people. They are people who made a very large financial decision on a number that turned out to be wrong by fifty thousand dollars, in the wrong direction, about their own lives. Separately, some chiropractic programs rank among graduate programs nationally with the highest debt to earnings ratios, based on 2024 data. Those two facts belong next to each other.

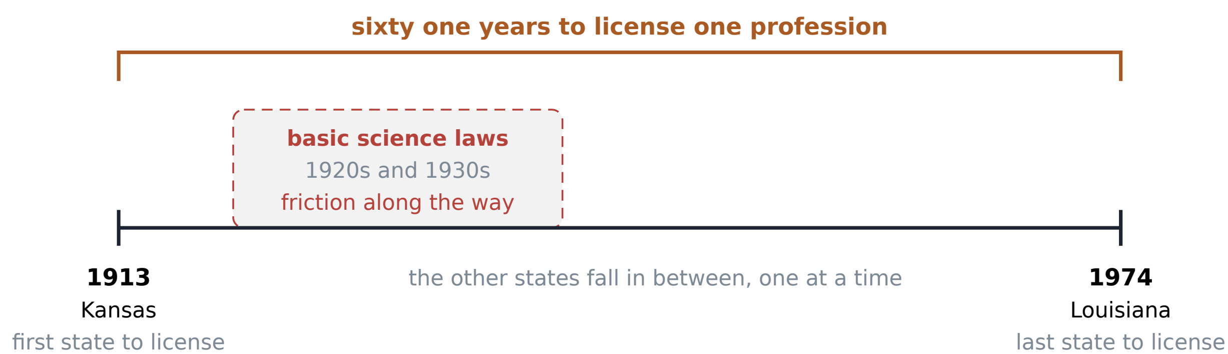
Graduates underestimated their own debt by roughly fifty thousand dollars, which means the most dangerous number in this chapter is not the size of the loan but the confidence people had about it.

Now the honest gap, because the brief for this book requires me to say where the evidence stops and this is one of those places. I am not going to give you a starting income figure. The reference file this book is built on does not carry an earnings number I can stand behind, and the numbers that circulate in recruiting material are usually either association surveys that oversample the successful or averages that conceal an enormous spread. I would rather tell you nothing than tell you something that turns out to be an advertisement.

What I can tell you is the shape of it from inside the profession, offered as my judgment rather than as a statistic. Income in the first several years depends far more on business circumstance than on clinical skill. Two graduates with identical grades and identical hands, one taking an associate position in a saturated suburb and one buying a retiring practitioner's practice with an established patient list in a town with two chiropractors, will be a long way apart by year three, and almost none of that distance will be about how well they adjust. That is not a comfortable thing to say to people who just spent four years learning to adjust. It is true, and it is the single most useful thing in this half of the chapter.

Some context on the size of the field. The Bureau of Labor Statistics estimated approximately 61,700 chiropractors employed in the United States as of 2025, and projects 9 percent employment growth from 2025 to 2035. The American Chiropractic Association puts the number at about 70,000 practitioners plus roughly 3,000 in academic and management roles, which is a broader count that likely includes licensees who are not clinically active. Sixty thousand to seventy thousand is the defensible range. However you count it, this is a small profession. Your reputation in it will be local and it will travel further than you expect.

One span, state by state



From the first prerequisite credit to a license in hand is a span of roughly seven years of full time study, and the money accumulates across the whole span rather than only during the four year program.

A note on doing the arithmetic before the campus tour

Here is a concrete assignment for anyone seriously considering this. Go to each program you are considering and find its published cost of attendance for the current year. Multiply by four. Add anything you will need to live on that the institution's estimate leaves out, which is usually more than you think. Look up the current federal graduate loan interest rate and estimate what the balance becomes by the time you are practicing, since interest accrues during the program. Then say the total out loud to someone who loves you and who is not enrolled in anything. If it still makes sense after that sentence has been said in a room, go. This exercise costs an afternoon and I have never once had a student regret doing it.

The first year out

Now the second half. You have the degree, you have the license, and for the first time since you were eighteen nobody is going to tell you what happens next. There are essentially three doors, and they trade income, autonomy and hours against each other in different proportions.

The associate position. You work in someone else's clinic. They own the schedule, the phone, the lease, the equipment, the billing system, and the patient relationships. You see patients and get paid, usually on some combination of a base and a production percentage. What you gain is enormous and undersold: a full schedule from day one, a working front desk, someone down the hall who has seen this presentation before, and the chance to make your early mistakes inside a structure that will catch them. Volume is how you get good. An associate position hands you volume you could not generate yourself for two years.

What you give up is autonomy, and more of it than the interview suggests. It is their technique culture, their expectations about visit frequency, their fee schedule, their documentation system, their hours, and often the shifts the owner does not want, which are evenings and Saturdays. Some of this is fine and some of it is not, and the difference is usually about clinical decisions rather than convenience. If a clinic expects a predetermined number of visits per patient regardless of what the patient in front of you is doing, that is not a scheduling preference, that is a clinical decision being made by someone who is not examining the patient, and you are the one who signed the note. Ask about it directly in the interview. Ask what happens when you tell a patient they are done.

Read the contract, and read the restrictive covenant in particular, because it determines where you are allowed to practice when this ends. A covenant with a wide radius and a long term can effectively mean that leaving the job means leaving the metropolitan area. People sign these at twenty six without reading them and discover them at twenty nine. Have someone who reads contracts for a living read yours. It costs a few hundred dollars and it is the best money you will spend that year.

Buying a practice. You purchase an existing clinic: the patient list, the lease, the equipment, the staff, the phone number people already dial. It is the fastest route to a full schedule that exists. It is also a second loan stacked on the loan you already carry, and the thing you paid most for, the goodwill, is attached to a human being who is about to leave. Patients were loyal to a person, not to a suite number. A transition period in which the selling practitioner is visibly present and visibly handing you off is worth more than any clause in the contract, and you should insist on one. Assume some attrition in year one, because there will be some.

Building your own. The maximum autonomy option, and it is real autonomy: your hours, your fees, your technique, your patient population, your walls. It is also the option where you are the marketing department, the billing department and the janitor, and where the first months contain something nobody warns you about. You will have an empty schedule. You spent four years and a quarter of a million dollars learning to put your hands on people, and you will spend a good part of your first six months not doing that. You will be printing things, meeting people, sitting in a quiet room at eleven in the morning. That stretch breaks more new practitioners than any clinical difficulty does, and knowing in advance that it is normal is most of the defense against it.

Underneath all three doors is one fact that the curriculum does not weight the way the job does. Most chiropractors are running a small business, whether or not they set out to. Even the associate is inside one and being shaped by its economics. You did not enroll to learn accounting, payroll, lease negotiation, hiring, or marketing, and you will do all five. I say this without complaint. It is genuinely part of the appeal for a lot of us, since owning your own days is only possible if you own the thing that generates them. But you should know it is coming, and you should stop imagining that the business part is a distraction from the real work. For most practitioners it is roughly a third of the real work.

What a full day actually contains

Let me describe an ordinary Tuesday, since that is what the student in the hallway is really asking about.

A busy practice runs established patients in short slots, often ten or fifteen minutes, with longer blocks for new patients and re examinations. That cadence sets the rhythm of everything else. You are on your feet essentially all day, moving between two or three rooms, and the movement itself matters, because it is what keeps a hundred repetitions from landing on the same tissue in the same posture.

Here is a single visit, at the resolution the job actually has. Dana is fifty two, on her third visit for low back pain that started when she moved a couch, and she is on the schedule at 9:20. Before you open the door you look at what you wrote on Friday, because you have seen forty people since then and you do not remember her. You ask what has changed, and you listen to the answer rather than to the answer you expected. You re examine the things you examined before, because a comparison is worth more than a finding. You tell her what you are about to do and why, out loud, in words she can repeat to her husband. You position her, take up the tissue slack, and deliver the thrust, which is described properly in chapter 3 and which takes a fraction of a second at the end of a setup that took thirty times longer. You tell her what she may feel this afternoon and this evening. You tell her what you want her to do between now and Thursday. Then she leaves, and you write the note, and the note takes about as long as everything else in the room combined.

Multiply that by thirty or forty and you have the clinical part of the day. Now add the rest of it, which is what nobody describes.

Documentation. Every visit generates a note in the standard subjective, objective, assessment and plan format, the same format the national practical examination gives you six minutes to produce. It records what the patient reported, what you found, what you did, and what the plan is. Insurers require it. Your board requires it. And you require it, because across a caseload of several hundred active patients your memory is not a clinical record and will not survive contact with a Thursday. Notes written the same day are notes. Notes written on Saturday for the week are fiction with good intentions. Everyone knows this and about a third of the profession does it anyway.

Insurance. Depending on your payer mix, some part of every week goes to prior authorizations, visit limits, documentation of medical necessity, denied claims, resubmissions, and the persistent gap between what you billed and what arrived. This is administrative work with clinical consequences, since an authorization that runs out mid course changes what care a patient can actually afford to continue. How chiropractic came to be inside the insurance and Medicare system at all is chapter 6. What it costs you in hours per week is this chapter, and the honest answer is more than you would guess and less than the internet claims.

The no shows. The two o'clock does not come. Then the two fifteen does not come. In a business with fixed rent and a paid front desk an empty slot is not neutral, it is a small loss, and a day with six of them is a bad day financially no matter how well the other thirty went. Practices manage this with reminder systems, cancellation policies, deposits for new patients, and a certain amount of strategic overbooking. What nobody tells you is the emotional part: for the first year or so you will take it personally, as a verdict on you. Then you will stop, because you will have understood that people miss appointments for reasons that have nothing to do with you, and that understanding is one of the small quiet reliefs of getting a few years in.

The front desk. I will say something that sounds like an exaggeration and is not. The person who answers your phone determines more of your income than your technique selection does. They decide whether a first time caller becomes a patient. They decide whether the schedule is a solid block or a sieve. They handle the money conversation, which is the conversation you are worst at. Hiring that person, training them, paying them enough to keep them, and covering the desk yourself when they are out with a sick child is part of your job now. So is payroll, so is scheduling their time off, so is the awkward day you have to correct them.

Marketing. Community talks, screenings at a race or a company health fair, relationships with physicians and therapists and trainers who might send someone your way, a website that loads on a phone, and reviews, which you will read more often than is good for you. Most of it produces less than you hope. The one that works is slow: a patient who got better and tells three people. It is the only marketing that compounds and the only one you cannot buy.

Then there is the thing that surprises people most, the switching cost. Inside twenty minutes you go from a torn hamstring, to a new patient who is frightened and needs you to slow down, to a denied claim, to an employee asking about her hours, and back to a table. Each requires a different part of you. Clinicians in large institutions have layers of other people absorbing three of those four. You will not, and by five o'clock the fatigue is at least as much from the switching as from the adjusting.

> **A note on what practitioners actually spend their week on**
>
> If I could show a new graduate one honest chart, it would be a breakdown of the working week. The hands on portion is the part everyone trains for and it is not the majority. Documentation, insurance administration, staff and payroll, marketing and community work, continuing education, and the ordinary friction of running a room where people come and go take up a large share of it. I am giving you shape rather than percentages here on purpose, because reliable time use figures for this profession are not something I can source, and this book does not invent numbers. But the shape is not in dispute among practitioners I know: if you want a job that is ninety percent hands on people, an associate position in a busy clinic is the closest version of it that exists, and even there it is not ninety percent.

The same degree, a different job

Here is the structural fact that a prospective student should understand before choosing a school, and that most understand only after they have chosen one. The degree is national and the examinations are national, but the job is defined by the state. Scope of practice, which is the legal description of what you are permitted to do with your license, is set by each state's statute and board, and it varies substantially.

A survey of state jurisdictions reported the broadest scopes in Missouri, New Mexico, Kansas, Utah, Oklahoma, Illinois and Alabama, and the most restrictive in New Hampshire, Hawaii, Michigan, New Jersey, Mississippi and Texas. I want to flag the quality of that source honestly: I reached it through a secondary summary, its exact publication year is not confirmed, and scope statutes change year to year. So use the shape of that finding, which is that scopes genuinely differ by a lot, and do not use the ranking as though it were current law. Check the board.

Some of the boundary is consistent nationally. Ninety percent or more of surveyed jurisdictions do not include prescription rights, obstetric care, or minor surgery within chiropractic scope. So the outer edge of the profession is fairly stable even though the interior varies. The variation lives in a set of specific hotspots, and those hotspots are exactly the things a student with a particular career in mind should be asking about.

Needling. Texas permits needle use only to draw blood for diagnostic purposes and for acupuncture. Other states set their own different rules for dry needling, which is the insertion of a fine needle into muscle tissue to treat a taut painful band, and for needle electromyography, which is a diagnostic test that reads electrical activity inside a muscle through a needle electrode. If you are interested in that kind of work, the state you license in decides whether you may do it at all.

Advanced imaging. Ordering or performing computed tomography and magnetic resonance imaging is allowed in some jurisdictions, Connecticut among them. Ohio requires diplomate status from the American Chiropractic Board of Radiology for some advanced imaging privileges, which is a postgraduate credential taking years to obtain. So the same clinical question, can I get this patient the imaging I think they need without sending them elsewhere, has different answers in different states, and one of those answers is a multiyear training commitment.

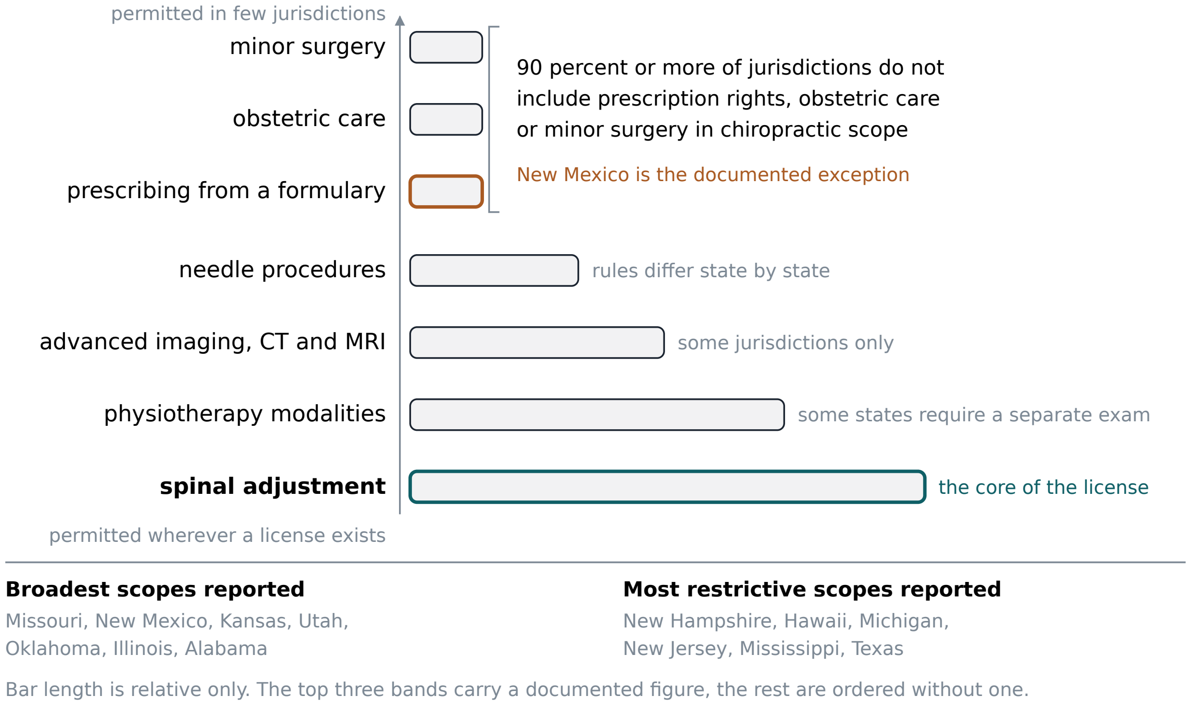
Physiotherapy and modalities. This is where the optional PHT examination stops being optional. Some states require it for a scope that includes physiotherapy modalities and some do not. Eligibility to sit it requires 120 hours of physiotherapy instruction, which means the question is not only which state you want but whether your program will actually give you those hours. Ask the school. Get the answer in writing.

The clearest and best documented example of an expanded scope state is New Mexico, and it repays a close look. New Mexico statute creates a credential called certified advanced practice chiropractic physician, which authorizes prescribing, administering and dispensing from a defined formulary. That formulary includes herbal and homeopathic medicines, over the counter drugs, vitamins, minerals and enzymes, glandular products, amino acids, dietary supplements, bioidentical hormones, sterile water and saline, Sarapin, caffeine, procaine, oxygen, epinephrine, and vapocoolants. Dangerous drugs, controlled substances and injectable drugs require separate approval from the state pharmacy and medical boards. That is a substantially different professional life from what the same degree buys in New Hampshire.

And here is the part that makes New Mexico a lesson rather than a destination. That statute is currently scheduled to sunset, with repeal effective July 1, 2028. A sunset provision is a built in expiration date that requires the legislature to affirmatively renew the law or watch it lapse. So the single clearest example of an expanded chiropractic scope in the country has a deadline attached to it. Scope is not a fact about the profession. It is politics, revisited periodically, in fifty separate rooms.

One degree, a different legal job in each state

What a chiropractic license permits is set state by state, so a student should decide where they intend to practice.



Scope of practice is best understood as a ladder of privileges rather than a map, since the outer boundary is fairly consistent nationally while a handful of specific hotspots such as needling, advanced imaging, and prescribing account for most of the variation between states.

The practical instruction is simple to state and hard to follow at twenty three. Decide where you intend to practice before you choose a school. Not with certainty, since nobody has that, but with enough seriousness that you have read two or three state statutes and know what they permit. The school shapes what you are trained to do. The state decides whether you may do it. If those two answers do not line up you will find out during your first year of practice, at which point you have very little room to move.

Risk, and the records that protect you

Chapter 5 owns the clinical risk conversation: adverse events, the dissection literature, red flags that require referral rather than treatment, and informed consent as a professional obligation. I am going to stay out of that territory and say only what belongs in a chapter about the working life.

Carry malpractice coverage from your first day of practice, including your associate year, and know whether the policy you are covered under follows you or follows the clinic. Those are different things, and the difference matters most at exactly the moment you stop working there. Know what the policy covers, what it excludes, and what your obligations are if a claim is made. Nobody enjoys this conversation. Have it once a year anyway.

The more important protection is not insurance, it is documentation. A note written the day of the visit, recording what the patient told you, what you found on examination, what you told them, what you did, and what you planned, is the only version of that day that will exist in three years. Memory will not help you and neither will sincerity. This is the practitioner's own protection, and it is worth understanding that way rather than as a compliance chore imposed by an insurer. When something goes wrong, and across a long career something will, the record you kept when nothing was wrong is what speaks for you.

The same logic applies to informed consent. Treat it as a habit rather than a form. A signature at intake is a legal artifact; saying out loud, before each significant intervention, what you are about to do, what it may feel like, and what the alternatives are, is the actual practice. The national practical examination scores exactly this and calls it verbalized intent. That is not a coincidence. The profession decided that the habit was worth testing, and it was right.

The body is the instrument

Manual work is physical labor. This is the part of the life that recruiting material never touches and that every practitioner over fifty will talk to you about for an hour if you let them.

Across a full day you may perform a hundred or more setups. The load lands in predictable places. The base of the thumb and the wrist take contact forces repeatedly. The shoulders take repeated loading across a table, often at the end of range. The low back takes the accumulated hours of bending over a surface that is a little too low, because you did not change the table height when the last patient was shorter than this one. None of these produce an injury on any given Tuesday. They produce an injury in year nine, and by then the habit that caused it has ten thousand repetitions behind it.

Here is what practitioners actually do about it, offered as practice wisdom rather than as research, because a controlled study of chiropractor longevity is not something I can hand you.

Set the table height for your body, every time, and accept the four seconds it costs. This is the single most common correction and the one most often skipped. Drive the thrust through your legs and body weight rather than your arms and shoulders; every technique instructor says this, and most new graduates ignore it for about two years and then start listening to their shoulders. Vary the load across the day by rotating among methods, since instrument assisted and drop table approaches, which chapter 3 describes, put a very different demand on the hands than a manual thrust does. Alternating them is not a philosophical compromise about technique purity. It is a decision about whether you would like to be working at sixty. Cluster and cap the volume where you can, because back to back adjusting for eight hours a day, five days a week, with no variation, is the pattern that breaks people. Train the tissues you actually use: grip, forearm, hips, and the whole posterior chain. Sleep, which sounds like advice from a poster and is the strongest recovery intervention available to you.

And get treated yourself. Practitioners are famously terrible patients, and the reason is not mysterious: seeing someone about your shoulder means admitting the instrument is wearing, and the instrument is also your income. Watch the early signals rather than waiting for the injury. The thumb that aches on waking. The wrist that clicks on a particular contact. The shoulder that will not let you sleep on that side. Those are not nuisances, they are the opening line of a conversation about the second half of your career, and the people who answer it early are the ones still adjusting at sixty five.

Your hands are the only instrument in the room you cannot replace, and nothing in a four year curriculum will make you take that seriously as fast as one bad thumb will.

Careers shape themselves around this in different ways, and I want to present the range without pretending I know the proportions. Some practitioners work well into their seventies with intact hands and a full manual practice. Some shift in their forties toward lower force methods, or toward rehabilitation and exercise prescription, or into teaching, or into ownership of a practice where they see fewer patients and manage more. Some leave. All of those are real careers and the second one is not a defeat. What I do not have, and will not manufacture, is a figure telling you how many land in each group. If someone gives you that number, ask where it came from.

Burnout, and what keeps people

People do leave this profession, and I want to describe what actually drives it, because the assumption students bring is usually wrong. They assume it is the patients. It is almost never the patients.

It is the business. It is the denied claim you appeal twice and lose. It is payroll landing on a week the schedule was thin. It is the empty afternoon in month four. It is the structural fact that a solo practice only produces income while you are physically standing in it, which means that a vacation, an illness, a new baby, or a parent who needs you in another state all convert directly into lost revenue in a way that a salaried position does not. That last one is the quiet grinding pressure of self employment and it does not show up in any brochure.

Then there is isolation, which I think is underrated as a cause. In solo practice you can go a full week without a professional conversation with anyone who does what you do. There is nobody down the hall to say what would you do with this to, so every difficult case is yours alone, and the accumulation of that is heavier than any single case. Add the patients who do not improve, which chapter 5 takes up seriously, and you have the actual list.

What practitioners do about it is mostly structural rather than attitudinal. Hire help earlier than feels affordable, because the alternative is that you are the front desk and the biller and the clinician, and that arrangement has a maximum duration. Get into a study group or a local association, anything with other practitioners in it, so the difficult case has somewhere to go. Build referral relationships across professions, which chapter 6 discusses properly and which matters for your morale and not only for your patients. Take actual time off with the schedule closed. And be willing to change the model, since a practitioner exhausted by a high volume insurance practice often has a good career available on the other side of a change in how the practice is built, and leaving the profession is a large solution to what is sometimes a smaller problem.

Now the other side, which is the reason I am still doing this. I have asked a lot of practitioners with twenty and thirty years in what keeps them, and the answers are almost identical, which surprised me the first several times and stopped surprising me after that. It is always some version of three things.

The work is hands on. You are not typing all day, you are not in meetings, you spend your working life in physical contact with other human beings solving a physical problem, and that turns out to satisfy something in people that a screen does not touch. The feedback is immediate. You examine, you find something, you do something, and often within a minute and almost always within a week you know whether it helped. Very few jobs give you that loop; most people work for months without ever learning whether what they did mattered. And you own your own days. You decide who to see, how long to spend with them, what to charge, when to close, whether to take Friday afternoon off to watch your daughter's game. For a certain kind of person that autonomy is worth more than a larger income with a supervisor attached, and if you are that kind of person you probably already know it.

I want to be careful about what those three claims are and are not. None of them is an argument that the treatment works. That is chapter 5's job and it is a separate question with its own evidence, some of it good and some of it thin. These three are claims about what a working life feels like from inside, and they are the honest answer to why a practitioner stays for thirty years.

What I wish someone had told me

So here is the part I would say to you if you stopped me in the hallway, in the order I would say it.

Do the arithmetic before the campus tour, not after. The mean debt at graduation in the study I cited was $176,297 and the mean total cost of the education was $260,642, and the people carrying those numbers underestimated their own by fifty thousand dollars. Be the person who did the multiplication.

Decide where you want to live before you decide where to study. Your license is a state license, the scope varies substantially between states, and the New Mexico statute with a repeal date in 2028 should teach you that even the good answers are temporary.

Accept early that you are going to be a small business owner. Take a bookkeeping course. Learn to read a lease and a contract, or learn who to pay to read them for you. You will use those skills more weeks of the year than half the technique electives you will agonize over.

Learn to write a fast, complete, honest note while you are still in school, when being slow at it is free. The habit is worth more to your career than any single technique you will learn, and the national practical examination gives you six minutes for a reason.

Say out loud what you are about to do, every time, to every patient. It will feel redundant by the four hundredth repetition. Do it anyway. It is how consent actually works and it is how you find out, occasionally, that the patient had something to tell you that would have changed the plan.

Protect your hands from the first week rather than from the first injury. Table height, legs not arms, vary the methods, watch the early aches. The version of you who is fifty five is a real person and is depending on the version of you who is twenty six.

Find people to ask. If I look at my own cohort and try to see what separates the ones still enjoying this at year fifteen from the ones who are not, the single clearest difference is not talent and it is not location. It is whether they had two or three people they could call about a case or a decision, and whether they kept calling them.

Learn to say I do not know, and to say this is not mine, let me send you to someone. New graduates think those sentences make them look weak. They are among the strongest things a clinician says, patients can tell the difference, and every good practitioner I know says both regularly.

And last, the one that is specific to this profession rather than to any health career. You are joining a field that argues with itself and gets argued about from outside. Some of the criticism is unfair and some of it is accurate, and chapter 5 will give you the evidence honestly, including the parts that are thinner than the marketing. If you need everyone to agree that what you do is legitimate before you can be at peace doing it, this will be a hard life. If you can hold a contested position honestly, state clearly what the evidence supports and where it stops, and still do careful, useful work on a Tuesday morning for the person on your table, then you will be fine, and the profession will be better for having you in it.

I would do this again. I say that knowing the debt figure, knowing what my hands feel like some mornings, and knowing exactly how many hours of my life have gone to insurance authorizations. I would do it again because of the three things practitioners always name, and because a job where a person walks in unable to turn her head and walks out able to is a good way to spend a working life. That is my answer, and it is worth what any one person's answer is worth. Yours has to be your own, and the point of this chapter is that you now have enough of the real numbers, and enough of the unglamorous detail, to build one.

Terms from this chapter

Council on Chiropractic Education

The accrediting body for Doctor of Chiropractic programs in the United States, recognized by the US Commissioner of Education in 1974. Its standards set the admission requirements and the minimum instructional and clinical hours every accredited program must meet.

Semester credit hour

The standard unit of undergraduate coursework. The accreditor's alternative to a bachelor's degree for admission is 90 of them, which is roughly three years of full time study.

Cost of attendance

An institution's published estimate of what one year of enrollment costs, including tuition and an allowance for living expenses. It is the number to multiply when estimating what a program will actually cost you.

National Board of Chiropractic Examiners

The organization that administers the national examinations, Parts I through IV plus the optional Physiotherapy examination, that essentially every state board requires for licensure.

Part IV

The practical examination taken within six months of graduation, consisting of seven case management stations and one technique station, video recorded and scored on safety, positioning, contact points, stabilization, and verbalized intent.

Verbalized intent

Saying out loud, before you act, what you are about to do and why. It is a scored criterion on the practical examination and the working form of informed consent.

Physiotherapy examination

The optional fifth national examination, 90 questions in 75 minutes, requiring 120 hours of physiotherapy instruction to sit. Some states require a passing score for a scope that includes physiotherapy modalities and some do not.

Scope of practice

The legal description, set by each state, of what a licensee is permitted to do. It varies substantially between states even though the degree and the national examinations are the same everywhere.

Sunset provision

A built in expiration date written into a statute, requiring a legislature to affirmatively renew the law or let it lapse. New Mexico's expanded scope statute carries one with a repeal effective in 2028.

Associate

A licensed chiropractor employed in someone else's clinic, trading autonomy over schedule, fees and clinical culture for patient volume, infrastructure, and someone experienced to ask.

Goodwill

In a practice purchase, the portion of the price attached to the existing patient relationships and reputation rather than to equipment or the lease. It is the part attached to a person who is leaving.

Restrictive covenant

A contract clause limiting where and when a departing associate may practice. Its radius and duration decide where you can work next, which is why it should be read before signing rather than after leaving.

SOAP note

The standard visit record, organized as subjective, objective, assessment and plan. It is required by boards and insurers, and it is the practitioner's own protection because it is the only version of that visit that will still exist in three years.

Prior authorization

An insurer's advance approval for a course of care, usually for a capped number of visits. Managing it is part of the administrative load of practice and it has direct clinical consequences when the authorization runs out mid course.

Before you move on

1. What are the two admission routes the accreditor recognizes, and what grade point conditions attach to the route that does not require a bachelor's degree?
2. State the minimum total instructional hours and the minimum supervised patient care hours required of an accredited program, and explain what the clinical hour requirement does to the shape of the second half of the program.
3. Name the four required national examinations and the optional fifth, and say when in the program each is taken.
4. Give the mean and median student loan debt at graduation reported in the survey cited here, and explain why the total cost of education figure is roughly a hundred thousand dollars higher than the typical debt figure.
5. A student wants a practice built around rehabilitation and modalities, and eventually wants to order advanced imaging without referring out. What specific questions should she ask a program, and what should she look up about a state, before she chooses either? Explain why the order of those two decisions matters.
6. You are offered an associate position that pays better than the others, in a clinic that expects a set number of visits per patient regardless of presentation. Say what you would want to know before signing, and name what would make you decline. Defend your answer.
7. This chapter declines to give a starting income figure and explains why. Was that the right call for a chapter meant to help you decide whether to enter the profession? If you think it was not, say what source you would trust to supply the number and how you would check it.

CHAPTER 5

In the Clinic

> *When a patient asks whether this is going to help, what does the evidence actually entitle you to say?*

A Tuesday morning in a general practice looks like this. A warehouse supervisor whose back seized four days ago moving a pallet jack. A retired teacher with a dull ache in her lower back that has come and gone for eleven years. A man with a stiff neck and a headache that starts at the base of the skull. A new mother told by a friend that an adjustment will settle her six week old's crying. A man who booked because a pharmacy kiosk said his blood pressure was high. And somewhere in that morning, a patient whose problem is not a chiropractic problem at all, which a large part of the job consists of noticing before you put your hands on anybody.

This chapter is about what you can honestly say to each of them. Not what you can get away with saying, and not the safest thing you could say to avoid criticism, but what the published evidence supports, where it runs out, and where it is genuinely argued about. Chapter 2 gave you the anatomy and chapter 3 the technique; this chapter asks what the evidence says when those hands meet a complaint. A practitioner who cannot tell those three apart will eventually say something in a treatment room that is not true.

One thing first, about case mix. I have no sourced number for what fraction of a chiropractic practice is back pain, so I will not invent one. From my own schedule: the great majority of my patients come because something in the spine or the muscles around it hurts, and the ones hoping for something else are a small minority who account for most of the arguments about this profession.

Compared with what

Before any of the evidence, one habit. If you take nothing else out of this chapter, take this. When a study reports that a treatment made a difference, the sentence is incomplete until you know what the difference was measured against. Three comparisons get made constantly in this literature and they mean entirely different things.

The first is a comparison against **sham**. A sham manipulation is built to look and feel to the patient like treatment while withholding the part thought to be active, usually the thrust: same room, same table, same hands, same time spent, no meaningful force delivered to the joint. If a real adjustment beats a sham, something specific to the adjustment is doing work, over and above being touched, being attended to, and expecting to improve.

The second is a comparison against **no treatment**, or a waiting list. If a treatment beats nothing, you have learned that the whole package helps: the treatment, the attention, the reassurance, the expectation, and whatever the patient does differently because someone told her she was being cared for. That is useful to a patient deciding how to spend a Tuesday afternoon. It is not evidence that the thrust is what worked.

The third is a comparison against **other conservative care**: exercise therapy, physical therapy, advice to stay active, heat, massage, usual medical management. This matters most to a patient with a choice and to an insurer paying for one of them. If manipulation performs about as well as exercise therapy, that is not a failure. It means the choice can turn on cost, preference, availability, and risk instead of on effectiveness.

A difference against sham, a difference against no treatment, and a difference against other conservative care are three different claims, and only the first tells you the hands are doing something the hands alone can do.

Marketing collapses the three. A brochure saying manipulation is proven effective for chronic low back pain is usually resting on the comparison against no treatment, the easiest bar in the set. A critic saying manipulation is no better than placebo is usually resting on the comparison against sham, the hardest. Both can be quoting the same review.

Low back pain

Start where the evidence is thickest. In 2017 the American College of Physicians published a guideline titled "Noninvasive Treatments for Acute, Subacute, and Chronic Low Back Pain" in the Annals of Internal Medicine. It is quoted at you constantly, by both sides, and usually inaccurately.

For acute and subacute low back pain, the guideline recommends starting with nonpharmacologic treatment and puts spinal manipulation on that list alongside superficial heat, massage, and acupuncture. Its framing throughout is that most patients recover regardless of what is done, so the reasonable move is to choose something with few harms. On the evidence for manipulation specifically: "Based on evidence of low quality, spinal manipulation results in minimal improvement in function compared with sham manipulation; data were insufficient to make conclusions about how it affects pain."

Read that slowly. It is a positive finding and a small one. Minimal improvement in function, low quality evidence, against sham, and no conclusion at all about pain. That is much weaker than "the American College of Physicians recommends chiropractic," which is roughly what it gets turned into.

For chronic low back pain the guideline again recommends nonpharmacologic treatment first, from a long list: exercise, multidisciplinary rehabilitation, acupuncture, mindfulness based stress reduction, tai chi, yoga, motor control exercise, progressive relaxation, biofeedback, low level laser therapy, operant therapy, cognitive behavioral therapy, or spinal manipulation. On the sham comparison it is blunter: "Compared with sham therapies ... spinal manipulation does not result in a difference in pain."

So the honest summary is that manipulation appears in a list of roughly equally supported first choices, none dramatic, all preferred over starting with medication because the harms are lower. The guideline does not single out manipulation, and certainly not chiropractic, as superior to anything else on that list. Tell a patient the guideline endorses what you do and you have overstated it. Tell her it includes what you do among the reasonable first things to try, and you have said something true.

Now the systematic reviews. A systematic review is an attempt to find every trial that has asked a question, judge how well each was done, and report them together instead of cherry picking the flattering ones. Cochrane reviews are a family of these with a published method and a reputation for being hard to please. Two matter here.

For **acute** low back pain, the Cochrane review by Rubinstein and colleagues pooled 20 randomized trials covering 2,674 participants. Its conclusion: "SMT is no more effective in participants with acute low back pain than inert interventions, sham SMT, or when added to another intervention." SMT there is spinal manipulative therapy. The review adds that referral decisions should therefore weigh cost, patient and provider preference, and relative safety against the other options, and flags its own weakness, that few studies are available for any single comparison.

That is a negative result and you should be able to say it out loud without flinching. For a back that seized four days ago, the trial evidence does not show that manipulation beats a convincing fake, an inert intervention, or the same care without it. It also does not show that manipulation is harmful or useless as part of a package a patient wants. Acute low back pain has a strong natural history: most episodes settle within weeks whatever anyone does. That is why sham is so hard to beat here, and why a practitioner sees so many grateful patients.

For **chronic** low back pain, the Cochrane review by de Zoete and colleagues found a different pattern, and this is where you practice the habit from the last section. Against sham, manipulation "may result in a small improvement in pain and medium improvement in functional status". Against no treatment, a medium improvement in pain and a large improvement in function. Against other conservative interventions, little to no difference in pain and a small improvement in function. Three comparators, three differently sized answers, one review.

Certainty was rated low to very low, mostly because the trials differed so much in technique, dose, and frequency that pooling them is a strain. No serious adverse effects were found. Notice what the pattern means: the largest apparent benefit is against doing nothing, the comparison that includes every non specific effect there is, and the smallest is against other conservative care, the comparison a patient with a physiotherapist down the road actually faces.

A note on why people get better anyway

Three things make a treatment look effective when it is not, and all three are strongest in the conditions chiropractors treat. Natural history: most episodes of back and neck pain resolve on their own within weeks, so whatever was done during that window gets the credit. Regression to the mean: patients book when the pain is at its worst, and the worst day is likely to be followed by better ones. Non specific effects: being examined carefully, being told what is wrong in plain language, being touched, and being expected to improve all measurably change reported pain. None of this is fraud and none of it is nothing. A patient who feels better feels better. It does mean your own clinical experience, however long and sincere, cannot tell you whether the thrust is what did it.

Neck pain

The Cochrane work on mechanical neck disorders, by Gross and colleagues, has a shape worth remembering. Manipulation or mobilization used **alone** showed no benefit for pain relief compared with placebo, control, or other treatments, whether the patient had one session or a course of three to eleven, and the two did not differ from each other. Combined **with exercise**, the evidence showed measurable benefit for pain, function, and the patient's overall sense of improvement in persistent neck disorders. For neck pain with radicular symptoms, meaning pain travelling down the arm along a nerve root, there was not enough evidence to conclude anything.

I want to flag a limitation in my own sourcing rather than hide it. The version I am describing is an earlier iteration of that review, covering 33 trials. Cochrane has since published an update and I have not confirmed whether it changed these conclusions. Pull the current version and read it before you quote the neck pain evidence.

Taken at face value: hands plus exercise beat hands alone. That is an argument for care in which the adjustment is one component and the patient leaves with something to do between visits.

Headache and migraine

These two get run together in conversation and they should not be. Different conditions, and the evidence points in different directions.

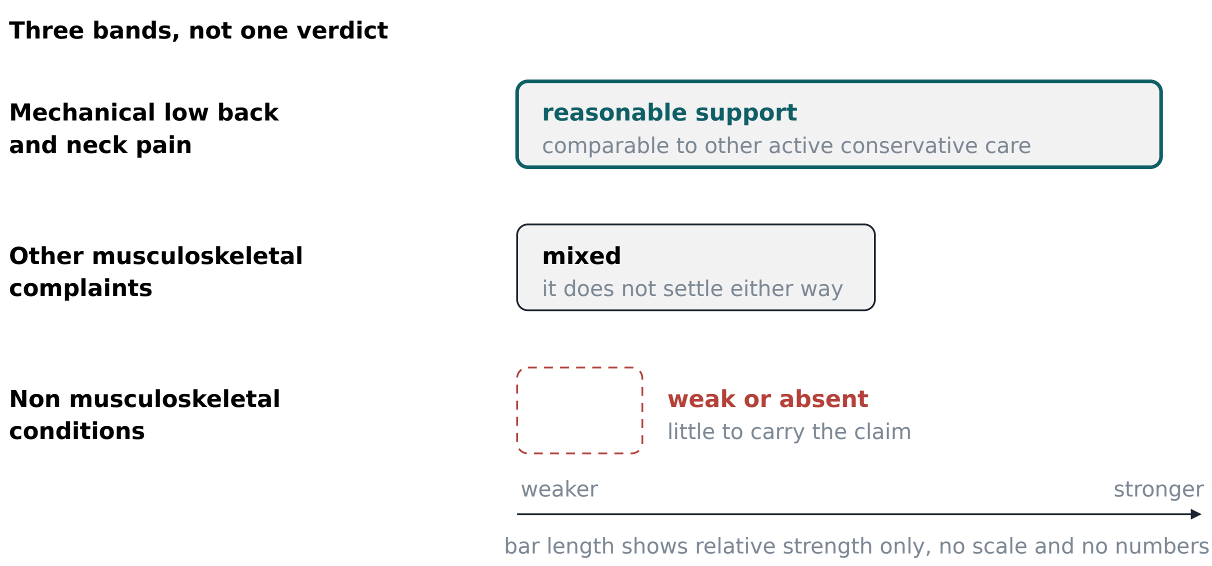
Cervicogenic headache is head pain generated by structures in the neck, typically one sided, often provoked by neck movement or posture. A 2020 systematic review and meta analysis by Fernandez and colleagues, pooling 7 randomized trials, found that spinal manipulative therapy produced small but statistically significant short term improvements in pain intensity, pain frequency, and disability compared with other manual therapies, and no effect on how long each headache lasted. At intermediate and long term follow up, no significant effects on intensity or frequency remained. The authors call it an effective short term option and say plainly that high quality evidence is lacking.

Notice the comparator: other manual therapies, not sham and not nothing. And notice the follow up window: the benefit is there early and gone later. Both facts belong in the sentence you say to a patient.

Migraine is a different matter. A 2024 updated systematic review and meta analysis pooled 6 randomized trials covering 645 patients. On intensity or severity, spinal manipulation showed no effect, with a standardized mean difference of minus 0.22 and a 95 percent confidence interval running from minus 0.65 to 0.21. On duration, nothing meaningful, minus 0.10 with an interval from minus 0.33 to 0.12. On emotional quality of life, likewise nothing significant. All of it was rated very low certainty. The authors' conclusion: "The effectiveness of SMT for the treatment of migraines remains unproven", and manipulation "cannot be considered an evidence based therapy for this condition."

A word on those confidence intervals, since they carry the argument. A confidence interval is the range of values compatible with the data. When it crosses zero, as both of these do, the data are compatible with a benefit, with no effect, and with a small harm. That is what no effect looks like in this literature: not a flat zero, but a range that includes zero comfortably.

The same review found adverse events more common with manipulation than with controls, at a risk ratio of 2.06 with a 95 percent confidence interval from 1.24 to 3.41, giving a number needed to harm of 6: for roughly every 6 patients treated, one additional adverse event occurs that would not have occurred in the control group. Be precise about what those events were. In these trials they were predominantly mild and transient, the soreness and stiffness that follow being handled. The authors separately argue that even a very small risk of arterial dissection would push the balance of harms further toward the negative where the benefit is unproven. That is reasoning under uncertainty, not a measured stroke rate.



The conditions people bring to a chiropractor sort into bands of evidence strength rather than into works and does not work, and which band a condition sits in depends as much on the comparator as on the result.

The harder territory

Everything above concerns the musculoskeletal system, where the treatment is applied to the tissue that hurts. The historically distinctive chiropractic claim, the one chapter 1 traces to 1895, is larger: that interference with nerve transmission at a spinal segment can cause illness elsewhere in the body, and that correcting it can relieve conditions with no obvious mechanical component. That claim generated the profession, and it is also the claim with the thinnest evidence behind it. I will report what the reviews found and let the findings sit where they land.

Infantile colic. The Cochrane review by Dobson and colleagues found the available studies generally small and prone to bias, which made a definitive conclusion about effectiveness impossible. Then it found something more interesting than a null result. Five of the six trials measured the outcome by asking parents, who knew whether their baby had been treated. Pooled, those unblinded parent reported outcomes suggested a statistically significant reduction in crying of roughly one hour and twelve minutes a day. When the analysis was restricted to trials at low risk of performance bias, meaning those where the person recording the outcome did not know which group the infant was in, the effect no longer reached statistical significance. Adverse event data appeared in only one of the six trials, so no safety conclusion could be drawn at all.

Sit with that pair of findings. Parents who knew their baby had been adjusted reported over an hour less crying a day. Observers who did not know reported no significant difference. The gap between them is what blinding is for, and it is not a small gap. It is the whole effect.

This is also where the profession's own literature and the independent literature tell different stories from the same kind of data. The American Chiropractic Association's statement on pediatric care cites a 2012 study reporting a 48 percent decrease in daily crying time for chiropractic treated colicky infants against 18 percent for untreated infants. That is a real study and those are its numbers. It is also, in kind, exactly the parent reported evidence that did not survive blinding in the Cochrane analysis. Neither side is fabricating anything. They are selecting differently, and now you can tell which is which.

Otitis media. A 2007 systematic review by Ernst on chiropractic care for non musculoskeletal conditions concluded there is "little data to suggest improvement of otitis media, asthma, nocturnal enuresis or attention deficit hyperactivity disorder." A later pediatric review found the chiropractic literature on acute middle ear infection consists mainly of case reports and case series, with "no credible solid evidence upon which to make recommendations." It also noted that chiropractors typically do not receive training in otoscopy sufficient to diagnose middle ear disease reliably in the first place. A treatment claim you cannot verify the diagnosis for is not a claim you should be making.

Asthma. Reviews report that manipulation has been associated with subjective symptom improvement in some studies but with little impact on objective markers of the disease, such as measured airway function. One review suggests the subjective difference may come from improved mobility of the thoracic cage rather than any effect on the airway obstruction that defines the disease. A randomized trial published in the New England Journal of Medicine in 1998 found chiropractic manipulation provided no benefit for pediatric asthma beyond usual care. Subjective and objective outcomes diverging is the colic pattern by another route.

Hypertension. Here I have to tell you what I do not have. There is a 2012 systematic review on spinal manipulation for hypertension that I could not obtain and read in full, so I will not quote its conclusion; quoting a conclusion you have only seen summarized is how errors propagate. What I can state with confidence is that hypertension is not an accepted primary indication for chiropractic care in any mainstream clinical guideline, and that the trials that exist are small, heterogeneous, and insufficient to establish a clinically meaningful effect on blood pressure. If a patient books because a kiosk gave him a high reading, the useful thing you do is make sure he sees a physician.

A note on how this chapter quotes

Two conventions. First, this book writes compound terms without hyphens, and that convention is applied inside quotations as well, so a source that hyphenated low back pain, non musculoskeletal, or evidence based appears here unhyphenated. Nothing else in the quoted wording is altered. Second: several sources behind this chapter were read as summaries rather than as full original papers, and I say so at each point where that is true. That is not pedantry. A number that has passed through two summaries has had two chances to change meaning, and in this literature it often has.

Children, and what a parent is owed

Now the part people fight about. My colleague who teaches this course with me has spent thirty years in pediatric and perinatal practice and will tell you things from that experience that I cannot. What I can do is lay out the disagreement accurately, refuse to pretend it is settled, and show you what a practitioner owes a parent standing in the room.

Here is the professional position, in its own terms and at its strongest. The American Chiropractic Association states that pediatric chiropractic care, when administered properly, is "effective, safe and gentle". It cites a 2014 literature review spanning more than a century of practice concluding that serious adverse events in infants and children are "exceedingly rare", with no deaths reported in the academic literature to date. It notes that chiropractors who treat children pay the same malpractice insurance rates as those who treat adults only, an argument from what insurers, who have every reason to price risk accurately, actually charge. The International Chiropractic Pediatric Association frames the question around consent, stating that it respects each individual's right to make informed, conscious health care choices.

Here is the independent position, also at its strongest. A 2010 review in the chiropractic and manual therapies literature states that "the efficacy of chiropractic care in the treatment of non musculoskeletal disorders has yet to be definitively proven or disproven" in children, and criticizes some practitioner claims about pediatric benefit for non musculoskeletal conditions as unsupported, at times to the point of overstatement. Critical reviewers also point to a systematic review reporting a small number of manipulation related pediatric injuries, including serious cases and deaths. I am deliberately not giving you that count, because I could not trace it to the primary review it is attributed to, and a casualty count must not be repeated on secondhand authority in either direction. The same literature quotes academic chiropractors observing that pediatric health claims are still supported by only low levels of scientific evidence.

What can be said as settled is less than either side would like. High quality randomized evidence for chiropractic treatment of non musculoskeletal conditions in children, meaning colic, otitis media, asthma, attention deficit hyperactivity disorder, and bedwetting, is lacking or inconsistent. Serious adverse events from pediatric manipulation appear rare in the literature that exists. But that literature is thin, largely case reports, and private practice has no mandatory adverse event reporting system, so absence of documented events is not demonstrated safety. Both statements are true and they do not resolve each other.

So do not go looking for the verdict. Go looking for what you owe the parent, because that question does have an answer. You owe her a plain account of four things: what the evidence supports, what it does not, what the risks are believed to be, and how confident anyone is entitled to be about that last part. Not a leaflet, not a form. A conversation in her vocabulary, with room for her to say no and stay your patient afterward. Here is roughly how mine goes.

I tell her: I want to be straight with you about what we know, because you are the one deciding. For the crying itself the studies are mixed, and the way they are mixed matters. When parents who knew their baby had been treated reported on the crying, they reported a real improvement, more than an hour a day. When the outcome was recorded by someone who did not know which babies had been treated, that difference did not hold up. I cannot tell you which is closer to the truth for your daughter. I can tell you I am not able to promise you the hour.

I tell her: on safety, the literature says serious problems in infants are very rare. Hear the exact shape of that. It is rare in the reports that exist, and no system requires practitioners to report problems, so nobody can put a firm number on it. Rare in the literature we have is a weaker statement than proven safe, and I am not going to upgrade it for you.

I tell her: what I would do with a baby is nothing like what you have seen me do with an adult. There is no thrust and no popping sound. It is sustained light contact, about the pressure you would use to test a ripe tomato. That is not a claim that it works. It is a description of what would happen, so you can picture it.

And I tell her: there are things that would make me send you to your pediatrician today instead of treating, and I will check for them first. Fever, poor feeding, weight that is not going up, blood in the stool, a baby who has become floppy or hard to rouse. Persistent crying is usually just crying. Occasionally it is the first sign of something that needs a physician, and I would rather find that than fix your afternoon.

Then I stop talking and let her decide. Sometimes she books. Sometimes she would rather wait, and I tell her that is a completely reasonable choice given what I have just said, because it is. A conversation that only ever ends in a booking was not a consent conversation. It was a sales close wearing consent vocabulary.

Be honest about the disagreement inside this, too. There are practitioners of long experience and real skill, including the colleague teaching this course with me, who would say that account undersells what they have seen work in thirty years, and that a review which cannot detect an effect is not the same as an effect that is not there. That is a fair objection, and the one you will hear from the best people in the field rather than the worst. My answer is that the strength of a practitioner's experience cannot tell you which of the three explanations in the note above is producing it, and the parent is entitled to hear that limitation from the person she is trusting.

Risk, and how large it is

Every intervention has a harm profile and it is professionally embarrassing not to know your own. The common adverse events after manipulation are local soreness, stiffness, headache, and tiredness, generally starting within hours and settling within a day or two. The chronic low back pain review found no serious adverse effects across its trials, and the migraine review's doubled adverse event rate was overwhelmingly this same category. Warn patients about soreness. Patients who are warned report it as expected; patients who are not warned report it as something having gone wrong.

Now the serious end. A large United Kingdom prospective national survey by Thiel and colleagues collected data on 28,807 treatment consultations involving 50,276 cervical manipulations across 19,722 patients, and recorded zero serious adverse events. Because you cannot compute a rate from zero events, the authors used a standard device called the rule of three to estimate an upper bound on the risk consistent with observing none: roughly 1 serious adverse event per 10,000 treatment consultations immediately after treatment, about 2 per 10,000 within seven days, or about 6 per 100,000 individual cervical manipulations. They characterized the risk as low to very low.

Understand what kind of number that is, because it is routinely misquoted. It is not an observed event rate. Nobody observed those events; they observed none. It is a ceiling: the highest risk still reasonably compatible with having seen zero events in that many treatments. The true rate could be anything from that ceiling down to zero. It also reflects United Kingdom registered practice. A 2012 meta analysis by Haynes and colleagues concluded that a risk of vertebral artery dissection following manipulation cannot be ruled out, and noted the true rate may be underestimated because private practice has no mandatory reporting requirement. One finding says the ceiling is low; the other says our ability to see through the floor is poor.

The dissection question

Cervical artery dissection is the serious adverse event this profession is asked about, so discuss it precisely. A dissection is a tear in the inner lining of an artery in the neck. Blood enters the wall of the vessel through the tear, and the resulting flap or clot can obstruct flow or throw a clot downstream into the brain, causing a stroke. It is rare. It occurs spontaneously, and it has been reported after coughing, after sports, after a visit to a hair salon, after ordinary neck movement. Its earliest symptoms are frequently neck pain and headache, often severe and often unlike a person's usual headaches.

Read that last sentence again, because everything difficult about this literature follows from it. The early symptom of the catastrophe is the exact complaint that brings people to a chiropractor.

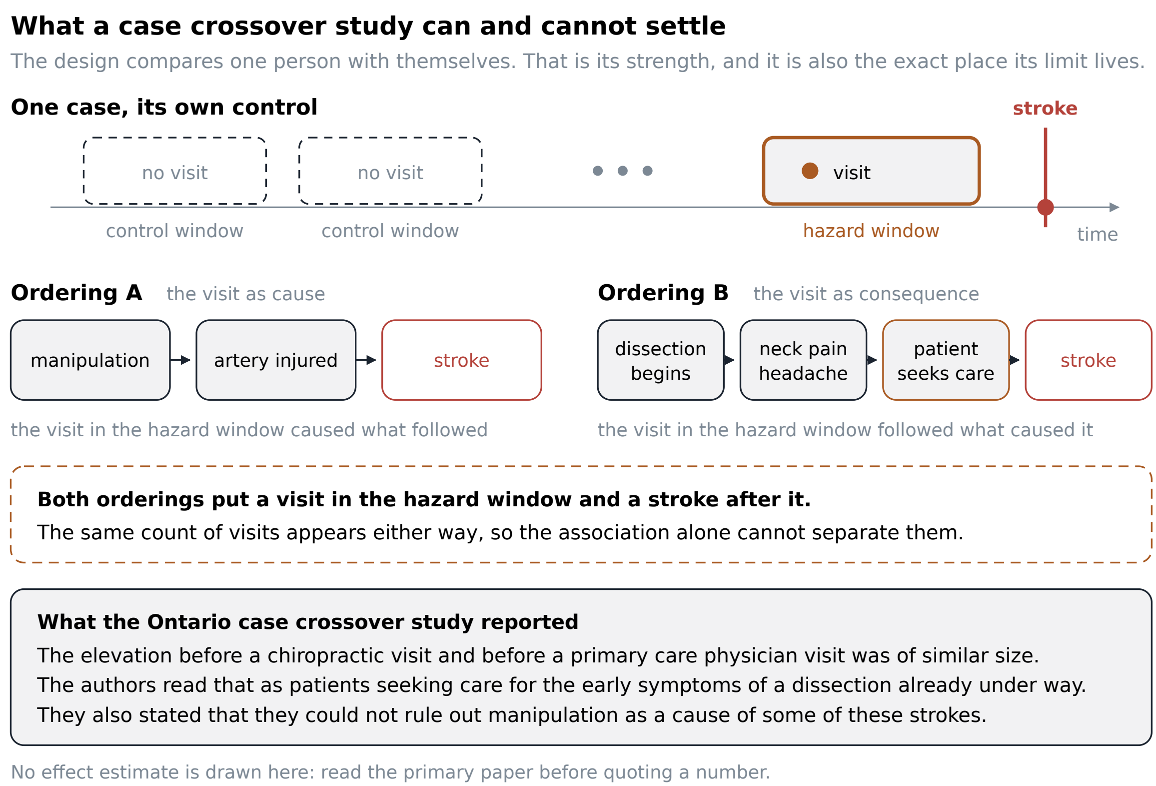
The best known epidemiological work here is a population based Ontario study by Cassidy and colleagues, published in 2008, which examined rates of chiropractic visits and of primary care physician visits in the days and weeks preceding vertebrobasilar artery stroke, a stroke in the arterial system supplying the back of the brain. One of its two designs was a **case crossover** design, and understanding that design is the point of this section.

A case crossover study uses each case as his own control. You take only the people who had the event. For each, you ask whether he was exposed to the thing in question during a window just before the event, and whether he was exposed during an earlier window of comparable length when nothing happened. Then you compare exposure in the hazard window against the control windows across all the cases. Because each person is compared only against himself, every stable characteristic is automatically controlled: age, sex, genetics, blood pressure, smoking, whether he is the kind of person who seeks care. Those cannot confound the comparison because they are identical in both windows by construction.

What the study found, in direction, was that visits were more common in the window shortly before the stroke than in earlier windows, and that this held for both chiropractic visits and primary care physician visits at broadly similar magnitudes. The authors' stated conclusion: "We found no evidence of excess risk of VBA stroke associated with chiropractic care compared to primary care." Their interpretation was that the elevated association after both kinds of visit is likely due to patients with headache and neck pain from an already occurring dissection seeking care before their stroke.

I am not going to give you the odds ratios from that study, and I want to explain why rather than just leave them out. I could not verify the numbers against the original paper; what I have is secondary commentary. The qualitative conclusion and the stated limitations I am confident in. The effect estimates I am not, and an effect estimate quoted from a summary is exactly the number that arrives in a lecture hall slightly wrong and stays wrong for a decade. To cite a number, read Cassidy and colleagues in Spine, 2008, volume 33, supplement 4, pages S176 to S183.

Here is the confounding problem the whole literature turns on. Suppose manipulation causes some dissections. Then you would expect treatment to cluster in the days before strokes. Now suppose manipulation causes none, but a dissection in progress causes the neck pain and headache that sends people to seek manual treatment. You would expect exactly the same thing. The two stories make the same prediction about the data, and an association equally consistent with both causal orderings cannot by itself tell you which one you are looking at. No amount of statistical care inside that design fixes this, because the problem is not noise. The design measures timing, and here timing is symmetrical between cause and consequence.



A case crossover design can establish that treatment and stroke cluster in time, but it cannot distinguish manipulation causing a dissection from a dissection already underway causing the patient to seek manipulation.

What partly breaks the symmetry is the comparison group. If treatment caused dissections, you would expect the clustering before a chiropractic visit to exceed the clustering before a visit to a physician, who does not manipulate the neck. Finding similar clustering before both is the strongest argument in this literature that the visits are markers of an event already underway rather than its cause. That argument, rather than the odds ratios usually quoted, is the substance of the finding.

It is not proof of zero risk, and the authors said so: they could not rule out neck manipulation as a potential cause of some strokes, for instance by dislodging a clot already formed. Later commentary raised a design limitation, that a same day physician comparison cohort was excluded from the published comparison, which limits how directly same day causation can be tested. Later population studies are reported to show broadly consistent patterns. There is also a biomechanical argument, mentioned here with a caution attached: a study of vertebral artery length change during simulated manipulation has been reported, via a professional association's evidence review, to show elongation falling far short of what earlier testing associated with injury. I could not verify those measurements against the primary paper, so I give you the argument and no numbers.

So what do you say to a patient? This, and say it before you touch a neck rather than after. Cervical artery dissection is a rare and serious condition that happens spontaneously and after all kinds of ordinary movement. The large population studies have not been able to show that chiropractic treatment raises the risk of this stroke beyond the risk found after seeing a physician for the same early symptoms, which many read as evidence that manipulation is not a major independent cause. That is not proof that the risk is zero in any individual case, and nobody can honestly give you that. Then the operational half, which matters more day to day than the epidemiology: a sudden severe headache or neck pain unlike anything you usually get, especially with dizziness, double vision, slurred speech, difficulty swallowing, or weakness on one side, means an emergency room immediately, and tell them you have had neck treatment.

What remains unresolved, plainly: whether cervical manipulation can precipitate dissection in rare susceptible individuals is not settled by the epidemiology we have, and given how rare the event is and how impossible it would be to randomize, it may not be settled by epidemiology at all. Practitioners who say the question is closed, in either direction, are ahead of the evidence.

The findings that mean stop

Every patient gets screened for the conditions that make manipulation the wrong thing to do and a physician the right one. This is the part of the job with the least glamour and the highest stakes. Nobody has ever complimented me on a referral. I have made several I still think about.

An international framework review of low back pain guidelines organizes the warning signs, usually called red flags, into four categories. Note the review's own critical finding first: guidelines vary substantially in which flags they prioritize, and the diagnostic accuracy evidence behind many individual flags is limited. A red flag is not a diagnosis. It is a reason to stop, think, and often to send.

Cauda equina syndrome. The cauda equina is the bundle of nerve roots continuing below the end of the spinal cord in the lower back. Compression of it, usually by a large disc herniation, is a surgical emergency measured in hours. Across the guidelines reviewed, the two most consistently cited signs were saddle anesthesia, meaning numbness in the area that would contact a saddle, the perineum and inner thighs, and sudden onset bladder dysfunction, meaning new retention or incontinence. Ask about both, in words the patient understands, and do not let embarrassment stop you. Numbness between the legs and a bladder that has stopped working, in a patient with back pain, is an emergency department today.

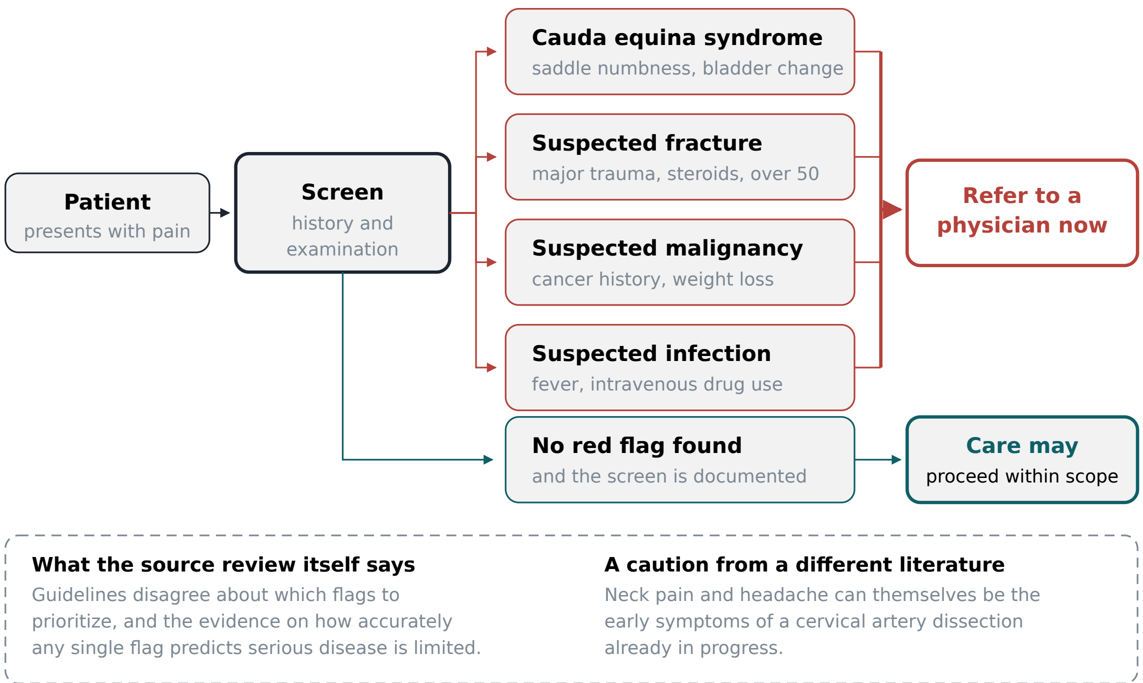
Fracture. The flags are major or significant trauma, use of steroids or immunosuppressant drugs, age over fifty, and female sex, the last three combining as markers of reduced bone density. What raises this in a real room is a mismatch between force and consequence. A twenty five year old with back pain after a car accident is one thing. A seventy year old on long term steroids with sudden severe mid back pain after lifting groceries is another, and she needs imaging before anybody thrusts on anything.

Malignancy. The two most consistently endorsed flags are a history of cancer and unexplained or unintentional weight loss. Others include pain at rest, pain at night, elevated inflammatory markers, and loss of appetite. The pattern to watch for is pain that does not behave mechanically: it does not ease with rest or position, it wakes the patient at night, and it worsens steadily rather than fluctuating. Pain that ignores position deserves a second look.

Infection. Fever, intravenous drug use, corticosteroid or immunosuppressant therapy, night pain, and bone tenderness over the lumbar spinous processes. Spinal infection is uncommon and the consequences of missing it are severe. Fever plus focal spinal tenderness is not a presentation to work through with a course of care.

The referral decision

Screening is not a formality. It is the one thing separating a safe practitioner from a dangerous one.



Each red flag category routes the patient off the treatment table onto a branch with its own urgency, from a same day emergency referral to a phone call to a physician this week.

Two things about screening. It is not a formality and it is not the intake form. The form is where you learn the patient had breast cancer nine years ago. The conversation is where you learn the pain wakes her at three in the morning and her weight is down without her trying. And screening fails predictably: at four in the afternoon on a busy day, with a patient you have seen eleven times, when something new has come up and you are already thinking about the next room. Nobody misses a red flag during a careful first examination. People miss them on the twelfth visit.

The clearest line between a safe practitioner and a dangerous one is not technique. It is whether the screening still happens when you are tired, behind schedule, and treating someone you know well.

Referral is not a defeat and not the end of the relationship. Chapter 6 deals with how this profession works alongside the others and how a referral letter is written and read. What matters here is the clinical trigger: these findings mean this patient needs a physician now, and your job is to be the person who noticed.

How to read a trial without being fooled

You will spend a career being handed studies by patients, colleagues, sales representatives, and strangers on the internet who are angry with you. Here is a working method: seven questions, in order.

Who was in it? Look at the population first. A trial of manipulation in patients with acute low back pain of under six weeks, no prior surgery, and no radiating leg symptoms tells you little about the patient in front of you with nine years of pain and a fusion at L4 to L5. Trials get clean results partly by excluding complicated people, and clinic patients are complicated. Ask whether yours would have been eligible.

Were they randomized, and how? Randomization is what makes the groups comparable in everything except the treatment, including things nobody thought to measure. If patients chose their group, or the practitioner chose, the comparison is worthless however large the study, because the sicker or more hopeful patients will have sorted themselves. Ask also whether allocation was concealed, meaning the person enrolling patients could not see which group came next.

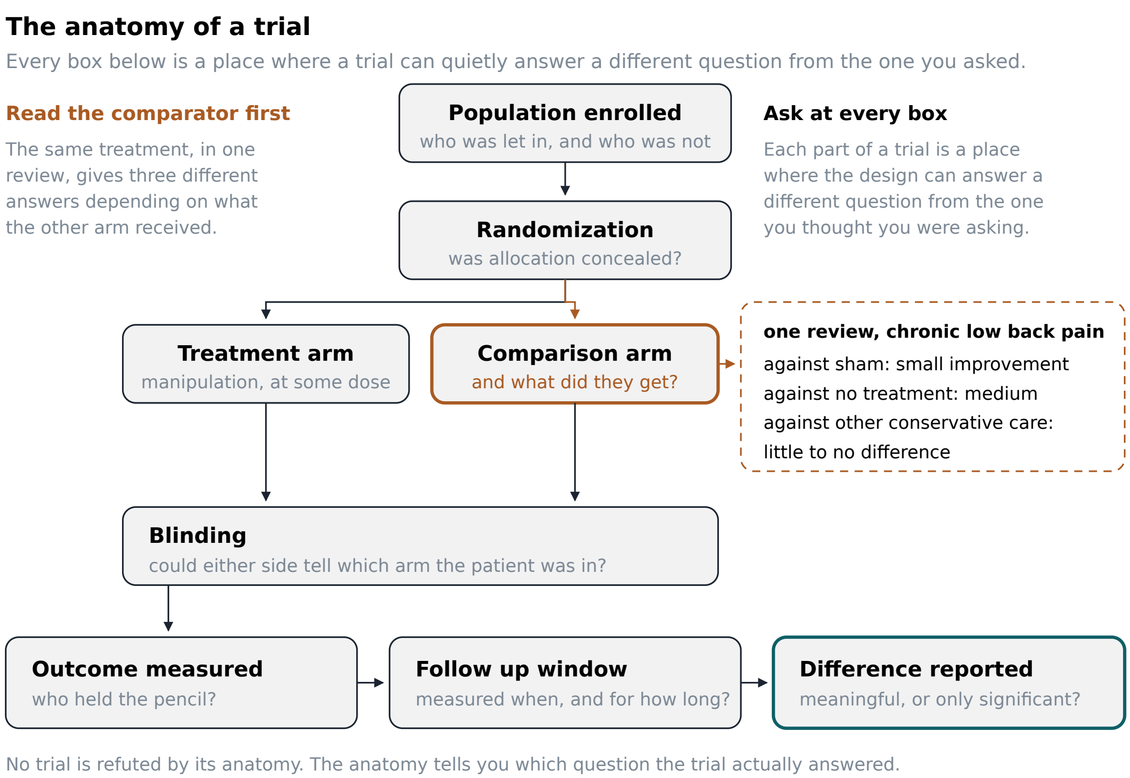
What did the comparison arm actually receive? This is the habit from the start of the chapter and the question most often skipped. Read the methods, not the abstract, and find out what happened to the control patients: sham manipulation, a detuned machine, a booklet, a waiting list, physiotherapy, or nothing. Then reread the conclusion with that in mind. Half the overclaiming in this field lives in the space between the conclusion and the control arm.

Who was blinded, and could they have been? Blinding means keeping people unaware of which group a patient is in: the patient, the practitioner, and the person recording the outcome. For a pill this is easy. For manual therapy it is genuinely hard, and that is a real methodological difficulty rather than an excuse. The practitioner always knows, because you cannot deliver a thrust without knowing you delivered it. The patient often knows, because a real cervical adjustment produces a noise and a sensation a sham does not. What is achievable, and what separates a good trial from a weak one, is blinding the assessor. The colic result is the whole argument for why that matters.

Who reported the outcome? Almost everything that matters in musculoskeletal care is patient reported: pain on a scale, a disability questionnaire, a global impression of change. That is not a flaw, since pain is what we are treating and there is no blood test for it. But a patient reported outcome from an unblinded patient measures the treatment plus the patient's beliefs about it, and the two cannot be separated afterward. Where a study also has an objective outcome, such as return to work or medication use, check whether it moved in the same direction. When the subjective improves and the objective does not, as in the asthma literature, that divergence is itself the finding.

How long did they follow people? Many manual therapy benefits are short term and shrink toward nothing over months, which is what the cervicogenic headache review found. A trial that measures at two weeks cannot tell you about three months, and cannot distinguish a treatment that speeds recovery from one that changes the eventual outcome. Ask when the last measurement was taken, and how many patients were still in the study by then.

Is the difference big enough to matter? Statistical significance means a difference is unlikely to be chance alone. It says nothing about size. With enough patients, half a point on a ten point pain scale becomes statistically significant, and no patient alive can feel it. The idea you want is the minimal clinically important difference: the smallest change a patient would actually notice and value. For pain on a zero to ten scale it is usually taken to be around two points, though that depends on the scale and the population. Find the size of the effect and the interval around it, and ask whether the bottom end of that interval would be worth a patient's afternoon and copayment.



A clinical trial has seven places where a reader can be misled, and each carries a question answerable only from the methods section.

The patient who does not get better

This is the part I would most like you to remember in ten years, because it is where the difference between a practitioner and a technician shows.

A course of care is not open ended. It is a hypothesis with a deadline. You form an idea of what is wrong, propose a plan, and decide in advance, out loud, when you will look at the results and what would count as failure. My habit is to set a defined trial of care at the first visit and say what I expect by the end of it. Something like: I want to see you six times over three weeks, and by then I expect your pain meaningfully lower and you sleeping through the night. If that has not happened, we change what we are doing or send you to someone else.

Saying that at the first visit does three things. It tells the patient what success looks like, so improvement is not judged by whether she likes you. It commits you publicly to a review point, which is harder to slide past than a private intention. And it makes an eventual referral a fulfilled promise rather than an admission of failure.

At the review point there are only a few honest options. If she is meaningfully better, taper toward discharge and a plan for staying well rather than sliding into indefinite maintenance by default. If she is somewhat better but not enough, extend deliberately, once, with a stated new endpoint. If she is unchanged, your working hypothesis is probably wrong, and the answer is to examine her again from the beginning, screen again for the findings in the last section, and consider a different diagnosis rather than a more vigorous version of the same treatment. If she is worse, stop and reassess immediately.

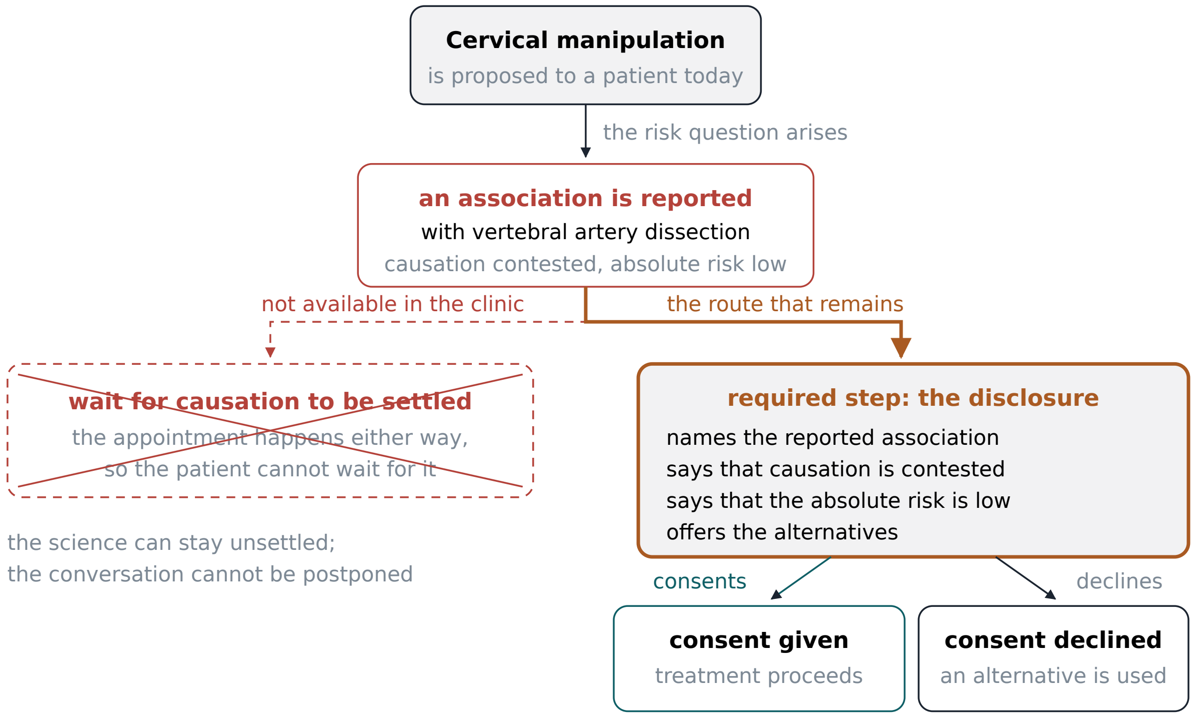
What must not happen is the thing that happens all the time, in every manual profession and not only this one. A patient comes twice a week for four months, does not improve in any way either of you could write down, and keeps coming because the visits have become a routine, because she likes you, and because nobody has looked squarely at the fact that nothing is changing. That is not care. It is a subscription, and the person responsible for noticing is you.

I have referred patients whose problem turned out to be an inflammatory arthritis, a hip that needed replacing, a compression fracture in a woman whose bone density nobody had checked, and once a depression presenting as intractable back pain. In none of those cases would more adjusting have helped. In all of them, what made me look again was a review point I had promised at the first visit and could not talk myself out of.

Consent is a conversation

Which brings us to the obligation everything in this chapter has been building toward. Informed consent is not the form. The form is a record that a conversation happened. If the conversation did not happen, the form records nothing, and you will both find that out at the worst possible moment. A signature collected at a front desk from a patient in pain who has not read it satisfies an administrative requirement and no ethical one.

What the conversation must contain is short enough to hold in your head. What I think is wrong, in plain language. What I propose to do, described physically, in the terms chapter 3 uses for the thrust, so she can picture what will happen to her body. What the evidence supports, honestly graded, including the phrase I do not know where it applies. What the alternatives are, including doing nothing and waiting, which for most acute back pain is a reasonable option and should be named as one. What the risks are, common and minor as well as rare and serious. And a check that any of it landed, which usually means asking the patient to tell you back what she understood.



Informed consent is a sequence of things said out loud and checked for understanding, and the signature is the last item in it.

The hard part is not the content. It is that a fully honest version of this conversation will sometimes talk a patient out of treatment, and every practitioner feels that pressure, especially in the early years when the schedule is thin and rent is due. Chapter 4 dealt with the business realities that make that pressure real. The practitioners I know who have lasted thirty years without becoming cynical are, without exception, the ones who tell patients the unflattering version early. It costs a booking now and again.

> **A note on writing it down**
>
> Document the conversation, not just the consent. A chart note saying patient consented tells you nothing in three years. A note saying discussed evidence for manipulation in chronic low back pain, discussed soreness as common and dissection as rare with the uncertainty stated, discussed exercise therapy as an alternative, patient elected to proceed and can describe the plan back to me, is a record of an actual professional act. It also happens to be the most useful thing in your file if a complaint is ever made, but that is a side benefit. Write it because it is what you did.

Here is the summary I would want you to be able to give, unprompted, at that dinner party. For low back pain and neck pain, spinal manipulation is one of several reasonable first choices with modest benefits and a low harm profile, performing best against doing nothing, less impressively against sham, and about the same as other conservative care. For cervicogenic headache there is a small short term benefit against other manual therapies that does not persist. For migraine the evidence does not support it. For non musculoskeletal conditions the evidence is thin, and where it looked positive it has often not survived blinding. Serious harm is rare, our systems for detecting it are weak, and the dissection question is unresolved rather than merely disputed.

That is not a triumphant summary and it is not a humiliating one. It is what the evidence says, and a practitioner who can say it in that form, without defensiveness and without embarrassment, will be trusted in a way no amount of enthusiasm will buy. Chapter 7 returns to how to say it when the person asking is hostile.

Terms from this chapter

Sham

A procedure built to feel like real treatment while withholding the part believed to be active, so that attention and expectation appear in both groups.

Comparator

Whatever the treatment group was measured against. The same result means different things against sham, against no treatment, and against other care.

Systematic review

A study of studies that searches for every trial asking a question, appraises each, and reports them together instead of selecting the favorable ones.

Confidence interval

The range of values compatible with the data. An interval crossing zero includes benefit, no effect, and small harm.

Minimal clinically important difference

The smallest change a patient would actually notice and value, a higher bar than statistical significance.

Performance bias

Distortion arising when patients, practitioners, or assessors know which group a patient is in and behave or report differently as a result.

Natural history

The course a condition takes on its own. Where it is favorable, as in most acute back pain, almost any intervention appears to work.

Case crossover design

A design in which each person who had the event serves as his own control, comparing exposure just before the event against exposure in earlier windows.

Reverse causation

When the apparent effect is really the cause: a dissection already underway producing the neck pain that sends a patient for treatment.

Cervical artery dissection

A tear in the inner lining of a neck artery, letting blood into the vessel wall, which can obstruct flow or send a clot to the brain.

Red flag

A finding that raises the possibility of serious pathology and calls for referral or investigation rather than a course of manual treatment.

Cauda equina syndrome

Compression of the nerve root bundle below the end of the spinal cord, signalled by saddle anesthesia and sudden onset bladder dysfunction, requiring emergency assessment.

Trial of care

A defined number of visits with a stated expected result, agreed in advance, so that failure to improve triggers reassessment rather than repetition.

Informed consent

The conversation in which a patient learns what is wrong, what is proposed, what the evidence supports, and what the alternatives and risks are. The signature records it; it is not the thing itself.

Before you move on

1. State what the 2017 American College of Physicians guideline says about spinal manipulation for acute low back pain, including how it rates that evidence and what it says about pain as opposed to function.
2. The Cochrane review of chronic low back pain reports three results against three comparators. Name them, and say which one matters most to a patient choosing between a chiropractor and a physiotherapist.
3. In the Cochrane review of manipulative therapies for infantile colic, what did parent reported outcomes show, what did blinded observer outcomes show, and what does the gap demonstrate?
4. Explain what a case crossover design controls for automatically, and what it cannot establish about manipulation and cervical artery dissection no matter how carefully it is conducted.
5. A mother brings a six week old who cries for hours every evening and asks whether you can help. Write out what you would say to her, in order, including what you would check first.
6. A patient with chronic low back pain has been seen eight times over five weeks with no measurable change, but she enjoys the visits and wants to continue. What do you do, and how do you justify it to her and to yourself?

CHAPTER 6

Alongside Everyone Else

> *If several professions put their hands on the same spine for the same complaint, why did it take a federal antitrust verdict to get them into the same building?*

A patient with low back pain has options, and she does not know most of them. She can call her primary care physician. She can look for a physical therapist, and in every state she can now do that without a referral from anyone. She can book a massage. She can see an acupuncturist. She can see an osteopathic physician who may put hands on her back or may write a prescription or may do both. Or she can come to me. From where she sits, these are five or six doors on the same hallway, and the sign on each one tells her almost nothing about what is behind it. She picks the door her sister recommended, or the one her insurance covers, or the one that can see her Thursday.

From the inside, those doors look nothing alike. Each opens onto a different training, a different legal scope, a different vocabulary for the same tissues, and a different set of assumptions about what is wrong with her. And for about fifteen years in the middle of the twentieth century, one of those professions ran an organized campaign to make sure that a physician who walked through my door, professionally speaking, would be committing an ethics violation. That is not a rhetorical flourish and it is not a grievance I am inviting you to inherit. It is a finding of fact by a United States District Court, affirmed on appeal, and this chapter will take you through it slowly, because it is the single piece of history that most shapes how a chiropractor is received by other clinicians and because it is the piece most often told badly by people on my own side.

We are going to do three things. First, look honestly at who else is in the building and what they are actually trained to do, because you cannot work with people you cannot describe. Second, work through the history of the boundary fight, in detail, using the court's own words. Third, come back to the present and to a skill you will use in your first month of practice, which is writing a referral that another clinician can act on. That last one sounds like paperwork. It is the whole of integration, compressed into a page.

Who else is in the building

Start with the physician, because the physician is the reference point everyone else is measured against, fairly or not. In the United States a medical physician holds either an MD or a DO degree. Both routes require a bachelor's degree, then four years of medical school, then residency training in a specialty, which the federal Bureau of Labor Statistics describes as lasting three to nine years depending on the specialty, with a further one to three years for those who subspecialize through a fellowship. Add it up and a general internist has spent something like eleven years in formal training after high school, and a spine surgeon considerably more. Licensure requires completing that residency and passing national examinations. What a physician's license permits is essentially unlimited: diagnosis of anything, prescription of any drug the law allows a prescriber to write, referral to any specialist, admission of patients to a hospital, and, with the appropriate surgical training, cutting.

The osteopathic physician deserves a slower look, because the DO is the profession whose history most closely rhymes with mine and whose present is most different. Andrew Taylor Still announced osteopathic principles in 1874 and opened his school three years before the events chapter 1 describes, and osteopathy arrived at the same place chiropractic did by a different route: structure governs function, disturbed structure produces illness, and the correction is manual. Where Palmer routed the harm through the nerves, Still routed it through the blood supply. That is one link apart in the reasoning and it was a century apart in professional outcome. Today a DO attends four years of osteopathic medical school which includes instruction in osteopathic manipulative treatment, sits a licensing examination sequence of its own, enters the same accredited residency programs as MD graduates, and holds full practice rights including prescribing and surgery in all fifty states. Osteopathy did not stay outside the house. It moved in.

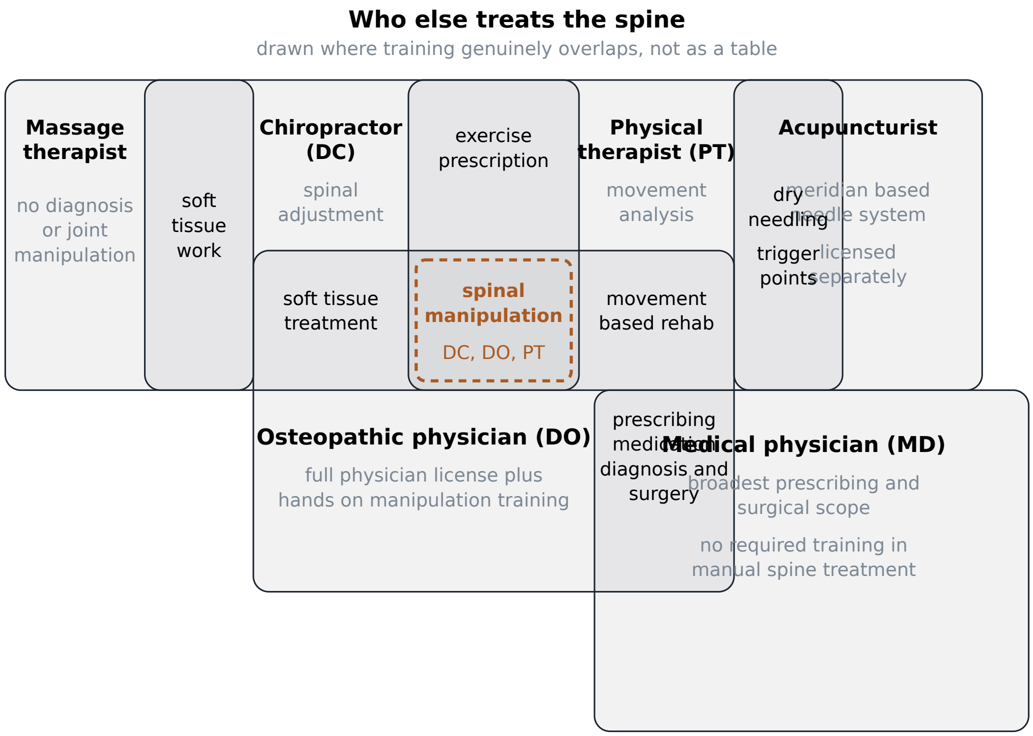
And here is the part that surprises students, and that ought to interest anyone thinking about what happens to a manual profession that integrates completely. Most osteopathic physicians do not use their hands. A survey reported in the early 2000s found that more than half of responding DOs used osteopathic manipulative treatment on fewer than five percent of their patients, and the number of osteopathic physicians who report consistently performing manipulation has been falling for decades even as the number of osteopathic graduates rose sharply. I want you to sit with that for a moment before we go further, because it is a genuine argument in both directions. One reading is that osteopathy achieved everything chiropractic wanted and lost the thing that made it distinct. The other reading is that osteopathy's hands were never the point, that the point was becoming a physician, and that the manual work was scaffolding. I do not know which reading is right. I do know that anyone in my profession who cites the DO story as a cautionary tale about integration ought to also notice that the caution comes with full prescribing rights and a hospital badge.

The physical therapist is the profession whose scope overlaps mine most directly, and therefore the one that produces the most friction. The entry level credential is now the Doctor of Physical Therapy, typically three to four years after a bachelor's degree, with substantial supervised clinical practice built in. Graduates sit a national licensure examination accepted in all fifty states. Since the beginning of 2015, all fifty states and the District of Columbia have allowed some form of direct access, meaning a patient can be evaluated by a physical therapist without a physician's referral first, although what that access permits varies considerably by state. Physical therapists diagnose movement problems, prescribe and progress exercise, use manual therapy on joints and soft tissue, and manage rehabilitation after injury and surgery. They do not prescribe drugs and they do not perform surgery.

Massage therapy sits on a very different regulatory footing, and students routinely overestimate how standardized it is. Forty six states and the District of Columbia require a massage credential; California, Kansas, Minnesota, and Wyoming have no statewide licensure at all. Minimum education runs roughly five hundred to a thousand hours, with most states landing between five hundred and seven hundred fifty. A national examination is accepted in nearly every regulated jurisdiction. What a massage therapist is trained to do is work soft tissue: muscle, fascia, and the tissue relationships around a joint. What a massage therapist is generally not licensed to do is diagnose, and in most jurisdictions is not permitted to perform a thrust to a joint. That last boundary is the one that matters clinically and the one patients understand least, because from the table the difference between deep soft tissue work and a manipulation is not obvious to the person face down on it.

The acupuncturist trains in a system with its own diagnostic vocabulary and its own theory of what is wrong with a patient, which is a genuinely different situation from the others on this list. National certification eligibility requires at minimum a three year master's degree of roughly one thousand nine hundred hours; herbal medicine credentials require substantial additional training on top of that. State licensure requirements vary and I am not going to pretend to summarize fifty of them. What matters for our purposes is that an acupuncturist's assessment framework does not map onto anatomical diagnosis in the way that a physical therapist's does, so the sentence "we found the same thing" is much harder to say honestly across that boundary than across the others.

Where do I fit? Chapter 4 gave you the road in detail, so I will only put the numbers beside the others for comparison. The accrediting standard for a chiropractic program requires a minimum of four thousand two hundred instructional hours, of which at least one thousand must be in supervised patient care. Entry requires a bachelor's degree or at least ninety semester credit hours of prior coursework, and licensure runs through the national board examination sequence plus whatever the individual state adds. That is a substantial education. It is also, in ninety percent or more of jurisdictions surveyed, an education that does not carry prescription rights, obstetric care, or minor surgery. New Mexico is the notable outlier with an advanced practice credential that permits a defined formulary, and that statute is currently scheduled to sunset. Everywhere else, if the patient in front of you needs a drug or an operation, you are not the person who provides it. Say it plainly in your own head now, because you will need to say it plainly to patients later.



The five professions a chiropractor works beside overlap heavily in the physical acts they perform on musculoskeletal complaints and separate almost entirely at prescribing and surgery, which is why the boundary disputes cluster where they do.

Now the overlap, stated without softening, because the size of it is the whole explanation for what follows. Spinal manipulation is not a chiropractic monopoly and never has been. Chiropractors are estimated to perform the large majority of manipulative treatments in North America and Europe, better than nine in ten by one commonly cited estimate, but osteopathic physicians perform it, physical therapists perform it, and osteopaths trained outside the United States perform it. Physical therapists have historically used a grading vocabulary in which what a chiropractor calls a manipulation appears as the highest grade of mobilization, which means the same physical act has been performed under two names in two professions for decades. Add in exercise prescription, soft tissue work, patient education, and rehabilitation, and the honest picture is that a chiropractor's daily work overlaps a physical therapist's daily work substantially and an osteopathic physician's manual work almost completely.

Professions do not fight hardest where they are most different. They fight hardest where they are most alike and the license says otherwise.

That principle explains the map. Nobody fights about whether an acupuncturist may prescribe antibiotics, because there is no ambiguity to exploit. But when two licensed professions perform the same act on the same joint for the same complaint and only one of them has the word for it in its statute, you get litigation. In

Alabama, the state chiropractic board sued a licensed physical therapist, arguing that his advertising and performance of spinal manipulation constituted the unlawful practice of chiropractic, and sought a court declaration that only chiropractic licensees may manipulate a spine. The trial court dismissed, and the state supreme court affirmed without writing an opinion. Here is where it gets instructive. The physical therapy side describes that outcome as a ruling that spinal manipulation falls within the physical therapy scope of practice. The Alabama chiropractic board, in its own newsletter, describes the trial court as having declined to rule on the ground that it lacked authority to decide the matter, and states flatly: "Neither the Circuit Court nor the Supreme Court by its affirmance specifically held that a physical therapist can lawfully perform spinal manipulations." Both descriptions are of the same docket. Neither is a lie. A dismissal that resolves nothing on the merits is genuinely available to be told either way, and each profession told it the way that helped.

> **A note on what a scope of practice actually is**
>
> Students arrive expecting scope of practice to be a description of competence. It is not. It is a description of legal permission, written by a legislature, and the two come apart constantly in both directions. There are acts you are legally permitted to perform that you personally have no business performing, and acts you could perform safely and well that your license does not cover. Scope also varies by state for the same degree, which means the identical education produces different legal practices depending on the address, a point chapter 4 develops. When you are working out whether something is yours to do, you have to answer two separate questions and not let either one answer the other: am I permitted, and am I competent. A yes to the first does not supply a yes to the second.

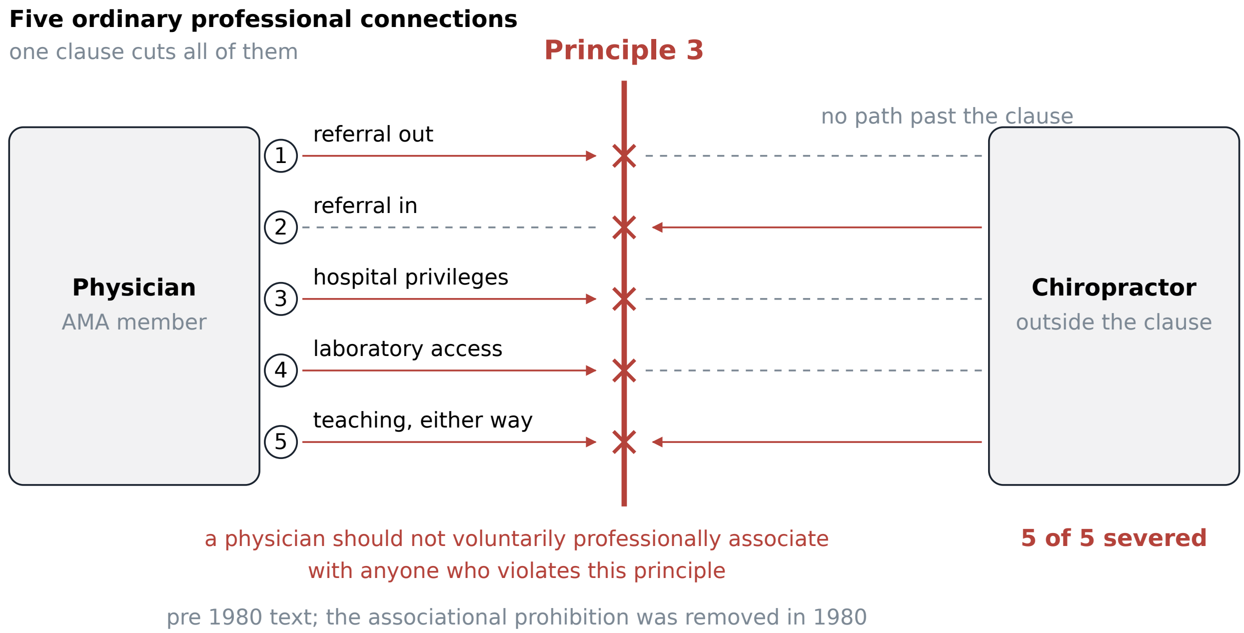
November 1963

In November 1963 the American Medical Association formed a body called the Committee on Quackery, under its Department of Investigation. The name tells you the framing. The committee's business was practices the association regarded as fraudulent, and chiropractic was its principal target. A federal court would later accept the plaintiffs' characterization of the committee's purpose in words that have been quoted ever since: the goal was to contain and eliminate the chiropractic profession.

The instrument was not a lawsuit and not a legislative campaign, though there were both of those elsewhere. The instrument was the association's own code of ethics, and specifically its Principle 3, which in the version of the Principles of Medical Ethics that stood before 1980 read:

> "A physician should practice a method of healing founded on a scientific basis; and he should not voluntarily professionally associate with anyone who violates this principle."

Read that twice, because on first pass it sounds unobjectionable. A professional body telling its members to practice scientifically is exactly what a professional body should do. The work is being done by the second clause, and by who gets to decide what counts as a scientific basis. If the association declares that chiropractic is not founded on a scientific basis, then the second clause converts that declaration into a prohibition on every form of professional contact between a physician and a chiropractor, and it does so without the association ever having to prove the declaration. The finding of fact was made in house. The consequence fell on everyone.



A single sentence in an ethics code severed four separate channels between the professions at once, which is why the effect was so much larger than an ordinary professional disagreement.

Now think about what "voluntarily professionally associate" covers, because this is the part that students underestimate. It covers referral: a physician cannot send a patient to a chiropractor. It covers consultation: a physician cannot ask a chiropractor's opinion on a shared patient, or accept a chiropractor's question about one. It covers hospital privileges: a chiropractor cannot practice inside an institution staffed by physicians who would then be associating with him, and cannot obtain imaging or laboratory work through that institution on the same terms as anyone else. And it covers teaching: a physician cannot lecture at a chiropractic college, which means the chiropractic curriculum cannot easily draw on the medical faculty available in its own city, and a chiropractic student's basic science instruction is cut off from the mainstream of it.

Those four channels are not four versions of the same thing. They are the four ways professions actually touch each other, and severing all of them simultaneously does something more than isolate a group. It makes the group's isolation self confirming. A profession locked out of hospitals will have less institutional experience with seriously ill patients, and can then be criticized for having less institutional experience with seriously ill patients. A profession that cannot recruit medical faculty will teach basic science from within its own tradition, and can then be criticized for teaching basic science from within its own tradition. A profession that receives no referrals sees the patients who select themselves, which shapes what its practitioners believe about who they can help. Every one of those effects is real, and every one of them was, for a period, partly manufactured. This is why I ask you to hold two thoughts at once here: the criticisms of mid century chiropractic were often accurate, and the conditions being criticized were in part produced by the people making the criticism. Both.

The machinery came apart before anyone sued over it. The Committee on Quackery disbanded in December 1974. Its parent Department of Investigation disbanded in May 1975, and the staff member who had led the effort left the association's employment. Principle 3 itself survived another five years and was removed from the ethics code in 1980. So the arc runs from November 1963 to 1975 for the committee structure, and to 1980 for the ethical rule it operated through.

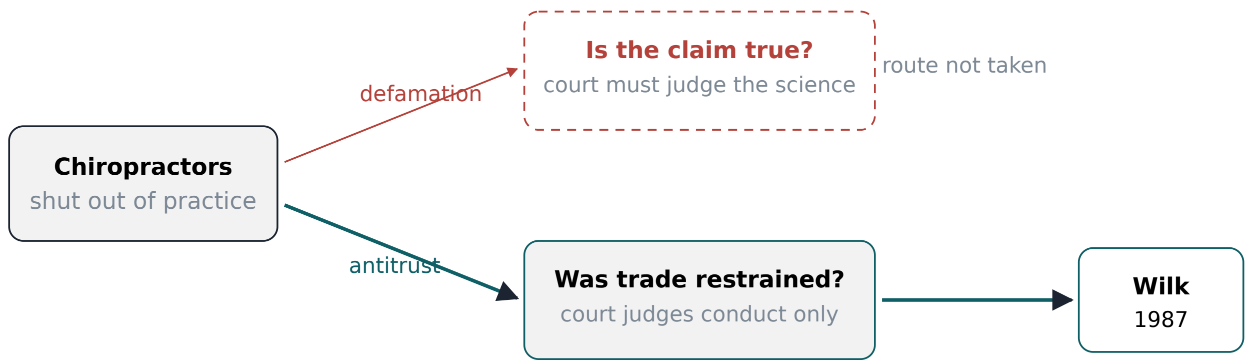
It is worth pausing on the year 1974, because three other things happened in it that had nothing to do with the committee and everything to do with why the committee stopped. Congress authorized Medicare payment for chiropractic services that year. The Council on Chiropractic Education was recognized as a nationally recognized accrediting agency by the US Commissioner of Education on August 26 of that year. And Louisiana is consistently reported as having become the fiftieth and last state to license chiropractic in 1974, though I would want a state record in front of me before I put a date on it. A boycott organized around the claim that a profession is illegitimate becomes very hard to sustain in the same twelve months that the federal government agrees to pay for it, recognizes its accreditor, and watches the last holdout state issue licenses. The committee did not disband because it was defeated in court. It disbanded because the ground had moved.

Why the case was brought as antitrust

Chester Wilk, a chiropractor, and several colleagues filed suit against the American Medical Association and a number of other medical organizations in 1976, after the committee was already dissolved. The legal theory they chose is the most useful thing in this chapter for a student who wants to think clearly, and I want to spend real time on it.

The obvious claim, the one a nonlawyer reaches for first, is defamation. The association had told its members and the public that chiropractors were unscientific practitioners. That is an accusation about the plaintiffs' work, it was published widely, and it plainly harmed them. Why not sue for that?

Because of what a defamation plaintiff has to prove. Broadly, you must show that the statement was false. Which means that a defamation suit brought on those facts puts one question to the court: is chiropractic in fact unscientific? To win, the plaintiffs would have had to persuade a judge or a jury that their profession rests on a scientific basis. To win, the defendants would only have had to persuade the same judge or jury that it does not. The entire case would have become a trial of chiropractic's validity, conducted in a courtroom, decided by people with no capacity to evaluate the underlying literature, and settled by whichever side put on the more compelling expert. And in 1976, with the evidence base that then existed, that was a fight the plaintiffs would very likely have lost. Worse, losing it would have produced a judicial finding that chiropractic is unscientific, which the association could then have cited forever.



A defamation claim would have required the court to decide whether chiropractic works, while the antitrust claim asked only whether a trade association may organize a boycott of a competitor, and that difference in the question is why the case was winnable.

The antitrust claim asks a completely different question. Section 1 of the Sherman Act prohibits contracts, combinations, and conspiracies in restraint of trade. Applied to these facts, the question becomes: did these organizations agree among themselves to refuse to deal with a group of competitors, and did that agreement restrain competition in the market for health care? Notice what has dropped out of the case. Whether chiropractic works is no longer an element of the claim. The plaintiffs do not have to prove their treatment is effective, and the defendants cannot win by proving it is not. What is on trial is the conduct of a trade association, judged by the rules that apply to trade associations generally.

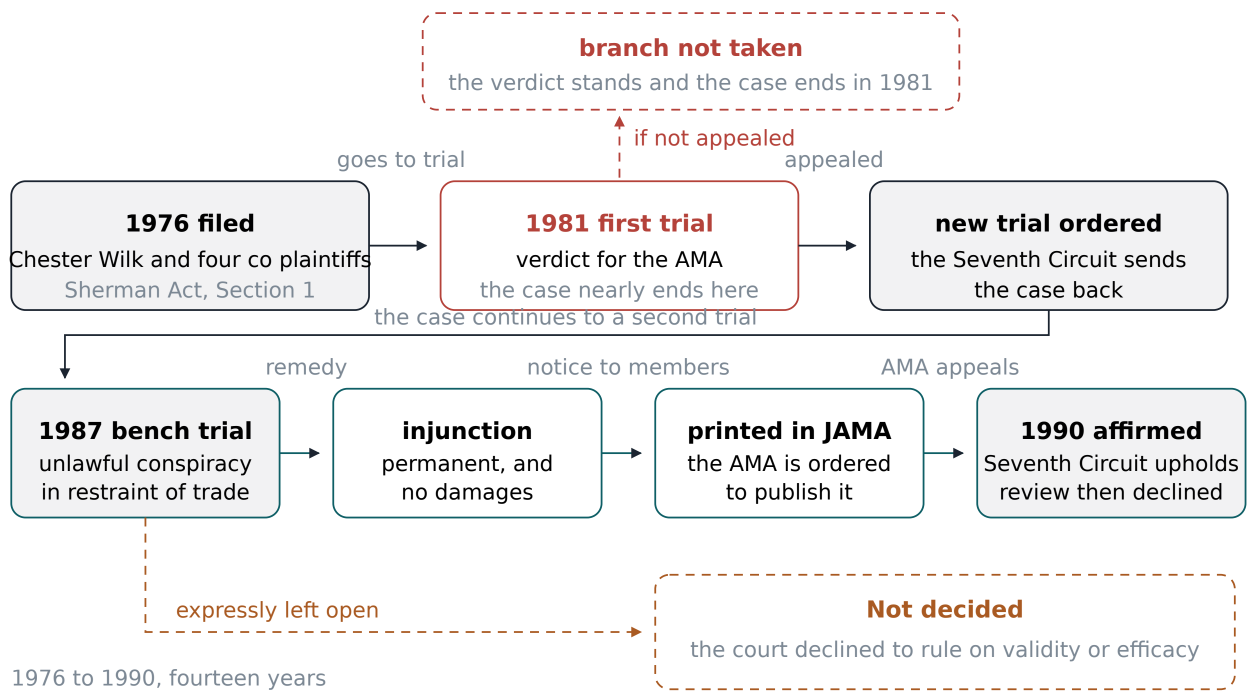
The association still had a defense available, and it made it. The argument was that this was not an ordinary commercial boycott but a patient care measure: physicians were being told to avoid chiropractors because the association genuinely believed chiropractic care harmed patients, and protecting patients is a legitimate purpose that ought to take the conduct outside the reach of antitrust law. That is a serious argument and you should be able to state it in its strongest form, because it is the argument any professional body would make in the same position, and sometimes it is correct. Professional associations do have legitimate reasons to warn members away from dangerous practices.

The court's answer to it is the part worth memorizing, because it is a general tool. The association carried the burden of proving that defense, and the court held it had not carried it, on the ground that the boycott was broader than the purpose required. If the concern is patient safety, there are less restrictive ways to pursue it. You can publish what you know. You can educate the public. You can tell your members what the evidence shows and let them decide who to refer to. What you may not do is organize a total refusal to deal, when a less restrictive alternative would have served the stated purpose. That test, less restrictive alternative, is the hinge of the whole case, and it is worth carrying with you well beyond this chapter. When any institution defends a sweeping restriction by pointing to a legitimate goal, the honest question is not whether the goal is legitimate. It is whether the restriction was the narrowest way to reach it.

The 1987 finding

Judge Susan Getzendanner of the United States District Court for the Northern District of Illinois issued her decision in 1987, eleven years after filing. The core finding reads, in the court's own words, with only the date phrase moved outside the quotation marks so that this book can keep its typographic rules: the AMA and its officials "instituted a boycott of chiropractors" in the middle 1960s "by informing AMA members that chiropractors were unscientific practitioners and that it was unethical for a medical physician to associate with chiropractors. The purpose of the boycott was to contain and eliminate the chiropractic profession."

The court found an unlawful conspiracy in restraint of trade in violation of Section 1 of the Sherman Antitrust Act, running from 1966 to at least 1980, the year Principle 3 came out of the ethics code. It found that the association had not met its burden of proving the patient care defense, for the reason given above.



Eleven years passed between filing and the district court finding and three more before the appellate affirmance, so a student who says the case took a decade is understating it.

The remedy was a permanent injunction against the association, entered under Section 16 of the Clayton Act, which is the provision that lets a private plaintiff obtain injunctive relief in an antitrust case. The court gave three reasons for granting it rather than concluding that the dissolved committee had made the problem moot. The effects of the conspiracy were still lingering. The association had never publicly acknowledged the unlawful conduct. And no affirmative corrective statement had ever been issued to its members. Note the structure of that reasoning: the court is saying that dismantling the machinery is not the same as undoing the effect, because the members who were told for fifteen years that association was unethical were never told anything else. On the necessity of the order the court wrote that "a proper exercise of the court's discretion permits, and in my judgment requires, an injunction."

Be precise about what she declined, because this is where partisan retellings go wrong. The court explicitly declined to grant the broadest injunctive relief the plaintiffs had asked for. They did not get everything they sought. What they got was an order barring the association from that conduct going forward, and a published opinion stating on the record what had happened. The United States Court of Appeals for the Seventh Circuit affirmed the judgment in 1990, fourteen years after the complaint was filed and twenty seven years after the Committee on Quackery was formed.

What the ruling did not decide

Now the paragraph I care most about in this chapter, and the one I would most like you to be able to reproduce under pressure.

Wilk did not decide that chiropractic works. It did not decide that chiropractic is scientific. It did not decide that spinal manipulation is effective for any condition whatsoever. It could not have decided any of those things, because none of them was an element of the claim, and the plaintiffs chose an antitrust theory precisely so that none of them would have to be. The court found that a trade association organized an unlawful boycott of a competitor. That is a finding about the association's conduct. It is not a finding about the competitor's merits, any more than a verdict against a company for price fixing is a verdict that its product is good.

I raise this with some urgency because the misuse of this case is common inside my own profession, and it is embarrassing when it happens in front of someone who has read the opinion. You will encounter the argument, in marketing copy, in a lecture, occasionally from a practitioner who should know better: chiropractic was vindicated in federal court, therefore the medical objections were shown to be baseless, therefore the treatment is proven. Every step of that is wrong, and the wrongness has a specific shape. It treats a ruling about process as a ruling about substance. It converts "they were not allowed to organize a boycott" into "they were wrong about the medicine."

And look at what that argument really is. It is exactly the error the association made, run in the opposite direction. The association declared, by internal vote and without demonstration, that chiropractic lacked a scientific basis, and treated its own declaration as though it settled a scientific question. A chiropractor who cites Wilk as proof of efficacy is treating a legal declaration as though it settles a scientific question. The institution differs. The move is identical. If you take nothing else from this chapter, take this: the question of whether an institution treated a group fairly and the question of whether that group's claims are true are separate questions, answered by separate methods, and no verdict on one is evidence about the other.

What the case does establish, and it is not small, is historical. It establishes as a judicial finding of fact, on a full record, that the isolation of chiropractic in mid century American medicine was in substantial part organized rather than incidental. That matters for how you read the history. It means that when you encounter a claim from that era that chiropractors could not get hospital privileges because they were unqualified, you are entitled to ask how much of the barrier was assessment and how much was policy. It does not mean the answer is always policy. Chapter 5 laid out what the evidence does and does not support for specific conditions, and the honest reading of that chapter is not flattering across the board. Both things are true, and a student who can hold both is worth ten who can hold either.

> **A note on how this story gets told**
>
> You will find versions of the Wilk story that are almost unrecognizable, in both directions. On one side, accounts that describe it as the moment organized medicine was forced to admit chiropractic is legitimate, usually with no mention of what the plaintiffs did not win and no mention that efficacy was never at issue. On the other, accounts that mention it only in passing as a technical antitrust matter of no continuing significance, which understates a finding that a professional association conspired to eliminate a licensed profession. The corrective for both is the same and it is available to you: the district court opinion is published, it is long, it is written in ordinary English, and you can read it. Anyone who summarizes a court decision for you without having read it is guessing, and that includes people who agree with you.

The present

Principle 3 has been gone since 1980 and the injunction has stood since 1987, so what does the landscape look like now? Better than the history would predict, patchier than the profession's own promotional material suggests, and enormously variable by geography and setting.

The most common form of integration is the multidisciplinary or integrated clinic, where a chiropractor practices in the same building and often under the same billing entity as physicians, physical therapists, massage therapists, and sometimes acupuncturists or behavioral health clinicians. These arrangements exist across the country and range from genuine shared management, with common records and case conferences, to what is really just a shared waiting room and a landlord. You should learn to tell the difference before you sign anything, and the diagnostic question is simple: does anyone in this building ever change their plan because of something a colleague found? If the answer is no, it is a rental arrangement wearing a clinical name.

A survey of chiropractors working inside integrated private sector facilities gives a picture of what that work looks like when it is real. Among those surveyed, forty percent were working in hospitals, twenty one percent in multispecialty offices, and sixteen percent in ambulatory clinics. Sixty eight percent were employees rather than owners, most on salary. More than sixty percent reported managing patients jointly with medical professionals, and most shared an electronic health record with the medical staff. Be careful with those numbers: that survey deliberately sampled chiropractors already known to be working in integrated settings. It tells you what integrated practice looks like, and nothing about how common it is. Anyone who quotes those percentages as if they described the profession as a whole has misread the study.

The clearest institutional example in the country is the Department of Veterans Affairs. Congress authorized the VA to begin providing chiropractic care in 1999, and since 2004 those services have been available to all eligible veterans. VA chiropractors are salaried clinicians integrated with primary care, rehabilitation, and pain management teams, working on shared records inside a referral system, treating musculoskeletal conditions that do not require surgery. The number of clinics on VA campuses grew from twenty seven to sixty five between fiscal years 2005 and 2015, and I do not have a reliable current figure, so I will not give you one. What makes the VA worth studying is not the size. It is that it is the one large American system where chiropractic operates under exactly the conditions Principle 3 forbade: a physician refers, a chiropractor evaluates and treats, both write in the same chart, and both are answerable to the same institution for the outcome. If you want to know what integration looks like when the incentives are aligned, that is where to look.

Who pays, and for what

Money shapes practice more than philosophy does, and coverage rules are where you will feel the boundaries of your profession most concretely.

Medicare Part B covers, in its own words, "manual manipulation of the spine by a chiropractor to correct a vertebral subluxation." That is the entire chiropractic benefit. Medicare does not cover other services or tests a chiropractor orders, including imaging, massage therapy, and acupuncture. The patient pays twenty percent of the approved amount after the deductible. Three things about that sentence deserve your attention.

First, the federal payer's coverage language is built around the word subluxation, which chapter 1 showed you is four different concepts wearing one word. A billing rule froze one usage of a contested term into federal payment policy in 1974 and it has stayed there. When a Medicare claim uses that word, it is not making a philosophical commitment; it is naming a billing category. Do not let anyone tell you Medicare's vocabulary is an endorsement of a theory.

Second, the benefit is narrower than the license. A chiropractor may lawfully perform an examination, order imaging, and provide soft tissue and rehabilitative care, and Medicare will pay for none of it from that provider. The payment system is willing to buy the thrust and not the thinking that decides whether the thrust is indicated. Every experienced practitioner I know has an opinion about that, and none of the opinions change the rule.

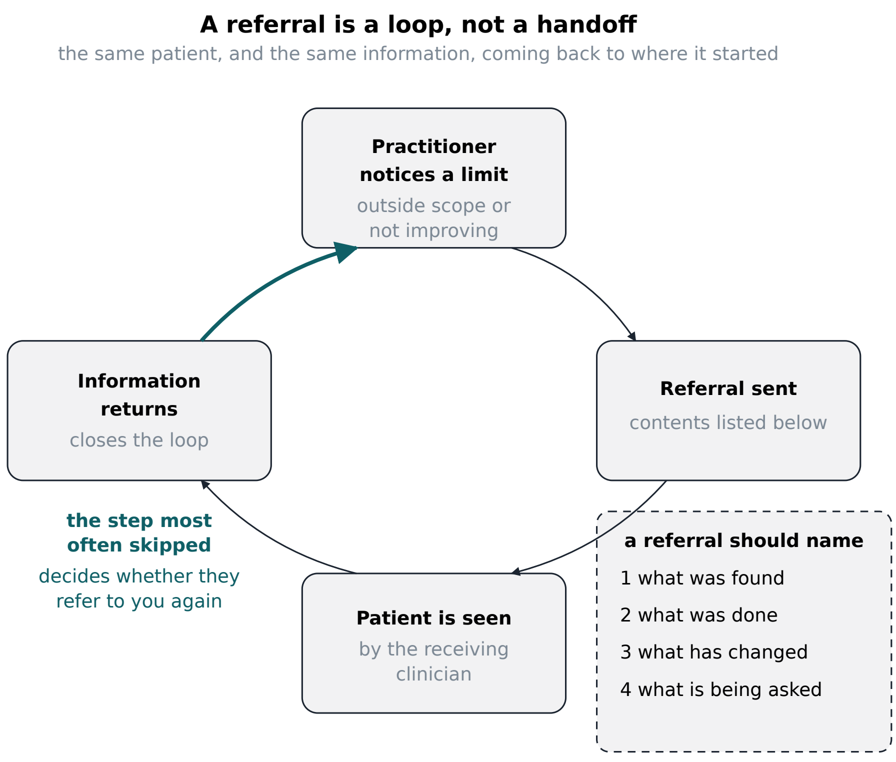
Third, Medicare distinguishes sharply between active corrective care and maintenance. Claims must carry a specific modifier indicating active treatment of an acute or chronic subluxation, and claims without it are treated as not medically necessary. The governing language is explicit: once the maximum therapeutic benefit has been achieved for a given condition, ongoing maintenance therapy is not considered medically necessary. Whatever a practitioner believes about the value of ongoing supportive care, and there is a real argument there, Medicare does not buy it. Knowing that in advance is the difference between a conversation with a patient about paying privately and a refund with an apology two months later.

Private coverage varies more and generalizes less. The Affordable Care Act does not name chiropractic among the essential health benefits, so coverage on individual and small group plans depends on the individual state's benchmark plan and any state benefit mandates. Where chiropractic is covered, it is typically capped at a set number of visits per year and conditioned on medical necessity, and maintenance or wellness visits are generally excluded. In practice this means the answer to "does insurance cover this" is genuinely different in two neighboring states and sometimes between two employers in the same town, and a practitioner who guesses will be wrong often enough to damage a patient relationship. Verify before the first visit, in writing, every time.

Writing a referral somebody can use

Everything above is context. This is the skill.

A referral is not an administrative act and it is not a handoff. It is a clinical communication with a specific reader and a specific request, and most of them are written as if neither of those things were true. I have received referrals consisting entirely of a patient name and the phrase "please evaluate." I have also received a page and a half of a colleague's internal shorthand describing findings in a technique specific vocabulary I could not parse. Both are failures of the same kind. Neither writer thought about the person opening the envelope.



A referral is a loop rather than a line, and the return leg carrying information back to the sender is the segment most often omitted and the one that determines whether a second referral ever arrives.

A referral that another clinician can act on answers four questions and asks one. Write it in that order and you will not go far wrong.

What was found. The history in two or three sentences, the relevant examination findings, and your working diagnosis stated as a working diagnosis rather than as a fact. Include what you looked for and did not find, particularly the red flags chapter 5 covered, because a receiving clinician needs to know whether malignancy, fracture, infection, and cauda equina signs have already been screened for or whether that work is still hers to do. "No history of cancer, no unexplained weight loss, no night pain, no saddle anesthesia or bladder change" takes one line and saves the reader twenty minutes. Use shared anatomical vocabulary, the kind chapter 2 gave you. Do not use technique internal terminology. If your listing shorthand means nothing outside your own technique's community, it means nothing in a referral.

What was done. The dates, the number of visits, and what you actually did on those visits, described concretely enough that another clinician can picture it. "Nine visits over five weeks, side posture lumbar manipulation, soft tissue work to the gluteal musculature, and a home program of two exercises" is useful. "Adjusted as indicated" is not.

What changed. This is the section most often skipped and it is the one that carries the most information. Give something measured. Pain rating at intake and now. Range of motion at intake and now. What the patient can do now that she could not do at the first visit, or still cannot do. If nothing changed, say nothing changed, and say it in exactly those words. A referral that reports no improvement after an adequate trial is far more useful to the receiving clinician than one that reports vague partial benefit, because it narrows what she has to consider.

What you are asking for. Be specific, and ask a question rather than transferring a problem. There is an enormous difference between "referring for further evaluation" and "I am asking whether advanced imaging is indicated given six weeks without change and a persistent L5 sensory deficit." There is a difference between "please see" and "I am asking you to assume care of this condition; I have taken it as far as I can." State plainly whether you are asking for an opinion, asking for a procedure or prescription outside your scope, or handing the patient over entirely. The reader cannot infer it and will guess wrong.

And then, the fifth element, which is not a question but a courtesy that functions as a clinical safeguard: say what you are doing in the meantime. Are you continuing care while she waits for the appointment, are you holding treatment pending the opinion, or have you discharged her? A receiving clinician who does not know whether the patient is still being treated cannot reason about what she is seeing.

One more thing belongs in a good referral and almost never appears in a bad one. Say what you do not know. "I cannot account for the fact that her symptoms are worse in the morning and better with activity, which does not fit the mechanical pattern I would expect" is an enormously valuable sentence. It tells the reader you have noticed the anomaly rather than papered over it, it tells her where to look, and it tells her something about you as a colleague. Practitioners who write only their conclusions are read as salesmen. Practitioners who write their uncertainties are read as clinicians.

The return leg

Now the part that separates the practitioners other clinicians refer to from the ones they do not.

A referral is a loop. Information goes out, a patient is seen, and information comes back. The return leg is the one that gets dropped, and it gets dropped in both directions constantly. A physician sends me a patient and hears nothing further. A chiropractor sends a patient for imaging and never learns what the report said. Everyone is busy, the patient is doing fine, the note never gets written, and the loop stays open forever.

Here is why that matters more than it looks like it should. When you refer a patient to someone and never hear back, you have learned nothing. You do not know whether your suspicion was right. You do not know whether the colleague found your note useful or useless. You cannot calibrate. And the next time you are deciding whether to refer to that person, you have no information to decide with, so you refer to someone else or you do not refer at all. Multiply that across a professional lifetime and you can see how referral relationships live and die. They are not built on credentials. They are built on whether the last three loops closed.

Nobody refers twice to a practitioner who never wrote back.

So close the loop, deliberately, as a policy rather than a mood. When I receive a referral I write back twice. The first is short and goes out the same week: I saw her, this is what I found, this is what I am doing, thank you for sending her. That note takes four minutes and it does something out of proportion to its length, because it tells the sender the patient landed safely. The second goes at the end of the episode of care or at a defined interval if care is ongoing: this is what happened, this is the outcome, this is what I would suggest going forward, and here is what I could not resolve. If the outcome was poor, that note is more important, not less. A colleague who learns from you that your treatment did not help her patient learns that you can be trusted to tell her the truth, which is worth considerably more than a good outcome she cannot verify.

The same discipline applies when you are the sender. If you refer and do not hear back within a reasonable window, ask. And when you get the answer, tell the other clinician what you learned from it, especially if you were wrong. The colleague who hears "you were right, it was not mechanical, thank you" from a chiropractor is a colleague who will take the next call.

A note on what a referral costs you

New practitioners sometimes resist referring because it feels like admitting failure or like giving away a patient, and I want to name that feeling rather than pretend nobody has it. When you are building a practice and every patient matters to the month's numbers, sending someone out the door is not costless. But consider what you are actually trading. A patient who does not improve and is not referred does not stay a patient; she leaves, tells people, and is gone. A patient who is referred appropriately usually comes back, and often brings the referral relationship with her. In more than one case I have gained a steady source of new patients specifically because a physician noticed that I sent someone back rather than keeping her. Referring well is not the opposite of building a practice. Over a career it is one of the more reliable ways to do it.

What you owe a colleague in another profession

I want to close on obligation rather than on strategy, because the strategic case is easy and the obligation is the part that will be tested when it is inconvenient.

What you owe another clinician is, first, clarity about what you did. Not a defense of your profession, not an explanation of your philosophy, not a paragraph on the body's capacity for self regulation. What you found, what you did, what changed. If a physical therapist can read your note and reconstruct the encounter, you have met the standard. If she has to translate it first, you have not. This matters more for us than for most, because our profession carries an internal vocabulary that is a century old and does not travel. Chapter 1 traced where that vocabulary came from and chapter 3 showed you how much technique specific language exists inside it. All of it is legitimate within the profession. Almost none of it belongs in a note that leaves the building.

Second, you owe honesty about what you do not know, and this is the harder one. It is uncomfortable to write "I do not know why this patient is not improving" to a physician you suspect is already skeptical of your training. The instinct is to project confidence into the gap, and the instinct is exactly wrong. Every clinician in every profession has patients who do not improve and cannot be explained. What distinguishes practitioners is not the absence of those patients but whether they say so. And there is a specific version of this that belongs to us: when a patient's presentation is outside what manipulation can plausibly address, say so early, and do not treat past that point because a referral feels like a loss. Chapter 5 gave you the red flags. The professional obligation attached to them is not merely to recognize them but to act on them promptly and to say plainly why.

Third, and I say this with the history of this chapter fully in view, you owe your colleagues the restraint that was not extended to us. There is a temptation, in a profession that was boycotted, to run a small version of the same play: to tell patients that medical care for their condition is dangerous, to disparage a colleague's judgment in front of the patient, to expand claims into territory the evidence does not support because someone else once denied us territory we deserved. Wilk is not a license for any of that. It is, read properly, an argument against it. The court's reasoning was that a professional body may not use a total refusal to deal when public education would serve, and the principle underneath that reasoning does not belong only to the side that won. If you believe another profession is getting something wrong, the remedy available to you is the same remedy the court said was available to them: say what you know, publish what you have, educate the patient in front of you, and let the evidence do the work.

The patient in the first paragraph of this chapter is still standing in the hallway looking at doors. She does not care about our boundary disputes and she should not have to. What she needs is for whichever door she picks to open onto someone who knows what he can do, says plainly what he cannot, and knows the person behind the next door well enough to walk her there. That is a modest description of integration and it is achievable in any town in the country, by any practitioner willing to write a decent note and answer the phone. It took a federal court fourteen years to make it legally possible. Making it actually happen is a matter of habit, and the habits are yours to build starting now.

Terms from this chapter

Scope of practice

The set of acts a license legally permits a practitioner to perform, defined by statute and varying by state. It describes legal permission, not personal competence, and the two frequently diverge.

Direct access

The ability of a patient to be evaluated by a clinician without a referral from a physician first. All fifty states and the District of Columbia have allowed some form of direct access to physical therapists since the beginning of 2015.

Osteopathic manipulative treatment

The manual therapy taught in osteopathic medical schools. Most practicing osteopathic physicians report using it on only a small fraction of their patients, and reported use has declined for decades.

Committee on Quackery

A body formed by the American Medical Association in November 1963 under its Department of Investigation, whose purpose a federal court later described as containing and eliminating the chiropractic profession. It disbanded in December 1974.

Principle 3

The provision of the pre 1980 AMA Principles of Medical Ethics stating that a physician should practice a method of healing founded on a scientific basis and should not voluntarily professionally associate with anyone who violates that principle. Removed from the code in 1980.

Boycott

A concerted refusal to deal with a person or group. In antitrust law, an agreement among competitors to refuse to deal with another party can be an unlawful restraint of trade regardless of whether the participants' opinion of that party is correct.

Sherman Act, Section 1

The federal statute prohibiting contracts, combinations, and conspiracies in restraint of trade. It was the provision the Wilk plaintiffs sued under, and it makes the defendants' conduct the issue rather than the plaintiffs' merits.

Clayton Act, Section 16

The federal provision permitting a private plaintiff in an antitrust case to obtain an injunction. It was the basis for the permanent injunction entered against the AMA in 1987.

Less restrictive alternative

The test the court applied to the AMA's patient care defense: a broad restriction is not justified by a legitimate purpose if a narrower measure, such as public education, would have served that purpose.

Category error

Treating an answer of one kind as though it settled a question of another kind, as when a legal finding about an institution's conduct is offered as evidence about whether a treatment works.

Integrated clinic

A practice in which clinicians from different professions share a facility and, in the genuine version, share records and management of patients. The test of whether the integration is real is whether anyone changes a plan because of what a colleague found.

Active care versus maintenance care

The distinction Medicare draws between treatment aimed at correcting an acute or chronic problem, which is covered, and ongoing supportive treatment after maximum therapeutic benefit has been reached, which is not.

Essential health benefits

The categories of coverage required of individual and small group health plans under the Affordable Care Act. Chiropractic is not named among them, so coverage depends on each state's benchmark plan and mandates.

Return leg

The information sent back to a referring clinician after a referred patient has been seen. It is the most frequently omitted part of a referral and the part that determines whether a referral relationship continues.

Before you move on

1. Name the three professions in this chapter that perform spinal manipulation, and name the two things a chiropractor's license does not permit that a physician's does.
2. Quote or closely paraphrase Principle 3, and then list the four separate channels of professional contact that its second clause severed.
3. Explain why the Wilk plaintiffs sued in antitrust rather than for defamation. What would each theory have required them to prove, and what would each have put on trial?
4. The court found that the AMA had not met its burden on the patient care defense. What test did it apply, and what less restrictive alternative did it name?
5. A classmate says the 1987 ruling proves chiropractic works. Write the two or three sentences you would say in reply, and explain why the mistake mirrors the AMA's own.
6. Medicare covers manual manipulation of the spine and essentially nothing else a chiropractor does. Argue both sides of whether that rule is defensible, using what chapter 5 established about the evidence.
7. You have treated a patient for six weeks with no measured change. Draft the referral, covering what was found, what was done, what changed, and what specifically you are asking for. Then say when and in what form you expect to hear back, and what you will do if you do not.

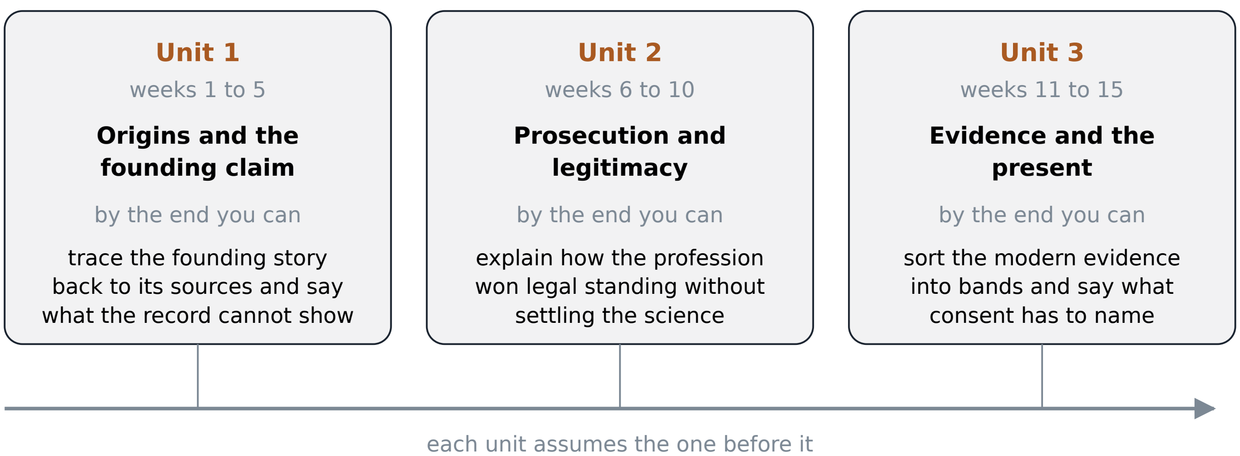
CHAPTER 7

What You Do With This

> *Seven weeks in, what does a graduate of this course now know how to do that she could not do on the first day?*

Six weeks ago this book asked you to hold open a question about what actually organizes a living body, then walked away from it to do six weeks of other work: the marketplace Palmer walked into and the philosophy he built in chapter 1, the anatomy in chapter 2, the mechanics of the adjustment and the uncomfortable reliability findings in chapter 3, the long road to licensure and what the working life actually contains in chapter 4, what the evidence supports and where it stops in chapter 5, and how this profession sits next to the rest of medicine, including the AMA boycott and the Wilk case, in chapter 6. There is no new material in this chapter. Its whole job is to take what you now know and turn it into something you can use: a position you can state plainly, a claim you can defend without flinching, and an honest way to talk about the parts that are not settled, whether the person across from you is a skeptical relative at a holiday table, a patient who wants more certainty than you can honestly give her, or yourself, if chiropractic school is somewhere in your future.

Fifteen weeks, three units



Each of the six preceding chapters supplied one layer of what this closing chapter now asks a graduate to use at once: history, anatomy, technique, licensure, evidence, and integration, combined into a single defensible position.

The question chapter 1 left open

Chapter 1 asked which of three readings of Innate Intelligence you leaned toward. Literal, in which a real organizing intelligence runs through the body and coordinates its healing by way of a nervous system that works better when it is not interfered with. Metaphorical, in which the phrase names something real, self regulation, homeostasis, the nervous system's own coordinating capacity, but claims no more for it than physiology already grants. Discarded, in which the concept is a historical artifact useful for understanding where the profession's vocabulary came from and no longer doing any explanatory work in how a practitioner actually reasons about a patient. Chapter 1 promised to ask again. Here is the second asking, and this time the point is not to pick a side in the abstract. The point is to notice that each answer is a set of instructions for what you actually say and do in front of a patient, and that a graduate of this course who has not decided which one she holds is not being appropriately humble. She is unequipped.

Take a concrete case, one that every practicing chiropractor eventually meets in some form. A patient comes in with a cough that has gone on ten days, a fever, chest pain on breathing. You suspect pneumonia. What do you say and do?

The practitioner who reads Innate Intelligence literally does not, in any serious modern form of the position, tell that patient an adjustment will cure the pneumonia in place of antibiotics. That caricature is worth naming and setting aside, because it is not what the literal reading actually commits a working practitioner to. It survives mostly as something critics say chiropractors believe rather than something chiropractors say. What the literal reading does commit her to is a claim about mechanism underneath the referral: that the body possesses an intelligence organizing its own defenses, that this intelligence works through a nervous system which functions better when it is not mechanically interfered with, and that removing that interference is worth doing alongside, not instead of, whatever medical treatment the case requires. She refers immediately, because the literal reading has never denied that lungs full of bacteria need antibiotics. A compromised nervous system was never claimed to substitute for one. What changes is the story she tells herself, and sometimes tells the patient, about why the adjustment she still offers for that same patient's unrelated low back pain matters: it is not merely palliative to her, it is participating in a coordinating capacity of the body that she takes to be literally real.

The practitioner who reads it metaphorically arrives at the identical referral by a different route, and notably says less. She does not reach for the vocabulary of intelligence at all when she explains the pneumonia. She talks about the immune system, inflammation, oxygen exchange, the ordinary language of physiology, because for her Innate Intelligence was always a nineteenth century name for facts physiology now describes more precisely: homeostasis, neural feedback, the body's documented and bounded capacity to regulate itself. She still believes those regulating capacities are worth respecting, and that a nervous system under less mechanical stress probably does its regulating work somewhat better, which is why she still practices. But she does not experience herself as invoking a distinct organizing intelligence when she says so. Ask her whether Innate Intelligence is real and she will likely say the question is not well formed. Something real is being pointed at, self regulation, but the word intelligence claims a kind of agency the physiology does not require.

The practitioner who has discarded the concept entirely does not use it in the exam room or in her own reasoning at all. She treats the patient's low back pain with a biomechanical and neurophysiological model, adjusts for measurable and functional reasons, and regards the pneumonia as wholly outside her scope, to be handled by someone else, full stop. If you asked her what Innate Intelligence is, she could describe Palmer's idea accurately, the way you can describe a theory you no longer hold, but she would tell you it does no work in her clinic. For her, the value of chapter 1 was historical: understanding where the profession's vocabulary came from, not carrying that vocabulary forward into practice.

All three of these are held, right now, by licensed and competent chiropractors, and this book has not tried to talk you into any of them. What it has tried to do is show you that each one is a real, defensible position with its own internal logic, held by people who can state it in its strongest form, and that a graduate who cannot say which one she holds, or why, has not finished the course even if she has finished the reading. Practicing without a position is not neutrality. It is having an unexamined position, which is worse, because it will surface under pressure, in front of a patient, in a form you did not choose.

Practicing without a position is not neutrality. It is having an unexamined position that will surface under pressure, in a form you did not choose.

> **A note on what having a position does not mean**
>
> Stating a position is not the same as being unable to update it, and it is not permission to argue your metaphysics with every patient who has not asked. I know practitioners who hold the literal reading and rarely mention Innate Intelligence to a patient across an entire career, because it is not what she came in for. Having a position means you know what you would say if asked, and that what you would say has been thought through rather than improvised in the moment. It does not mean turning every visit into a lecture.

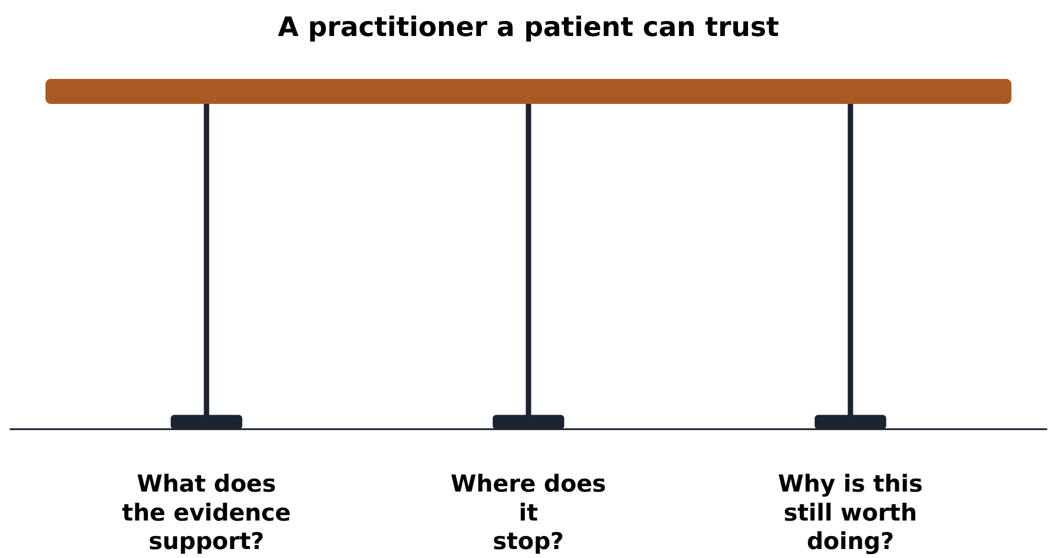
What a graduate of this course should be able to do

There are three things this course has been trying to build in you since week one, and they are worth stating as a short, memorizable list, because you will need to produce all three under pressure, and pressure is not when new thinking happens well.

First, you should be able to say, specifically and without hedging, what the evidence supports. For low back pain, chapter 5 walked through this in detail: the major reviews and the 2017 American College of Physicians guideline place spinal manipulation among several reasonable first line options for acute and chronic low back pain, with evidence quality generally rated low, and manipulation performing comparably to, not dramatically better than, the other conservative options on that list. For neck pain, manipulation used alone has not shown benefit over placebo in controlled trials, though manipulation combined with exercise has. For cervicogenic headache, a real but small and short lived effect. That is not a weak position to hold. It is a specific, defensible one, and specific claims are easier to hold under cross examination than vague ones.

Second, you should be able to say, just as plainly, where the evidence stops. For migraine, the honest answer from the most recent pooled trial evidence is that manipulation has not shown a meaningful effect, and that adverse events were more common in the treated group than in controls, mostly mild and transient, but more of them all the same. For most conditions outside the musculoskeletal system, colic, ear infections, asthma, high blood pressure, the evidence is thin, inconsistent, or absent, and chapter 5 asked you to notice that the gap between what professional bodies claim and what independent reviewers find is exactly here, in this territory. You should also be able to state clearly what the reliability findings from chapter 3 do and do not imply: that practitioners agree reasonably well with one another on where a patient reports pain, and agree poorly on which specific segment is restricted, and that this is a real problem for the assessment step of the traditional model, not proof that treatment does not work, since those rest on different bodies of evidence entirely.

Third, and this is the one students find hardest to say out loud without either overclaiming or apologizing, you should be able to say why the work is still worth doing. Not because the evidence is stronger than it is. Because a hands on, unhurried, musculoskeletal focused visit helps a large number of people with a common and disabling complaint, at low risk when red flags are respected and referrals are made when they should be, in a health system that mostly does not have time to offer that kind of visit anywhere else. That is a real reason, and it does not require pretending the evidence says more than it says.



A practitioner's position is stable only when she can answer all three questions, what the evidence supports, where it stops, and why the work is worth doing regardless, since missing any one produces its own specific and predictable failure.

Now watch what happens when one leg of that stool is missing, because this is where the three questions stop being an academic exercise and start being a description of practitioners you will actually meet.

A practitioner who can only do the first, who can rattle off what the guidelines say and nothing else, is a technician. Ask her why she chose this field instead of physical therapy, which the same guidelines treat as a comparable option for many of the same complaints, and she has no answer beyond credentialing, because she never worked out for herself what is distinctive about the practice she is in. She will not last, or she will last unhappily.

A practitioner who can only do the third, who talks fluently about touch and time and the value of being heard, and who has never actually sat with the trial data, is a believer, and believers get caught out. Sooner or later a patient's relative who works in medicine, or a patient who read the Cochrane review herself, will ask a specific question, and a practitioner who cannot answer it specifically loses the room, deservedly.

A practitioner who cannot do the second, who cannot say plainly where the evidence stops, is the most common failure and the most avoidable, because it does not look like a failure in the moment. It looks like confidence. It is the practitioner who tells a parent that a course of pediatric adjustments will help a child's recurring ear infections, because a colleague swears by it, without saying that the controlled evidence for that claim is thin and that mainstream and independent reviewers read it very differently, as chapter 5 laid out plainly. That practitioner will eventually be embarrassed in front of a patient, and the embarrassment will be earned, because the patient did the reading the practitioner should have done first.

The relative at the holiday table

You will have this conversation. Not might, will. A cousin, an uncle who is a nurse, a physician you meet at a party, a patient who read something the night before her appointment and arrives half convinced you are a fraud. The instinct under challenge is to defend everything at once, the philosophy, the technique, the whole history, because it all feels like one thing under attack. Resist that instinct. It is exactly backward.

What actually works, and I have had enough of these conversations across enough years of holiday tables to trust this, is three moves, in order. Know what the evidence supports well enough to say it without hedging, so you are not scrambling for a number while your uncle watches you scramble. Concede plainly what is genuinely contested, rather than defending the whole enterprise as though every part of it stood or fell together, because conceding a real weakness is the single most credible thing you can do in that conversation, and most people have never watched a chiropractor do it. And keep the founding story and the evidence base in two separate mental boxes, because your skeptic will often blur them together for you, and if you blur them back apart, you have already won more of the conversation than you realize.

Here is roughly how it goes when it goes well. Your uncle says chiropractic is quackery, that a man in the 1890s made it up to cure deafness by hitting a janitor on the back, and that nothing about that has changed since. You do not defend the Lillard story, because chapter 1 already told you the documentary record on that story is genuinely a mess, three different accounts that do not agree with one another, and pretending otherwise here would be dishonest and would also lose the argument, since he may know some of this too. You say something closer to this: you are right that the origin story does not hold up as clean history, and that is a fair thing to bring up, and it is also a different question from whether manipulation helps a person with low back pain today, which is the question the actual trials were designed to answer, and there the honest answer is that it helps about as well as several other conservative options, with evidence quality that is generally rated low rather than strong, for a problem that is extremely common and expensive to leave untreated. Then you stop talking. You do not chase the founding story further, and you do not oversell the trials further either. You said what the evidence supports, you said where it stops, and you separated the founding claim from the clinical one instead of letting him keep them tangled together, which was what was actually doing the work in his objection.

That will not always convert your uncle. It is not supposed to. What it does is make you someone whose claims can be checked, which is a different and better thing to be than someone who won the argument.

A note on the other two conversations

The physician who is dismissive of chiropractic outright, sometimes in front of a shared patient, is a different conversation, and the wrong move is to argue chiropractic's case to her at all. Chapter 6 already gave you the history that makes some of that dismissal understandable even where it is unfair: a real antitrust boycott ran from the 1960s into the 1980s, and some of what you meet in practice is its residue rather than a considered judgment of the current evidence. The useful response is not a debate. It is a well written referral letter, on time, with a clear reason for referral and a clear statement of what you found and did not find, because that is the thing that reliably changes how a skeptical colleague treats the next patient you send her. And the patient who arrives with something she read online the night before deserves neither dismissal nor capitulation. Ask her what she read. Often it is a real finding, stated more starkly than the source intended, and the honest response is to tell her plainly which parts of what she read chapter 5 would confirm and which parts overreach, rather than deciding in advance that she was wrong to have looked at all.

The patient who wants more certainty than you have

The skeptic conversation is, in an odd way, the easier one, because the skeptic does not need anything from you emotionally. A patient does. She is in pain, she wants to know it will get better, and she wants to hear that from someone who sounds sure. That want is completely reasonable, and it creates the single strongest pull toward dishonesty you will feel in this profession, stronger than any pull to oversell to a journalist or a jury, because it happens quietly, one appointment at a time, with someone who is trusting you.

The temptation is to give her more certainty than the evidence gives you. To say this will fix it, rather than this helps most people with what you have, and here is what we will watch for. The trade looks free in the moment. She leaves reassured, you have had a good visit, everyone feels better. It is not free. Chapter 5 already asked you to sit with the patient who does not improve, and that patient exists in every real practice, not as a rare exception but as an expected fraction of the people who walk in. When she is the one who does not improve, the certainty you sold her becomes a debt. She does not conclude that spinal manipulation has a modest, well documented effect that did not happen to apply to her case. She concludes that you either did not know what you were talking about or were not honest with her, and both conclusions are reasonable given what you told her, and both cost you the trust of everyone she talks to afterward.

The honest version is harder to say, and it still leaves a patient feeling cared for, if you say it right. Something like this, to the patient in front of you with three months of low back pain and no red flags on your exam: most people with what you have improve substantially over the coming weeks regardless of which reasonable treatment they choose, and manipulation is one of several options with evidence behind it, generally modest evidence rather than dramatic evidence, and I think it is a reasonable one to try alongside staying active, and here is specifically what I will be checking at each visit to tell whether it is working for you, and here is what we do if it is not. Notice what that sentence does. It does not promise. It commits to a specific, checkable plan, and it tells her in advance what not working will look like and what happens next, which is exactly what removes the sting when not working turns out to be the answer. Patients do not actually need certainty from you. They need to know you have a plan, that the plan has a way of failing that you already thought about, and that you will not disappear if it does. That is a version of confidence built on the ground you actually have, a hands on visit, real attention, and a modest but genuine evidence base, rather than on ground you do not have.

Patients do not need certainty from you. They need to know you have a plan and a way of telling if it is failing.

If chiropractic school is where you are headed

Some of you are reading this book to find out whether you want this life, and this book has tried not to talk you into it or out of it anywhere in six chapters. It will not start now. What it can do is give you the questions worth sitting with before you commit four years and a large sum of money to finding out.

Sit with the cost honestly, because chapter 4 gave you real numbers and they are worth rereading before you apply rather than after you enroll. Recent graduates surveyed in 2024 and 2025 carried a mean student loan debt at graduation of $176,297, against a mean total cost of attendance across programs of $260,642, and graduates as a group have been shown to underestimate what they would actually owe by roughly $50,000 relative to what their own schools publish. That is not a reason to avoid the field. It is a number you should know cold before you sign anything, and you should ask a program directly, in writing, what its graduates actually earn and how long it takes them to reach a stable income, rather than accepting a national average that may not describe your state or the patient population you plan to serve.

Sit with the working life chapter 4 described honestly, not the version in the recruiting brochure. Most new graduates spend their first years as an associate before they are anywhere near running their own practice, if they ever do. A working day includes documentation, insurance paperwork, and patients who do not show up, in roughly the proportions any small medical business does, and the hands on part of the job, the part that probably drew you to it, is only part of what fills a week. Ask a working chiropractor, more than one, what an ordinary Tuesday actually looks like, not what a good Tuesday looks like.

Sit with the philosophical question this whole chapter has been asking you to answer for yourself, because you cannot borrow someone else's position and wear it convincingly for forty years. If you are drawn to this field because Innate Intelligence read literally speaks to something you already believe about the body, that is a real and legitimate reason to go, and you should look for a program that will let you develop that position rigorously rather than either mock it or leave it unexamined. If you are drawn to it because you read the evidence in chapter 5 and concluded that a hands on, evidence informed musculoskeletal practice is a genuinely useful thing to spend a career doing, whatever your view of Innate Intelligence ends up being, that is also a real and legitimate reason. What will not sustain you across a career is going because it sounded interesting once and you never got further than that, because the reliability findings in chapter 3, the scope of practice differences in chapter 4, and the contested territory in chapter 5 will all eventually land on your desk, attached to an actual patient, and you want to have already thought about them once, here, before they arrive.

> **A note on shadowing before you apply**
>
> Every piece of advice in this section is secondhand until you have spent time actually watching a practice, ideally more than one, in more than one setting: a solo practice, a multidisciplinary clinic, a practice built around a single technique family from chapter 3. Ask to see a full day, not a curated hour. Ask what the practitioner wishes someone had told her before she enrolled, since chapter 4 already told you what practitioners say when asked that question in surveys. Go hear it from someone in your own town, in person, before you commit four years to finding out for yourself.

The last assignment

The final assignment in this course asks you to explain chiropractic, in about five to seven minutes of video, to someone who has never heard of it. That sounds like the easiest task in the syllabus. It is the hardest, and it is the reason every other week existed.

Explaining something simply is not the same as knowing a simplified version of it. You cannot do it by memorizing a paragraph. To explain chiropractic in five to seven minutes you first have to have actually understood it, all the way from the marketplace Palmer walked into in chapter 1 to the referral letter a working chiropractor writes today in chapter 6, because a listener with no background will ask, without knowing it is a hard question, exactly the question your explanation is weakest on. Then you have to decide, out loud, in front of a camera, what matters most, which means leaving nearly everything out, since seven minutes will not hold six chapters, and the leaving out is itself a judgment about the field that only you can make and that will reveal, more honestly than any exam could, what you actually think this profession is for. And then you have to do the hardest part of everything this chapter has covered: be honest, inside a short, confident, watchable explanation, about what is settled and what is not, without either running from the uncertainty or hiding behind it.

That is not a video assignment about chiropractic. It is the whole course, compressed to the length of a conversation you are actually going to have, with a patient, a relative, or yourself, sooner than you think. Everything in the last seven weeks was preparation for being able to give that explanation and mean every word of it.

Terms from this chapter

Literal reading

The view that Innate Intelligence is a real, distinct organizing force working through the nervous system to direct the body's healing, claiming more than ordinary physiology already accounts for.

Metaphorical reading

The view that Innate Intelligence names something real, the body's own capacity to regulate and repair itself, without claiming any force beyond what physiology already describes.

Discarded reading

The view that Innate Intelligence is a historical concept worth understanding for context but doing no working role in present day clinical practice or reasoning.

Evidence base

The body of controlled research, guideline statements, and systematic reviews supporting or failing to support a given clinical claim, as distinct from tradition, testimonial, or founding narrative.

Founding story

A dated narrative of a profession's origin, which chapter 1 showed can do real institutional work regardless of how well it survives as history.

Red flag

A sign or symptom, such as unexplained weight loss or sudden bladder dysfunction, that indicates a serious underlying condition requiring referral rather than manipulation.

Referral letter

The written communication a practitioner sends to another provider, stating findings and reasoning clearly enough to be trusted by a colleague who may be skeptical of the sender's profession.

Informed consent

A patient's right to understand what a proposed treatment can and cannot be expected to do, including what is genuinely uncertain, before agreeing to it.

Reliability

A statistical measure, often expressed as kappa, of how consistently practitioners agree with one another or with themselves on a clinical finding; chapter 3 showed this differs sharply between pain findings and motion findings.

Debt to cost gap

The documented tendency of chiropractic graduates to underestimate, often by tens of thousands of dollars, what their education will actually cost them by the time they graduate.

Scope of practice

The specific procedures and treatments a state licensing board permits a chiropractor to perform, which vary substantially from state to state.

Position

A practitioner's considered, statable, and defensible stance on an unsettled question, held openly enough to survive being asked about directly.

Before you move on

1. Name the three readings of Innate Intelligence from chapter 1, and describe what each commits a practitioner to when a patient presents with a serious non musculoskeletal illness such as pneumonia.
2. What are the three things a graduate of this course should be able to state, and what specific kind of practitioner results when each one, in turn, is missing?
3. Which of the three readings of Innate Intelligence do you hold now, and has your answer changed since chapter 1? Defend it in the vocabulary of that position, in its strongest form.
4. What is the difference between conceding a genuinely contested point to a skeptic and defending the whole profession as though every part of it stood or fell together?
5. Write, in your own words, what you would say to a patient with three months of low back pain who asks whether an adjustment will fix her problem, using the honest uncertainty approach this chapter describes.
6. What two figures from chapter 4 should a prospective chiropractic student know before enrolling, and why does this chapter warn that graduates tend to underestimate one of them?
7. What is the hardest part of the final video assignment, and why does this chapter argue that answering it well requires everything covered in the previous six weeks rather than just this one?